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# Periodic billiard trajectories in polygons: generating mechanisms

**Ya.B. Vorobets, G.A. Gal'perin, and A.M. Stepin**

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## Introduction

In this article we discuss means for constructing and detecting periodic billiard trajectories in triangles, one of the main items being the study of periodic trajectories in *obtuse triangles*. As a result of the investigation there naturally arises the question of describing those trajectories that are not destroyed by an *arbitrary small deformation* of the triangle. We will call such trajectories *stable*, and for certain cases we will give a complete description of them. The definition of a stable periodic trajectory is proposed in this paper for the first time ever; it can be naturally given for an arbitrary polygon or even a polyhedron. Information on the presence of a stable trajectory in a given polygon allows us to indicate an *open set* of polygons in which there is

at least one periodic trajectory. The question of the existence of a periodic trajectory in an arbitrary polygon (triangle) is still open. The aim of the present paper is to indicate the set of triangles in which it is now known that periodic trajectories exist, and to give a constructive description of them.

We list the results contained in this paper.

In §1 and §2 we give the basic definitions and examples of billiards in various plane domains, and we give mechanical interpretations of periodic trajectories. §3 is devoted to an extremal property of billiard trajectories which can be visually demonstrated using convex billiards (Birkhoff) with smooth boundary and scattering billiards (Sinai) with piecewise smooth boundary, 'pressed' inside the domain. This section closes with the assertion, actually proved, that in contrast to the billiards just mentioned, for billiards in polygons there is no *unified* construction for periodic trajectories. In §4 we construct so-called 'perpendicular' periodic trajectories in triangles of special shape. In §5 we subsequently prove the existence of 'perpendicular' trajectories in 'rational' polygons and polyhedra.

From §6 onwards we study *stable periodic trajectories* in polygons, in particular in triangles. These are characterized by the fact that their combinatorial type does not change under a small deformation of the polygon, and, moreover, they remain periodic and close to their original position. We show that each periodic trajectory can be coded by a 'word', and we study the stability of the trajectory in terms of words. It turns out (§6) that in a 'typical' polygon *all* trajectories are stable (provided they exist), that there are stable perpendicular trajectories (§7), and that *the number of links of a stable perpendicular trajectory in an arbitrary polygon is a multiple of* 4 (§7). In §8 we consider *isolated trajectories*, consisting of an odd number of links. We prove that all isolated trajectories are stable, and that any rational polygon can contain only *finitely many* such trajectories. We also give an upper bound of this number, depending only on the geometry of the polygon.

§9 is devoted to isolated trajectories in *triangles*. We propose a construction for these trajectories, and give the bifurcation diagram of triangles containing isolated trajectories. In §10 we compute the limiting density of the triangles containing isolated trajectories, when their acute angles tend to zero. In §11 the theme of isolated trajectories is completed, generalizing the construction given in §9.

In the remaining part of the article we study the stability of periodic trajectories in *plane Weyl chambers*: in a rectangle, in a right-angled triangle with angles of 45°, 45°, in an equilateral triangle, and in a right-angled triangle with angles of 30°, 60° (§12), and also in a regular hexagon (§13).

It turns out that in a rectangle and in the two types of right-angled triangles indicated *any periodic trajectory* is *unstable*. In the case of an equilateral triangle and a regular hexagon all stable trajectories can be described in an invariant way. The proofs use a special technique expounded in this article.

In the Conclusion section we make some remarks and give additional assertions, state unsolved problems, and pose some conjectures on periodic trajectories in polygons.

## §1. Billiard trajectories in a plane domain

Let $Q$ be a figure in the plane whose boundary $\partial Q$ consists of finitely many smooth curves. A *billiard trajectory* in $Q$ is a polygonal line $\Gamma$ (in general, infinite on both sides) any two segments of which with common vertex $P \in \partial Q$ form *equal* angles with the tangent $l$ to the boundary at $P$ (Fig. 1). The segments of $\Gamma$ are called its *links*, the vertices *reflection* or *break* points, and the angles with the tangent at a reflection point *angles of incidence* or *reflection*. The figure $Q$ itself is called a '*billiard*' or '*billiard table*'. It is assumed that $\Gamma$ does not hit corner points of $Q$ (and if it does, then it should terminate at them). If $\partial Q$ contains a rectilinear segment $l$, the angles of incidence and reflection are the angles of the two adjacent links of $\Gamma$ with $l$ (Fig. 2).

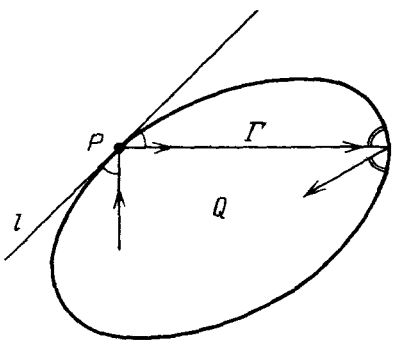



Fig. 1

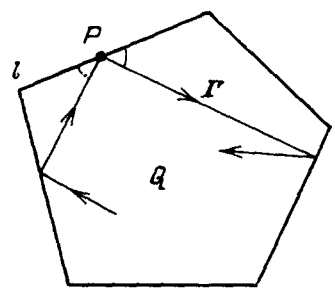



Fig. 2

A billiard trajectory $\Gamma$ is called *periodic* if it is closed: a billiard particle launched from an arbitrary point $q$ of this trajectory returns to $q$ after a certain time while moving in the initial direction (we take the modulus of the velocity of the particle constant and equal to 1). The various questions concerning the behaviour of billiard trajectories are related to problems in geometry, mechanics, geometrical optics, and other branches of modern mathematics and physics. In studying these problems a special role is played by periodic trajectories.

## §2. Fagnano's problem. Mechanical interpretations of periodic trajectories in triangles

The first question about periodic trajectories which arises immediately is: are there such trajectories at all in a given 'billiard' $Q$? In fact, it is precisely this question which we will discuss in the present paper, in the case when $Q$ is

a triangle or polygon (and, occasionally, a polyhedron). This question is a difficult one, and an exhaustive answer to it has not been given. However, for certain figures $Q$ we can not only prove the existence of periodic trajectories in $Q$, but even study the structure of these trajectories in some detail.

**A. A billiard in an acute triangle.**
The first example is a billiard in an *acute* triangle $ABC$. The simplest periodic trajectory $\Gamma$ in it has three links, whose reflection points $H_1$, $H_2$, $H_3$ are the bases of the altitudes $AH_1$, $BH_2$, $CH_3$ (Fig. 3). The trajectory $\Gamma$ is *unique* among the three-link trajectories in $ABC$, and has the property that its length minimizes the perimeter of the inscribed triangle in $ABC$. This assertion is known as the 'Fagnano problem' (see [7], [2]), while the triangle $H_1H_2H_3$ is known as the 'altitude' triangle or 'Schwarz triangle' (the solution of Fagnano's problem was first given by the German mathematician H.A. Schwarz (1843–1921)).

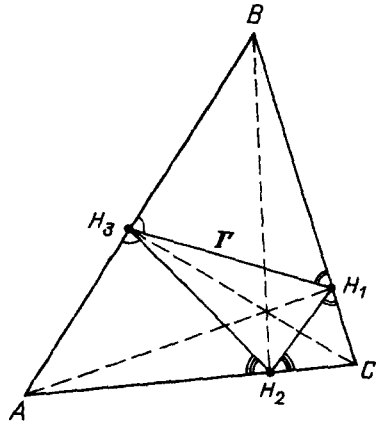



Fig. 3

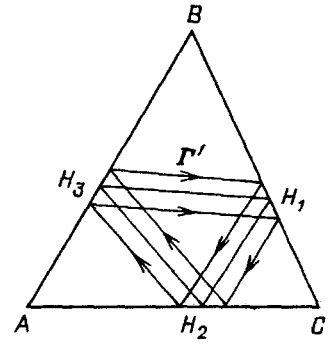



Fig. 4

If we send the billiard particle parallel to one of the links of the trajectory $\Gamma$ constructed above, then it describes a 6-link periodic trajectory $\Gamma'$, consisting of three pairs of links parallel to the sides of the Schwarz triangle $H_1H_2H_3$ (Fig. 4).

Hence there arises a whole family of 6-link periodic trajectories $\{\Gamma'\}$, 'parallel' to $\gamma$, while $\Gamma$ itself is naturally included in this family as a twofold 3-link trajectory. In the sequel we will call such families *pencils of parallel periodic trajectories*.

**B. Mechanical interpretations.**
It is not difficult to give a mechanical interpretation of the 3-link trajectory $\Gamma$ in an *acute* triangle. Suppose that on each side of $\Delta ABC$, which we assume to be a short wire, a ringlet is laid, where these ringlets may move freely along

the sides. Neighbouring ringlets are connected by a rubber band and are set free. When reaching the stationary state, the ringlets and rubber bands will form the vertices and sides of a triangle $H_1H_2H_3$ of least perimeter and inscribed in $\Delta ABC$. Moreover, the equilibrium forces acting on neighbouring ringlets are orthogonal to the sides of $\Delta ABC$ and are directed along its altitudes. This follows from the fact that the tensions of all ringlets in the stationary state are equal to one another. Thus, $H_1H_2H_3$ is a periodic billiard trajectory (Fig. 5a).

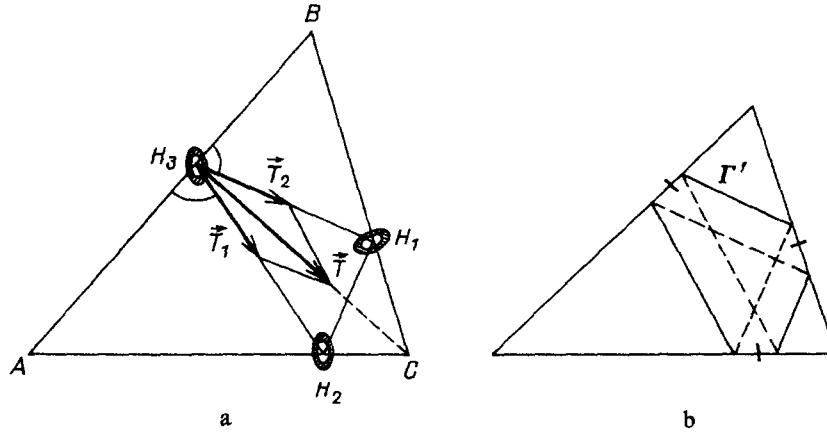



Fig. 5

Another mechanical interpretation is as follows. If the triangle $ABC$ is considered to be a plane smooth plate, then the twice-traversed 3-link trajectory $\Gamma$ and each 6-link trajectory $\Gamma'$ in the pencil of trajectories parallel to $\Gamma$ can be represented as an elastic closed thread winding around this plate and in turn passing from one side ('boundary') to another (Fig. 5b).

C. However, if $ABC$ is an *obtuse* triangle, then the first mechanical construction does not work. The problem here is that in this situation there is no triangle of least perimeter *inscribed* in $\Delta ABC$ (such a triangle degenerates to the altitude drawn from the obtuse angle). The attempt to find a periodic trajectory using the second construction, as an elastic thread winding around a triangular plate, is also doomed to fail. In fact, on a smooth plate the thread jumps off at the vertex of the obtuse angle.

The mechanical models given demonstrate the fact that periodic billiard trajectories in an obtuse triangle (more generally, in an arbitrary polygon) must be looked for not among the triangles inscribed in it (among the inscribed polygons with the same number of sides as the original), as we do in the case of an acute triangle, but among the wider class of inscribed self-intersecting polygonal lines.

## §3. An extremal property of billiard trajectories. Birkhoff's theorem. The non-existence of a unified construction of periodic trajectories in obtuse triangles

**A. A convex billiard with smooth boundary.**
Another example of a billiard in which there are obviously periodic trajectories is a convex bounded figure $Q$ with smooth boundary $\partial Q$. In the 20's the American mathematician G. Birkhoff proved that *in such a billiard there is an n-link periodic trajectory, for any integer* $n \geqslant 2$.

We give a short proof of Birkhoff's theorem. For $n = 2$ a periodic trajectory is given by the twice-traversed diameter $AB$ of $Q$. In fact, the segment $AB$, being the largest chord in $Q$, *is perpendicular* to the tangents $l_A$ and $l_B$ to the boundary $\partial Q$ at $A$ and $B$, since otherwise a small shift of one of the ends $A$ or $B$ would increase the length of the chord $AB$. For arbitrary $n > 2$ we consider the set of all inscribed closed polygonal lines in $Q$ with number of links not exceeding $n$. We choose from these the polygonal line $\Gamma_n$ of *maximal* perimeter (such a polygonal line exists by the continuous dependence of the perimeter on the line and the compactness of the set under consideration).

Since $Q$ is convex: 1) the *polygonal line* $\Gamma_n$ *has exactly n links*, since otherwise any link $AB$ of it could be replaced by two links $AC$ and $CB$, taking $C$ to be a point on the arc of $\partial Q$ between $A$ and $B$, thus increasing the number of links and the perimeter of $\Gamma_n$; 2) if $P_{k-1}$, $P_k$, $P_{k+1}$ are three adjacent vertices of $\Gamma_n$, then for all points $P$ on the arc between $P_{k-1}$ and $P_{k+1}$ the sum of the lengths of the chords $P_{k-1}P$ and $PP_{k+1}$ is *maximal* for the point $P = P_k$ (this follows from the maximality of the perimeter of $\Gamma_n$). And then in the family of confocal ellipses with foci at $P_{k-1}$ and $P_{k+1}$ there is exactly one ellipse tangent to $\partial Q$ at $P_k$ (again in view of the convexity of $Q$). The optical (billiard) property of ellipses implies that the segments $P_{k-1}P_k$ and $P_kP_{k+1}$ form equal angles with the tangent $l$ to $\partial Q$ at $P_k$ ($l$ coincides with the tangent to the ellipse). Hence, $\Gamma_n$ is a *billiard trajectory*, and Birkhoff's theorem has been proved. □

Birkhoff also obtained a lower bound for the number $P_n$ of $n$-link periodic trajectories: $P_n \geqslant \text{const} \cdot n^2$. This estimate has also been obtained recently by Treshchev [12], who used topological methods.

**B. Scattering billiards (Sinai billiards).**
Yet another example of a domain $Q$ for which the existence of periodic trajectories has been proved is the *Sinai billiard*: its boundary $\partial Q$ consists of finitely many smooth arcs, convex inside $Q$ (Fig. 6), For this billiard we can give the same mechanical interpretation of periodic trajectories as for an acute triangle (§2B). In fact, on the boundary components we put ringlets, connect them by elastic bands, and we let this construction move freely. The bands tighten, and form a periodic billiard trajectory. As in the case of an acute triangle, in scattering billiards the length of each periodic trajectory is *minimal*

among the lengths of all polygonal lines with vertices on the same boundary sides as the reflection points of the trajectory; moreover, this minimum is strict, that is, *unique*.

It was proved in 1989 by the Bulgarian mathematician L. Stoyanov that the number $P_n$ of $n$-link periodic trajectories in the Sinai billiard with $k$ sides does not exceed $k^n$. This bound is based on the following *uniqueness principle*: *there cannot be two distinct periodic trajectories with the same number of links and successively reflecting from the same boundary components.*

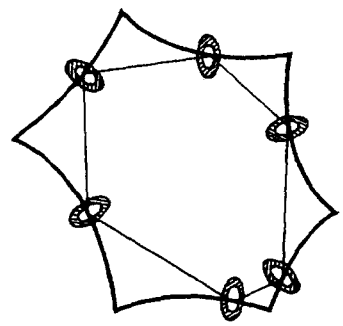


Fig. 6

**C. An extremal property of billiard trajectories.**

In turn, the uniqueness property for Sinai billiards follows from the Birkhoff *extremality property*: *every periodic trajectory has an extremal* (minimal or maximal) *length among all closed polygonal lines inscribed in the figure Q and with vertices close to the reflection points of the given trajectory.* This property also holds for Birkhoff billiards and for billiards in polygons: in the first case the length of a periodic trajectory is maximal, and in the second minimal.

For Sinai billiards Chernov[1] obtained a lower bound for the number of $n$-link periodic trajectories: $P_n \geqslant L^n$, where $h = \ln L$ is the entropy of the billiard. Moreover, he proved that *the periodic trajectories in a scattering billiard form an everywhere dense set in the phase space* [5].

**D. Periodic trajectories in obtuse triangles.**

Both for Birkhoff and Sinai billiards there is for $n \geqslant 3$ a lower bound for the number of $n$-link periodic trajectories (subsections A, C above). But already in the case of billiards in an obtuse triangle such a bound does not exist.

***Assertion.*** *For every n there is an obtuse triangle $\Delta(n)$ in which every periodic trajectory has more than n links.*

*Proof.* For a given $n$ we choose the acute angles $\alpha$ and $\beta$ of the triangle $\Delta(n) = \Delta ABC$ so small that $\min(-[-\pi/\alpha], -[-\pi/\beta]) > n$ ($[\cdot]$ denotes the integer part of the number; $N(\alpha) = -[-\pi/\alpha]$ is the maximal possible number

[1] N.I. Chernov, Topological entropy and periodic points of two-dimensional hyperbolic billiards, Funktsional. Anal. i Prilozhen. **25**:1 (1991), 50–57 = Functional Anal. Appl. **25** (1991), 39–45. MR **92c**:58101.

of reflections of a billiard particle in a sector of angle $\alpha$). Moreover, we require $\alpha$, $\beta$, and $\pi$ to be rationally independent:

$$(k,\ m,\ l \in \mathbb{Z},\ k\alpha + m\beta + l\pi = 0) \Rightarrow (k = m = l = 0).$$

(In other words, we require all angles of the triangle $\Delta(n)$ to be rationally independent.)

Let $\Gamma$ be a periodic trajectory in $\Delta(n)$. We show that at least one of its links $XY$ has endpoints on the lateral sides $AB$ and $BC$ ($\measuredangle B > \pi/2$) (Fig. 7).

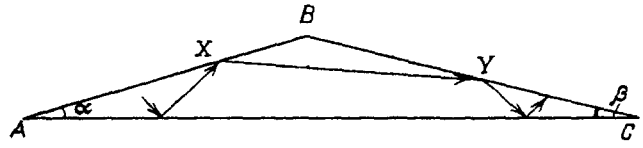



Fig. 7

*Proof* by contradiction: Suppose the billiard particle hits the base $AC$ after each reflection in a lateral side. Then any two angles of *reflection* in the base $AC$ are related by $\varphi' - \varphi'' = k\alpha + m\beta$ ($k, m \in \mathbb{Z}$), since adjacent angles of reflection in $AC$ differ in modulus only by $2\alpha$ or by $2\beta$. Hence, if the particle had started under an angle $\varphi_0$ with $AC$, then after the first reflection in $AC$ the angle of reflection would be $\varphi = \varphi_0 + k\alpha + m\beta$. Since $\Gamma$ is a periodic trajectory, for certain $k, m \in \mathbb{Z}$ the angles $\varphi$ and $\varphi_0$ differ by a number of full rotations:

$$\varphi = \varphi_0 + 2\pi s \Rightarrow k\alpha + m\beta = 2\pi s \Rightarrow k = m = s = 0$$

in view of the traditional independence of $\alpha$, $\beta$, and $\pi$.

So we have proved that $\Gamma$ contains a link $XY$, $X \in AB$, $Y \in BC$. This link forms so small an angle with the side $AC$ that after reflection in $AB$ and $BC$ at $X$ and $Y$ the particle has performed $N(\alpha)$ and $N(\beta)$ reflections inside the angles $A$ and $B$, respectively. Hence $\Gamma$ contains more than $n$ links, and the proposition is proved. □

**E. Discussion.**

The 'obstacle' hindering the transference of the proof of Birkhoff's theorem to *obtuse* triangles is the fact that for a given $n < \min(N(\alpha), N(\beta))$ the required $n$-link periodic trajectory has break points falling at a vertex of the triangle, while this was forbidden. If we smooth the boundary at all vertices, the Birkhoff theorem is valid for the convex domain $Q$ thus obtained; here all break points of the $n$-link trajectory will be concentrated on the smoothed parts, and only when $n > \min(N(\alpha), N(\beta))$ will the break points be on the sides of the triangle.

The considerations in subsection D above shows that possible mechanisms of emergence of periodic trajectories in an obtuse triangle are fundamentally different from the mechanism for their emergence in convex domains with smooth boundary, and they also show that *there is no unique construction of periodic trajectories in triangles.*

## §4. 'Perpendicular' trajectories in obtuse triangles of special shape

If we wind a thread on a smooth triangular plate in the manner indicated in Fig. 8, then the thread does not come off the plate. Of course, the obtuse triangle in Fig. 8 is not arbitrary, but special, that is, its acute angles $\alpha$ and $\beta$ are related in a certain way. A 6-link periodic trajectory in this triangle has the property that it leaves one side $(AC)$ perpendicular to it and falls on the other side $(BC)$ also perpendicular to it. A trajectory leaving one side orthogonally is naturally called a '*perpendicular*' trajectory. A careful consideration of Fig. 8 reveals an idea for constructing more complicated 'perpendicular' periodic trajectories.

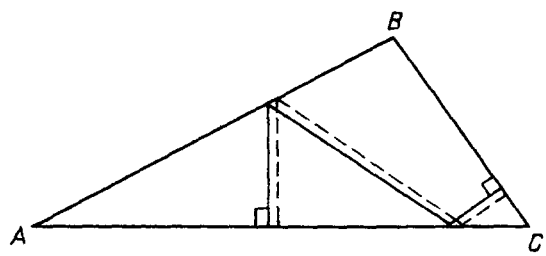



Fig. 8

**A. Periodic trajectories in triangles with a rational relation between the acute angles.**

***Assertion* 1.** *Suppose the acute angles $\alpha$ and $\beta$ of a triangle $ABC$ satisfy the conditions $k\alpha = l\beta < \pi/2$, where $k$ and $l$ are natural numbers. Then $ABC$ contains a 'perpendicular' periodic trajectory having $2(k+l)$ links.*

*Proof.* To prove this we use the well-known method of straightening (see, for example, [7]). We construct the chain of triangles

$$(1) \quad A_{-l}B_{-l}C_0;\ \ldots;\ A_{-1}B_{-1}C_0;\ A_0B_0C_0=ABC;\ \ A_0B_1C_1;\ \ldots;\ A_0B_{k-1}C_{k-1},$$

in which every two adjacent triangles have a common side and are positioned symmetrically with respect to this side (Fig. 9).

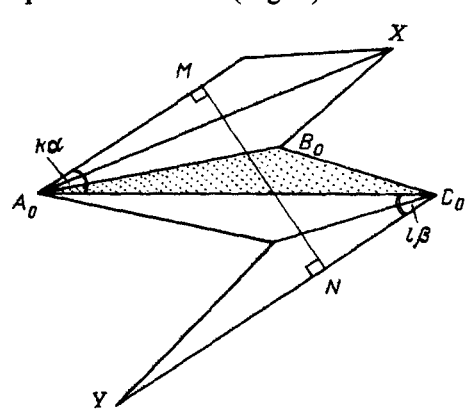



Fig. 9

For this we mirror-reflect the angle $B_0A_0C_0$ $k-1$ times *counterclockwise* around the vertex $A_0$, and the angle $B_0C_0A_0$ $l$ times also *counterclockwise* around the vertex $C_0$. One of the sides of the triangle $A_0B_{k-1}C_{k-1}$, call it $A_0X$, forms with $A_0C_0$ the angle $k\alpha$; similarly, one of the sides of $\Delta A_{-l}B_{-l}C_0$, call it $C_0Y$, forms with $C_0A_0$ the angle $l\beta$.

The condition $k\alpha = l\beta$ ensures that $A_0X$ and $C_0Y$ are parallel. We draw a common perpendicular $MN$ to $A_0X$ and $C_0Y$ separating the points $B_0$ and $B_{-1}$ (this can be done if $k\alpha < \pi/2$ and $l\beta < \pi/2$). The segment $MN$ is entirely contained in the corridor $\Phi$ formed by the triangles (1). The closed polygonal line obtained by traversing $MN$ twice is a periodic trajectory in $\Phi$. Suppose $MN$ intersects the sides of the triangles in the chain (1) at the points $N = M_{-l}, M_{-l+1}, \ldots, M_{-1}, M_0, M_1, \ldots, M_{k-1}, M_k = M$. We now 'fold' the corridor $\Phi$ 'like an accordion' so that we obtain $k+l$ triangular layers positioned above $\Delta A_0B_0C_0$ (the triangle $\Delta A_0B_0C_0$ itself is one such layer). Under this 'stacking' the segments $M_{i-1}M_i$ $(-l+1 \leqslant i \leqslant k)$ become the links of a 'perpendicular' periodic trajectory $\gamma$ in $\Delta A_0B_0C_0$. This $\gamma$ thus has $2(k+l)$ links (Fig. 10). Assertion 1 has been proved. □

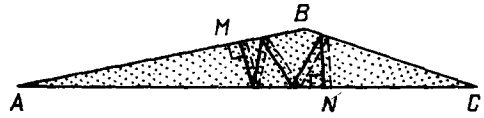



Fig. 10

*Remark.* In the particular case $k = l = 1$ we obtain a pencil of the simplest, 4-link, periodic trajectories in an *isosceles* triangle. By 'stacking' the isosceles triangle along its altitude we obtain from this pencil the pencil of 6-link periodic trajectories in a right-angled triangle. So we have proved that *in any isosceles and any right-angled triangle there are pencils of perpendicular trajectories.*

**B. Periodic trajectories in quasi-right-angled triangles.**

There are also 'perpendicular' periodic trajectories in obtuse triangles whose acute angles satisfy the relations $\alpha + k\beta = \pi/2$ $(k = 1, 2, \ldots)$, $2\alpha + \beta > \pi/2$.

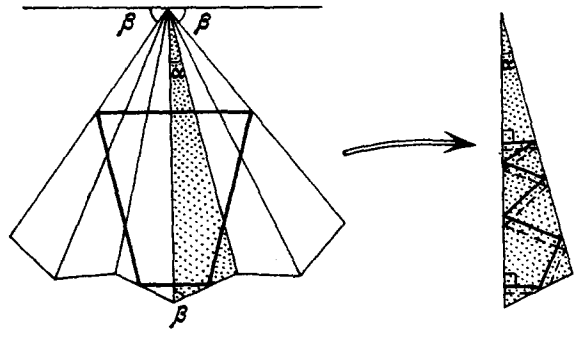



Fig. 11

Such triangles are naturally called '*quasi-right-angled*' (for $k = 1$ we have right-angled triangles). The construction of periodic trajectories in them is sketched in Fig. 11.

## §5. 'Perpendicular' trajectories in rational polygons and polyhedra

**A. Masur's theorem.**
In 1989, H. Masur proved the following theorem concerning periodic trajectories in 'rational' polygons.

A polygon is called *rational* if all its angles are commensurate with $\pi$. The group generated by the reflections in the sides of a rational polygon is finite. Because of this, all links of the trajectory of a billiard particle starting at an angle $\varphi_0$ with a fixed ('horizontal') side of the polygon will give only finitely many directions with respect to this side [9]. Billiards and rational polygons were first considered in [14]; below we will use some results of this paper.

***Theorem* 1** [Masur]. *Any rational polygon $Q$ contains infinitely many periodic billiard trajectories. The set of directions of all links of these trajectories is everywhere dense on the unit circle of directions $S^1$.*

Thus, we can 'launch' a billiard particle at an angle $\varphi_0$ with the horizontal such that it describes a periodic trajectory in $Q$, then we can subsequently change $\varphi_0$ by an arbitrarily small quantity $\varepsilon$ (smaller than an arbitrary $\varepsilon_0 > 0$) so that the particle launched at the angle $\varphi_0 + \varepsilon$ also describes a periodic trajectory in $Q$ (however, it may be necessary to launch the particle at another point of the polygon).

Masur's proof uses powerful analytical methods (Teichmüller theory) and does not give a *constructive* means for constructing the corresponding periodic trajectories.

**B. A theorem on pencils of 'perpendicular' trajectories.**
Nevertheless, there is a whole class of periodic trajectories in an arbitrary rational polygon which can be constructively described: these are the 'perpendicular' trajectories.

***Theorem* 2.** *Every rational $n$-gon $Q$ contains at least $n/2$ pencils of periodic trajectories.*

**C.** *Proof of Theorem* 2. We choose an arbitrary ('horizontal') side $AB$ of $Q$. From some point $C \in AB$ we launch a billiard particle, perpendicular to $AB$. We assume that the trajectory of this particle is not special, that is, it does not hit a vertex of $Q$. We prove that this trajectory is periodic.

In subsection D we shall prove a lemma asserting that at some moment $t_1$ after its first reflection, the particle will again move perpendicular to $AB$. If, at this moment, the particle moves *towards the side $AB$*, then after hitting $AB$ the particle will move along the same path as before the reflection. If, however, at time $t_1$ it moves *away from the side $AB$*, then before $t_1$ there has

been a moment $t_0$ (also falling after the first reflection following the start at $C$) in which the particle moved towards $AB$, and we are in the previous situation. In both cases the trajectory $\Gamma$ of the particle is periodic (the particle 'returns' after its second reflection in $AB$ along its own previous path, but in the opposite direction).

With each trajectory $\Gamma$ we have associated the pencil $\{\Gamma\}$ of trajectories parallel to it, uniquely determined by the periodic sequence $\{a_i\}$ of sides of $Q$ which the billiard particle has hit.

Such a pencil can be constructed for each side of the polygon, taking the side to be horizontal. However, the sets of sides $\{a_i\}$, defining the 'perpendicular' paths, may coincide, and then the pencils also coincide. Since an arbitrary trajectory cannot have links that are perpendicular to more than two sides of the polygon, we find that at least $n/2$ of the pencils constructed have different sets of sides $\{a_i\}$, and therefore all these pencils are distinct. Theorem 2 has been proved. □

**D. A lemma concerning the motion of billiard particles in a rational polygon.** To complete the proof of Theorem 2 it remains to prove the lemma promised in the previous subsection.

***Lemma.*** *A billiard particle launched at an angle $\varphi_0$ to the horizontal side $AB$ from a point $\boldsymbol{q}$ of a rational polygon $Q$ returns infinitely often to an arbitrary $\varepsilon$-neighbourhood of $\boldsymbol{q}$, moving at the angle $\varphi_0$ with $AB$.*

*Proof.* We consider the phase space $\Omega$ of the billiard $Q$. It is the 3-dimensional manifold obtained from the cylinder $Cy = Q \times [0, 2\pi)$ by the following 'billiard-wise' gluing of the boundary points of $Cy$: two points $(\boldsymbol{q}, \boldsymbol{v})$, $(\boldsymbol{q}', \boldsymbol{v}') \in Cy$ are glued together if and only if $\boldsymbol{q} = \boldsymbol{q}'$ while the vectors $\mathbf{v}$ and $\boldsymbol{v}'$ form an angle of incidence and an angle of reflection with the side to which $\boldsymbol{q}$ belongs.

The cylinder $Cy$ is fibred into 'floors' $Q(\boldsymbol{v}) = \varphi(\boldsymbol{v}) \times Q$, where $\varphi(\boldsymbol{v}) \in [0, 2\pi)$ is the angle between the velocity vector $\boldsymbol{v}$ and the horizontal side $AB$ (Fig. 12).

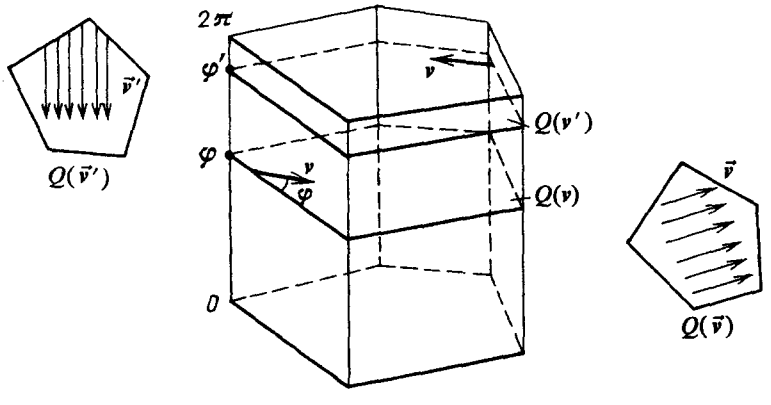



Fig. 12

The phase point $(\boldsymbol{q}, \boldsymbol{v})$ moves on the 'floor' $Q(\boldsymbol{v})$ with velocity $\boldsymbol{v}$ so long as it does not hit the boundary of the polygon $Q(\boldsymbol{v})$. Subsequently, returning from its boundary according to the billiard law, it moves with velocity $\boldsymbol{v}'$, but now on the 'floor' $Q(\boldsymbol{v}') = \varphi(\boldsymbol{v}') \times Q$, then on the 'floor' $Q(\boldsymbol{v}'')$, and so on. Since $Q$ is *rational*, the number of 'floors' $Q_0 = Q(\boldsymbol{v})$, $Q_1 = Q(\boldsymbol{v}')$, $Q_2 = Q(\boldsymbol{v}'')$, ... is finite. (In [14] it has been proved that after this gluing of the boundaries of the polygons $Q_0$, $Q_1$, $Q_2$, ... there results a two-dimensional surface $\mathcal{M}^2 \subset \Omega$ without boundary, which is invariant under the billiard phase flow $S_t : \Omega \to \Omega$. We will need this assertion in the sequel; for now it suffices to know that the set $Q_0$, $Q_1$, $Q_2$, ... is finite.)

The billiard phase flow $S_t$ preserves the measure $d\mu = dq d\omega$ in the phase space $\Omega$, where $\boldsymbol{q} \in Q$ and $d\omega$ is the uniform distribution on the unit circle $S^1$.

Hence, Poincaré's recurrence theorem (see, for example, [1], [11], [7]) is applicable. It implies that a phase point returns infinitely often to an arbitrarily small neighbourhood of the initial point $(\boldsymbol{q}, \boldsymbol{v}_0)$: for any $\varepsilon > 0$ there are infinitely many moments $t$ for which

$$|\boldsymbol{q}(t) - \boldsymbol{q}| < \varepsilon, \quad |\varphi(\boldsymbol{v}_i) - \varphi(\boldsymbol{v}_0)| < \varepsilon.$$

Let $\varphi_i = \varphi(\boldsymbol{v}_i)$ be the angle between the velocity vector $\boldsymbol{v}_i$ of the phase point moving on the 'floor' $Q_i$ and the line $AB$, where the subscript $i$ varies from 0 to the number of 'floors'. We put $\varepsilon < \min\{|\varphi_i - \varphi_j|\}$, where $i \neq j$, $i \geqslant 0$, $j \geqslant 1$. Then after finitely many 'jumps' from floor to floor the phase point is again on the initial floor $Q_0$, that is, $\varphi(\boldsymbol{v}_t) = \varphi(\boldsymbol{v}_0)$ for some $t > 0$; moreover $|\boldsymbol{q}(t) - \boldsymbol{q}| < \varepsilon$. The lemma has been proved. □

***Corollary.*** *A particle launched perpendicular to the side AB will move perpendicular to AB infinitely often.*

In fact, it suffices to put $\varphi_0 = \pi/2$ in the lemma.

**E. Periodic billiard trajectories in rational polyhedra.**

Masur's theorem (subsection A) does not allow us to generalize the result that the set of pencils of parallel periodic trajectories in a rational polygon is infinite to the multidimensional case of 'rational' polyhedra. The advantage of the geometrical proof is that it can be generalized to the multidimensional case.

A polyhedron $Q \subset \mathbb{R}^d$ is said to be *rational* if the group $G_Q$ generated by the reflections in all faces of $Q$ is finite. In other words, as the result of all possible mirror reflections in planes passing through the point $O$ in $\mathbb{R}^d$ and parallel to the faces of $Q$, as well as after reflections in the planes that are the images of these initial planes under infinitely many mirror reflections in these planes, only *finitely* many directions perpendicular to these planes arise. All rational polyhedra can be composed from both cubes and Weyl chambers (Coxeter polyhedra) (see [4]).

The group $G_Q$ is a subgroup of the group of isometries of the $(d-1)$-dimensional sphere $S^{d-1}$, generated by the isometries

$$\sigma_i(v) = v - 2(n_i, v) \cdot n_i, \tag{2}$$

where $v \in S^{d-1}$ and $n_i$ is the unit vector normal to the face with subscript $i$. Since $G_Q$ is finite, we find that by moving along billiard trajectories from a given initial direction we can obtain only finitely many directions. Theorem 2 from subsection B can be transferred to rational polyhedra almost word for word.

***Theorem* 3.** *In every rational n-hedron $Q \subset \mathbb{R}^d$ there are at least $n/2$ perpendicular periodic trajectories.*

*Proof.* The phase space $\Omega$ of the billiard in $Q$ is a $(2d-1)$-dimensional manifold obtained from the cylinder $Cy = Q \times S^{d-1}$ (with $S^{d-1}$ the $(d-1)$-dimensional unit sphere of directions in $\mathbb{R}^d$) by identifying equivalent boundary points: $(q, v) \approx (q', v') \Leftrightarrow v' = \sigma_i(v)$ [see formula (2)].

Since $Q$ is rational, a billiard trajectory $\Gamma$ launched *orthogonally* from a fixed face of $Q$ is situated in *finitely* many 'floors' of the cylinder $Cy$, that is in the $d$-dimensional polyhedra $Q = Q(\varphi_0), Q(\varphi_1), \ldots, Q(\varphi_m)$: in each of these the billiard particle moves in the directions $v_i = v(\varphi_i)$, $i = 0, 1, \ldots, m$, forming the angle $\varphi_i$ with the (fixed beforehand) boundary $M$ of $Q$.

Applying the Poincaré recurrence theorem and repeating the proofs of the lemma in subsection D and of Theorem 2 in subsection B, we find that the phase point will be arbitrarily close to the initial point, while their directions of motion *coincide*, that is, in $Q$ the particle will move perpendicular to the fixed boundary $M$ twice: in the direction *towards* it and in the direction *away from* it. This means that the trajectory $\Gamma$ is periodic.

There are at least $n/2$ such trajectories: one for each face, taking into account that a pair of parallel faces gives rise to only one trajectory (if no two faces are parallel, then there are at least $n$ periodic trajectories). Theorem 3 has been proved. □

*Remark.* In the spatial case we cannot speak of pencils of perpendicular periodic trajectories, since after 'straightening' a periodic trajectory $\Gamma$, the trajectories adjacent to it hit the same face, but rotated around the straightened trajectory $\Gamma$.

## §6. Stable trajectories

In this section we introduce the notion of a *stable periodic trajectory* in an arbitrary polygon, give a criterion for stability, and prove the stability of *every* periodic trajectory in a *typical* polyhedron.

**A. Definition. A criterion for the stability of a trajectory.**
Let $Q$ be an arbitrary polyhedron.

*Definition.* A periodic trajectory $\Gamma$ in $Q$ is called *stable* if it is not destroyed by *any* small deformation of $Q$.

In other words, by an arbitrarily small change (a deformation) of the angles and sides of $Q$ to form a new polygon $Q'$ there is a periodic trajectory $\Gamma'$ with the same number of links as $\Gamma$, and where each link is close to the corresponding link of $\Gamma$.

A typical example of a *stable* periodic trajectory is the 3-link 'altitude' trajectory in an arbitrary acute triangle (§1A). Examples of *unstable* trajectories are 'perpendicular' trajectories in a rectangle, in an isosceles right-angled triangle, and in a triangle with angles $\pi/6$, $\pi/3$, and $\pi/2$. This can be immediately verified (moreover, in general such triangles do not have a single stable trajectory at all, see §12).

We give an algorithmic description of a stable trajectory in a polygon $Q$. For this we use the well-known method of straightening a billiard trajectory (the '*method of Baron Munchhausen*', see [7]).

We fix a periodic trajectory $\Gamma$ in the polygon $Q$ and consider the polygons $Q_0 = Q, Q_1, Q_2, \ldots, Q_{N-1}, Q_N$ obtained by successive reflections of $Q$ in those sides $i_1, i_2, \ldots, i_N$ in which the billiard particle moving along $\Gamma$ is reflected. The trajectory $\Gamma$ is a segment in the 'corridor' $Q_0, Q_1 Q_2 \ldots Q_{N-1}$ thus obtained. Since $\Gamma$ is periodic, there is an $N$ such that $Q_N$ is obtained from $Q_0$ by parallel translation by a vector $\vec{a}$ : $Q_N = T_{\vec{a}} Q_0$; moreover, a particle starting at a point $A$ on the 'horizontal' side of $Q_0$ falls back to the same point $A$ on the 'horizontal' side of $Q_N$ (Fig. 13). Thus, every straightened periodic trajectory $\Gamma$ joins 'identical' points on the bases of the parallel polygons $Q_0$ and $Q_N$ and lies in the corridor formed by the successive reflections of $Q$ in those sides of $Q$ that intersect the straightened trajectory. Here, if the length of $\Gamma$ (its number of links) is *even*, then this length equals $N$; if it is *odd*, then $N$ is twice the length of $\Gamma$. In each case $N$ is even.

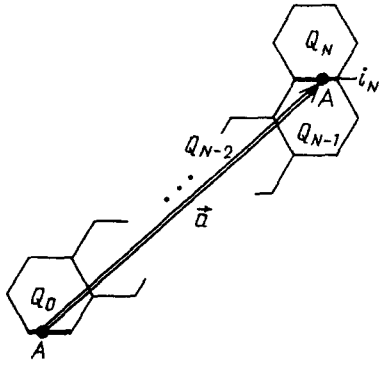



Fig. 13

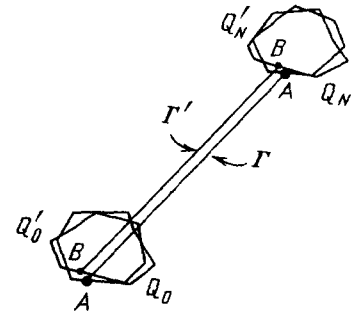



Fig. 14

Now let $\Gamma$ be a *stable* periodic trajectory. Then, by deforming $Q$ to $Q'$ and by performing $N$ reflections in the sides with the same numbers $i_1, \ldots, i_N$, we obtain a corridor $Q_0'Q_1' \ldots, Q_N'$ in which there lies a straightened trajectory $\Gamma'$, close to the straightened trajectory $\Gamma$; moreover, $Q_N' = T_{\vec{b}}Q_0'$, where $\vec{b}$ is some vector (Fig. 14). This means that the polygons $Q_0'$ and $Q_N'$ are parallel precisely because there is a unique relation between the variations $\delta\alpha_1, \ldots, \delta\alpha_n$ of the angles of $Q$ : $\delta\alpha_1 + \ldots + \delta\alpha_n = 0$.

In other words, the stability of $\Gamma$ is determined by a property of the sequence of numbers of the sides $i_1, i_2, \ldots, i_N$ with respect to which we reflect. The sequence $W = i_1 i_2 \ldots i_N$ is called the *code-word* (or simply the *word*) determining the trajectory $\Gamma$.

We mark on the unit circle of directions $S^1$ the directions of the sides $i_1, i_2, \ldots, i_N$. Superposition of the two reflections of $Q$ in the sides $i_k$ and $i_s$ is equivalent to rotation through an angle $2\alpha$, where $\alpha$ is the angle between the directions $i_k$ and $i_s$ (Fig. 15). We write this symbolically as $2(i_k - i_s)$.

What is the angle $\Phi$ through which $Q_0$ must be rotated to obtain $Q_N$? Answer: $\Phi = 2(i_1 - i_2) + 2(i_3 - i_4) + \ldots + 2(i_{N-1} - i_N)$. Since $Q_N$ and $Q_0$ are *parallel* we obtain $\Phi = 2\pi m$, $m \in \mathbb{Z}$, and since the deformation of $Q_0$ is *arbitrary* we obtain $\Phi = \text{const}$, whence $m = \text{const}$ (independent of the deformation). Hence, by considering the word $W = i_1 i_2 \ldots i_N$ abstractly, we find that *if* $\Gamma$ *is a stable trajectory, then the alternating sum of the letters in this word is equal to* 0:

$$i_1 - i_2 + i_3 - i_4 + i_5 - i_6 + \ldots \equiv 0 \tag{3}$$

(we recall that each $i_k$ is a letter from the alphabet of $n$ letters).

A word $W$ with the property (3) is called a *stable word.*

*Example.* The word $W$ corresponding to the altitude triangle (in an acute triangle) has the form $W = abcabc$. It is stable because $a - b + c - a + b - c \equiv 0$ (Fig. 16a). For a rectangle we have $W = acac$; however, $a - c + a - c = 2a - 2c \not\equiv 0$ (Fig. 16b).

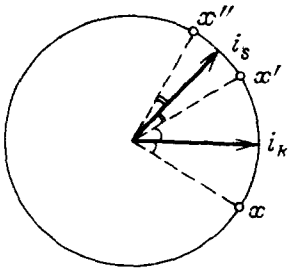



Fig. 15

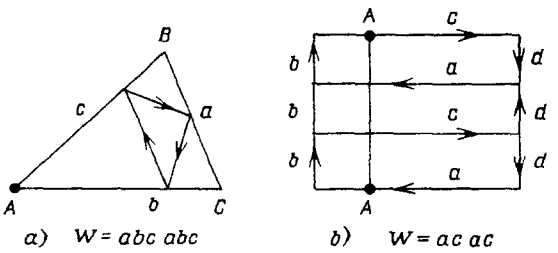



Fig. 16

We note that if the boundary of $Q$ is given an orientation, and $n$ distinct letters are distributed on its sides, then after an even number $N$ of reflections of $Q$ in the sides given by an arbitrary stable word $W = i_1 i_2 \ldots i_N$ we obtain a polygon $Q_N$ which is parallel to $Q$. However, this does not mean that the sequence of polygons $Q, Q_1, Q_2, \ldots, Q_N$ forms a corridor in which at least one straightened periodic trajectory $\Gamma$ can be completely 'located'. If, on the other hand, some segment $AA$ joining *identical* points $A$ on 'horizontal' sides of the parallel polygons $Q$ and $Q_N$ lies *entirely* in this corridor $QQ_1Q_2 \ldots Q_N$, then $AA$ is a straightened trajectory $L$, and $\Gamma$ is a *stable periodic trajectory*. As a result we obtain the following criterion.

*Criterion for the stability of a periodic trajectory $\Gamma$ in a polygon $Q$:*
($\Gamma$ *is stable in* $Q$) $\Leftrightarrow$ (*the code-word $W$ is stable and the corridor constructed from it contains the segment $AA$*).

As a result there arise a number of related problems.

*Problem* 1. Describe the set $\{W\}$ of all stable code-words, consisting of the letters of a given $n$-letter alphabet $\mathfrak{A}$, corresponding to the periodic trajectories in some polygon. In particular, describe the set of stable words in the alphabet $\mathfrak{A} = \{a, b, c\}$ that correspond to the periodic trajectories in a triangle.

*Problem* 2. For a given $n$-gon $Q$, describe the set $\{W\}$ of stable code-words, consisting of the letters of the alphabet $\mathfrak{A} = \{a_1, \ldots, a_n\}$, that correspond to the periodic trajectories in $Q$ (they will automatically be stable).

Problem 2 will be solved later for certain particular cases of polygons $Q$ (see §12, §13).

*Corollary-Remark.* Suppose we want to find a stable periodic trajectory with $n$ links that is strictly inscribed in an $n$-gon $Q$ (Berger [2] calls such a trajectory an *optical trajectory*). For odd $n$ such a trajectory is unique, while for even $n$ there is an optical trajectory only if the alternating sum of the angles of $Q$ vanishes ([2], Theorem 9.4.2.1). In particular, for $n = 4$ the quadrangle $Q$ must be *inscribed*. These assertions readily follow from the criterion for stability of trajectories.

**B. Stability of all periodic trajectories in a typical *n*-gon.**
A polygon $Q$ is called *typical* if all its interior angles $\alpha_1, \alpha_2, \ldots, \alpha_n$ are rationally independent:

$$(k_1, \ldots, k_n \in \mathbb{Z}:\ k_1\alpha_1 + k_2\alpha_2 + \ldots + k_n\alpha_n = 0) \Rightarrow (k_1 = \ldots = k_n = 0).$$

***Theorem.*** *Every periodic trajectory $\Gamma$ in a typical $n$-gon $Q$ is stable.*

*Proof.* Since the sum of the interior angles is equal to $\alpha_1 + \ldots + \alpha_n = \pi(n-2)$, we find that the numbers $\alpha_1, \alpha_2, \ldots, \alpha_{n-1}, \pi$ are rationally independent. Let $W = i_1 i_2 \ldots i_N$ be the code-word of even length constructed from the periodic

trajectory $\Gamma$ in $Q$. We straighten the trajectory $\Gamma$. Then the last polygon $Q_N$ in the resulting corridor of reflected polygons is parallel to $Q$.

We orient the boundary of $Q$ in the counterclockwise direction. Let $\beta_{i_k}$ be the angle between the side $i_k$ of $Q$ and a fixed direction $\vec{e}$ in the plane. Then the angle of inclination of $Q_N$ with respect to $Q$ is equal to

$$\Phi=2(\beta_{i_1}-\beta_{i_2})+2(\beta_{i_3}-\beta_{i_4})+\ldots=\sum_{i=1}^{n} 2m_i\beta_i,\ m_i\in\mathbb{Z}$$

(here $\beta_i$ is the angle of inclination of the side with number $i$ to the vector $\vec{e}$). We rewrite this equality as

$$\Phi=2m_1'(\beta_1-\beta_2)+2m_2'(\beta_2-\beta_3)+\ldots+2m_{n-1}'(\beta_{n-1}-\beta_n),$$

where $m_1' = m_1$, $m_k' = m_1+m_2+\ \ldots\ +m_k$ $(k = 2, \ldots, n-1)$, $m_n' = -m_n$.

We now note that for all $k = 1, 2, \ldots, n-1$ the equality $\beta_{k+1}-\beta_k = \pi-\alpha_k$ holds (Fig. 17).

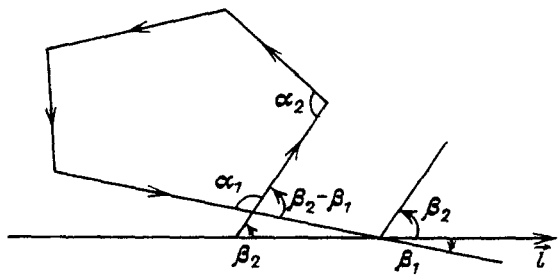



Fig. 17

Hence

$$\Phi=2m_1'(\pi-\alpha_1)+2m_2'(\pi-\alpha_2)+\ldots+2m_{n-1}'(\pi-\alpha_{n-1}).$$

However, since $Q$ and $Q_N$ are parallel, we have $\Phi \equiv 0 \pmod{2\pi}$. This means that $\sum_{i=1}^{n-1} m_i'\alpha_i \equiv 0 \pmod{\pi}$, or $\sum_{i=1}^{n-1} m_i'\alpha_i - k\pi = 0$ for some integer $k$. Since the angles $\alpha_1, \ldots, \alpha_{n-1}, \pi$ are rationally independent, this implies that $m_1' = m_2' = \ldots = m_{n-1}' = 0$, $k = 0$. Hence $m_1 = \ldots = m_n = 0$ and $\Phi = 0$. However,

$$\Phi=2(\beta_{i_1}-\beta_{i_2})+2(\beta_{i_3}-\beta_{i_4})+\ldots+2(\beta_{i_{N-1}}-\beta_{i_N})\equiv 0,$$

whence it follows that the alternating sum of the letters

$$i_1-i_2+i_3-i_4+\ldots+i_{N-1}-i_N\equiv 0,$$

that is, the word $W = i_1 i_2 \ldots i_{N-1} i_N$ is stable, as we had to prove. □

*Remarks.* 1) In proving the assertions in §3D we have in fact at the same time proved the stability of periodic trajectories with more than $n$ links.

2) Unfortunately, the theorem proved does not tell us anything about the existence of at least one periodic trajectory in a typical $n$-gon $Q$. The presence of such a trajectory in $Q$ would immediately give (by its stability) a whole neighbourhood of $n$-gons in each of which there would be at least one periodic trajectory. If it turns out that there is a periodic trajectory in *every* typical polygon, then this would imply that there would be periodic trajectories in at least an open everywhere dense subset of the set of all $n$-gons (see Conjecture 1 of the Conclusion).

3) On the other hand, in every rational polygon $Q$ there are infinitely many types of periodic trajectories (by Masur's theorem). However, a priori we cannot say that even one of them will be stable, and hence we cannot assert that there are periodic trajectories in polygons close to $Q$.

## §7. Stable perpendicular trajectories

**A. The bifurcation diagram of triangles.**
Every triangle is given, up to a similarity, by a pair of numbers $\alpha, \beta \in (0, \pi/2)$, its *acute* angles. Thus, each triangle becomes an interior point of a square $\boldsymbol{K}$ with side $\pi/2$. We mark those points of $\boldsymbol{K}$ that correspond to triangles having periodic billiard trajectories. The square $\boldsymbol{K}$ with these points marked is called the *bifurcation diagram of triangles.*

In Figs. 18a, b, c the known (above-described) sets of triangles with periodic trajectories are indicated. The superposition of these squares gives the bifurcation diagram of triangles of §4A (Fig. 18b), §4B (Fig. 18c), and §5 (Fig. 18a).

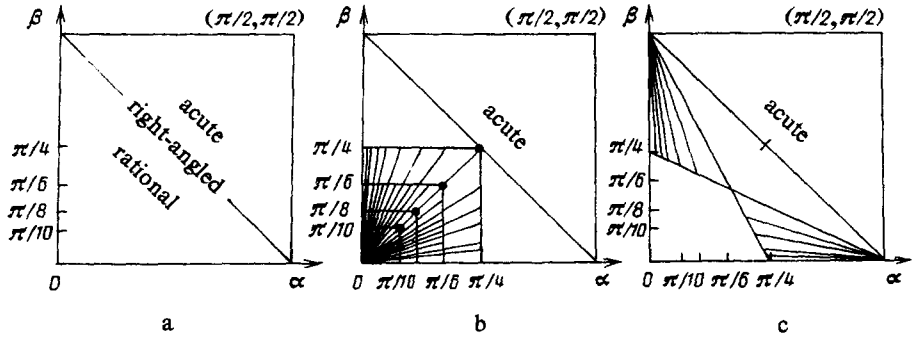



Fig. 18

**B. Stable periodic trajectories in the triangle (45°, 60°, 75°).**
Suppose that the triangle $ABC$ is such that $\measuredangle A = 60°$, $\measuredangle B = 45°$, $\measuredangle C = 75°$. We denote by $a$, $b$, $c$ the sides opposite to $A$, $B$, $C$. We consider the word $W = bacbacabcabc$ and construct from it the corridor starting with $\Delta ABC$. Let $D$ be the foot of the altitude $CD$, $D \in AB$. It is clear from

Fig. 19 that every perpendicular trajectory in the pencil of trajectories launched from the segment $AD$ lies completely inside the corridor formed above. Since $W$ is a stable word, we find that *all 12-link trajectories in this pencil are stable.*

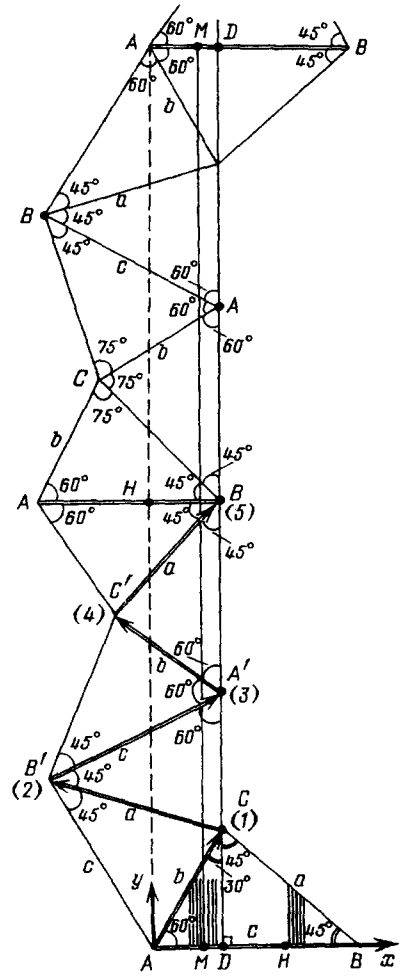



Fig. 19

**C. A neighbourhood of the triangle ($\pi/4$, $\pi/3$) in which there are perpendicular periodic trajectories.**

We change the angles of $\Delta ABC$ in subsection B, and assume that $\measuredangle A = \alpha$, $\measuredangle B = \beta$, $\measuredangle C = \gamma$. We introduce the orthogonal coordinate system with origin at $A$, $x$-axis along the line $AB$, and $y$-axis vertical, that is, perpendicular to $AB$. The $y$-axis will be parallel to the trajectory launched perpendicular to $AB$. In Fig. 19 we have drawn the vectors $\vec{0} = \overrightarrow{AA}, \overrightarrow{AC}, \overrightarrow{AC} + \overrightarrow{CB'}$,

$\overrightarrow{AC}+\overrightarrow{CB'}+\overrightarrow{B'A'}$, $\overrightarrow{AC}+\overrightarrow{CB'}+\overrightarrow{B'A'}+\overrightarrow{A'C'}$, and $\overrightarrow{AC}+\overrightarrow{CB'}+\overrightarrow{B'A'}+$ $+\overrightarrow{A'C'}+\overrightarrow{C'B''}$; we denote the endpoints of these vectors by (0), (1), (2), (3), (4), (5).

For the required straightened trajectory to exist, that is, to belong to the corridor, it is necessary that each of the orthogonal projections of (1), (3), (5) onto the $x$-axis lies to the right of each of the orthogonal projections of (0), (2), (4). (If we considered an inclined, and not orthogonal, straightened trajectory, it would be necessary to project onto the $x$-axis along it.)

It is easy to compute the $x$-coordinates of all these vectors:

$$(1)=b\cos\alpha;$$
$$(2)=b\cos\alpha+a\cos(\alpha+\pi-\gamma);$$
$$(3)=b\cos\alpha+a\cos(\alpha+\pi-\gamma)+c\cos(\alpha+\beta-\gamma);$$
$$(4)=b\cos\alpha+a\cos(\alpha+\pi-\gamma)+c\cos(\alpha+\beta-\gamma)+b\cos(\pi+\beta-\gamma);$$
$$(5)=(4)+a\cos\beta.$$

We will now write all inequalities (0), (2), (4) $<$ (1), (3), (5) pairwise. But before doing so we use the sine theorem, and assume that the diameter of the circumscribed circle of $\Delta ABC$ is equal to 1. Then

$$a=\sin\alpha,\ b=\sin\beta,\ c=\sin\gamma.$$

We have

$$(0)<(1)\Leftrightarrow b\cos\alpha>0\Leftrightarrow\alpha<\pi/2;$$
$$(2)<(1)\Leftrightarrow\alpha+\pi-\gamma>\pi/2\Leftrightarrow\pi/2<2\alpha+\beta<\pi;$$
$$(2)<(3)\Leftrightarrow-\pi/2<\alpha+\beta-\gamma<\pi/2\Leftrightarrow\pi/4<\alpha+\beta<3\pi/4;$$
$$(4)<(3)\Leftrightarrow\pi/2<\pi+\beta-\gamma<3\pi/2\Leftrightarrow\pi/2<\alpha+2\beta<3\pi/2;$$
$$(4)<(5)\Leftrightarrow\beta<\pi/2;$$
$$(0)<(3)\Leftrightarrow\pi/2<2\alpha+\beta<\pi;$$
$$(2)<(5)\Leftrightarrow\pi/2<\pi+\beta-\gamma<\pi\Leftrightarrow\pi/2<\alpha+2\beta<\pi;$$
$$(4)<(1)\Leftrightarrow0<\alpha+\beta-\gamma<\pi/2\Leftrightarrow\gamma<\pi/2.$$

Thus we have established that in a neighbourhood of the triangle $(\pi/4,\ \pi/3)$ a stable periodic trajectory can exist only in *acute* triangles.

$$(0)<(5)\Leftrightarrow\cos(\alpha+\pi-\gamma-\pi/2)\cdot 2c\sin\beta+c\cdot\cos 2\beta>0\Leftrightarrow$$
$$\Leftrightarrow 2\sin(2\alpha+\beta)\cdot\sin\beta+\cos 2\beta>0\Leftrightarrow\cos 2\alpha+\cos 2\beta>\cos 2\gamma.$$

Hence

$$(*)\qquad 2\cos(\alpha+\beta)\cdot\cos(\alpha-\beta)>2\cos^2(\alpha+\beta)-1.$$

Setting $\cos(\alpha+\beta)=X$, $\cos(\alpha-\beta)=Y$, we obtain

$$2XY>2X^2-1\Leftrightarrow(X-Y/2)^2<(Y^2+2)/4.$$

In the coordinate system $X$, $Y$ the curve $(X-Y/2)^2 = (Y^2+2)/4$ is a hyperbola.

Using all inequalities written down above except (*), we first find that these are satisfied for all points in the diagram $K$ lying inside the triangle with vertices $(0, \pi/2)$, $(\pi/3, \pi/3)$, $(\pi/2, 0)$. However, the sheet of the hyperbola (*) cuts off from this triangle a fairly large part around the vertex $(\pi/3, \pi/3)$. Finally, the neighbourhood of the point $(\pi/4, \pi/3)$ which we need is drawn in Fig. 20 in the form of a curvilinear trapezium with vertices $(0, \pi/2)$, $(\pi/4, 3\pi/8)$, $(3\pi/8, \pi/4)$, $(\pi/2, 0)$.

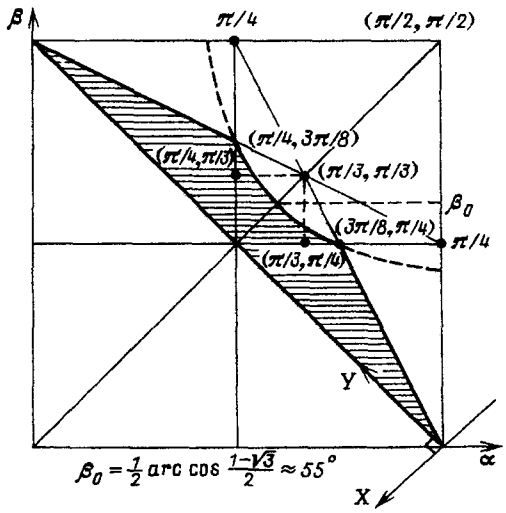



Fig. 20

**D. Stability of a perpendicular trajectory in an arbitrary rational polygon.** It is easy to prove (see §12) that every perpendicular trajectory in an equilateral triangle is *unstable*. On the other hand, we have just indicated (subsections B, C) a neighbourhood of the triangle (45°, 60°, 75°) in which there is a *stable* perpendicular trajectory. The question of the stability of a perpendicular trajectory in an arbitrary rational polygon naturally arises. The following theorem holds.

***Theorem.*** *If $\gamma$ is a stable perpendicular trajectory in a rational polygon $Q$, then the number $n$ of its links is a multiple of* 4.

*Remark.* The converse of this theorem is false. Example: the 4-link perpendicular trajectory in the equilateral triangle (the unstable word $W = bcba$ corresponds to it).

*Proof.* a) We first show that the property that a stable periodic trajectory be perpendicular to the boundary of $Q$ (at a certain moment) is preserved under small deformations of $Q$ (*preservation of perpendicularity*).

In fact, let $\gamma$ be a perpendicular trajectory one link of which is perpendicular to a side $a \in Q$. The word corresponding to $\gamma$ has the form

$$W(\gamma) = Xb\overline{X}a,$$

where $X$ is an 'intermediate' word, whose 'letters' number the sides of the reflected polygons that intersect the part of the straightened trajectory $\gamma$ between the two parallel sides $a$ and $b$, and $\overline{X}$ is the word $X$ written in reverse order.

Conversely, if the word $W(\gamma)$ has the form $W(\gamma) = Xb\overline{X}a$, then $\gamma$ is a perpendicular trajectory one link of which is perpendicular to the side $a$ while another is perpendicular to the side $b$, and $W(\gamma) = W$ (this can be readily proved by contradiction). Thus we have obtained a *criterion for perpendicularity* of $\gamma$ in terms of code-words:

$$(\gamma \textit{ is perpendicular}) \Leftrightarrow (W(\gamma) = Xb\overline{X}a).$$

We will now obtain a criterion for a trajectory to be *stable and perpendicular*. For this it is necessary and sufficient that the word $W(\gamma) = Xb\overline{X}a$ be stable. If $X = x_1x_2 \ldots x_k$, $\overline{X} = x_k \ldots x_2x_1$, then in the alternating sum of the letters of $W(\gamma)$ all the letters $x_1, x_2, \ldots, x_k$ occur with coefficient 2 or $-2$, while $a$ and $b$ occur with coefficient 1 or $-1$. Since this sum must be zero, the letters $a$ and $b$ must coincide, otherwise the sum cannot be cancelled and the coefficient 2 does not appear. Thus,

$$(\gamma \textit{ is stable and perpendicular}) \Leftrightarrow (W(\gamma) = Xa\overline{X}a).$$

It follows that since under a small deformation of $Q$ the word $W(\gamma)$ does not change, the trajectory $\gamma'$ in the deformed polygon $Q'$ is also perpendicular: the preservation of perpendicularity under a deformation of $Q$ has been proved.

b) We clarify how many letters can occur in the word $X$ that appears in the word $W(\gamma) = Xa\overline{X}a$ of a stable trajectory $\gamma$.

Let $X = x_1x_2 \ldots x_k$ with $k$ even, $k = 2s$. Then the alternating sum of the letters of $W(\gamma)$ is equal to

$$x_1 - x_2 + x_3 - \ldots - x_{2s} + a - x_{2s} + \ldots + x_3 - x_2 + x_1 - a =$$
$$= 2(x_1 - x_2 + x_3 + \ldots + x_{2s-1} - x_{2s}).$$

Since $\gamma$ is stable we have

$$x_1 - x_2 + x_3 - \ldots + x_{2s-1} - x_{2s} \equiv 0,$$

that is, $X$ must be a stable word.

We return to geometry: by the stability of $X$, the composite of the reflections in the sides $x_1, x_2, \ldots, x_{2s}$ is the parallel displacement by a vector $\vec{v}$, so the polygon $Q_{2s}$ arising as the final result of the reflections of the initial polygon in these sides is obtained from $Q$ by parallel displacement: $Q_{2s} = Q + \vec{v}$. However, because of the perpendicularity of $\gamma$ and the

arrangement of $W(\gamma)$, after reflection of $Q$ in the sides $x_1, x_2, \ldots, x_{2s}, a$ we obtain a polygon $Q_{2s+1}$ which can be obtained from $Q$ by parallel displacement by a vector $\vec{V}: Q_{2s+1} = Q + \vec{V}$.

However, $Q_{2s+1}$ is obtained from $Q_{2s}$ by reflection in their common side $a$. We have obtained a contradiction.

Hence the number of letters in the word $X$ cannot be even.

We now assume that $k = 2s+1$, $X = x_1x_2 \ldots x_{2s}x_{2s+1}$; then

$$W(\gamma) = x_1x_2 \ldots x_{2s}x_{2s+1}ax_{2s+1}x_{2s} \ldots x_2x_1a.$$

The alternating sum of the letters of $W(\gamma)$ vanishes:

$$2(x_1 - x_2 + x_3 - \ldots + x_{2s+1} - a) = 0.$$

Thus, the words $Xa$ and $\overline{X}a$ taken individually are stable.

In the case under consideration, the number of letters in $W(\gamma)$ is equal to $2(2s+2) = 4(s+1)$. On the other hand it is equal to the number $n$ of links of the trajectory $\gamma$. So $n$ is divisible by 4, as we had to prove. □

***Corollary.*** *The number of links of a stable perpendicular trajectory in an arbitrary polygon is at least* 12. *If it is equal to* 12, *then the word corresponding to the trajectory has the form* $W = bacbacabcabc$.

*Proof.* If $\gamma$ is a stable perpendicular trajectory, then for an appropriate choice of the initial link it has corresponding word $W(\gamma) = Xa\overline{X}a$, where both words $Xa$ and $\overline{X}a$ are stable (this follows from the proof of the theorem). But every stable word contains at least 6 letters, and the only 6-letter stable word is, up to renaming of letters, $abcabc$. Applying this to the word $Xa$ we obtain the assertion of the corollary. □

*Remark.* Thus, the 12-link stable perpendicular trajectories considered in subsections B, C are minimal (as regards the number of links) stable perpendicular trajectories.

## §8. Isolated trajectories

In this section we study *isolated* periodic billiard trajectories in polygons. We call a periodic trajectory *isolated* if its length is *strictly minimal* among all trajectories close to it in the pencil of parallel trajectories. Clearly, an isolated trajectory has an *odd* number of links; the periodic trajectories with an even number of links in a single parallel pencil have the same length, which can be seen by comparing the segments of the straightened trajectories between the two parallel polygons $Q$ and $Q_N$ ($N$ even).

Below we give a construction of isolated trajectories in rational polygons, and give an upper bound for their number.

**A. Stability of isolated trajectories.**

***Theorem.*** *In an arbitrary polygon every isolated trajectory is stable.*

*Proof.* Suppose that the isolated trajectory $\Gamma$ has $N = 2k+1$ links. After straightening $\Gamma$ we consider the $N$ reflections of $Q$ in the sides $i_1, \ldots, i_N$ in which a billiard particle launched at a point $M$ on the '*horizontal*' side $AB$ along $\Gamma$ is reflected (Fig. 21). The polygons $Q$ and $Q_N$ have opposite orientations, and can be obtained from each other by a rotational symmetry (the construction of the straightened trajectory $\Gamma$ is given in subsection B).

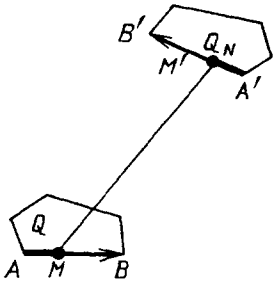



Fig. 21

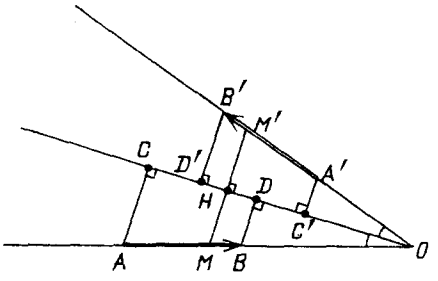



Fig. 22

After hitting $Q_N$, the particle starts to reflect in the same sides as before. Therefore the code-word of the trajectory $\Gamma$ in $Q$ is $W = i_1 i_2 \ldots i_N i_1 i_2 \ldots i_N$. Since $N$ is odd,

$$i_1 - i_2 + i_3 - \ldots + i_N - i_1 + i_2 - \ldots - i_N \equiv 0,$$

that is, $W$, and thus $\Gamma$, is stable. The theorem has been proved. □

**B. The transport lemma.**

***Assertion.*** *If $\Gamma$ is an isolated trajectory in a polygon $Q$ emanating from a 'horizontal' side $AB$ at a given angle, then it is unique and can be constructed as follows. Let $A'B'$ be the image of $AB$ under $N$ reflections in the sides $i_1, \ldots, i_N$ of $Q$ ($N$ being the number of links of $\Gamma$). Let $O$ be the point of intersection of the lines $A'B'$ and $AB$* (Fig. 22). *Construct the bisector of the angle $A'OA$ and project the segments $AB$ and $A'B'$ orthogonally onto it. Suppose that their projections, $CD$ and $C'D'$, have a common intersection, the segment $DD'$. Then the straightened trajectory $\Gamma$ is the perpendicular bisector $MM'$ of $DD'$.*

*Proof.* In fact, the figure $B'A'C'DBACD'B'$ has a 'quasi-centre' of symmetry, the midpoint $H$ of $D'D$. Hence the perpendicular $M'M$ ($M' \in A'B'$, $M \in AB$) to $D'D$ passing through $H$ has minimal length among all segments $NN'$ for which $AN = A'N'$. □

*Remark.* We call this a 'transport lemma' because it allows us to determine the shortest distance between two cars travelling from $A$ to $B$ and from $A'$ to $B'$.

**C. An upper bound for the number of isolated trajectories in a rational polygon.** In [8] there is mention of a polynomial bound for the number of isolated trajectories in a rational polygon (polynomial in their lengths, or number of links). Here we prove that in a given rational polygon $Q$ there are only finitely many isolated trajectories.

***Theorem.*** *Let $Q$ be a rational n-gon, $\alpha_i = \pi \cdot p_i/q_i$ $(i = 1, ..., n)$ its angles, and $q$ the least common denominator of the fractions $p_i/q_i$, $i = 1, 2, ..., n$. Then the number of isolated periodic trajectories in $Q$ is at most $2nq$.*

*Proof.* With each isolated trajectory $\Gamma$ in $Q$ we associate the maximally wide pencil of trajectories parallel to it. The boundaries of this pencil, parts of the so-called 'generalized diagonals', hit $\partial Q$ at the vertices of the polygon (the terminology 'generalized diagonals' was introduced in [8]).

We consider the left boundary of this pencil. Suppose that it is based at some vertex $A$ and has direction $\vec{v}$. We associate with $\Gamma$ the pair $(A, \vec{v})$ (Fig. 23).

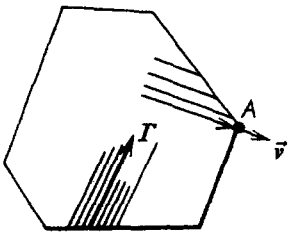



Fig. 23

Let us turn to the paper [14]. The three-dimensional phase space $\Omega$ of the billiard in $Q$ is obtained from the cylinder $Cy = Q \times [0, 2\pi)$ by gluing, as indicated in §5D above. The whole phase space $\Omega$ can be fibred into surfaces that are invariant under the billiard phase flow

$$M_s = \{(q, \vec{v}) | F(\vec{v}) \equiv |\varphi| (\mathrm{mod}\ 2\pi/q) = c\},$$

glued together from $2q$ floors of the polygons $Q(\varphi_s^{\pm})$, where

$$\varphi_s^{\pm} = \pm c + s \cdot (2\pi/q),\ s = 0, 1, ..., q-1.$$

We can visualize these $2q$ floors as consisting of two sets, each of which consists of floors having equal distance $(2\pi/q)$ from one another: the 'base' floor in the first set is $Q(\varphi_0^+) = Q(c)$, and that in the second is $Q(\varphi_0^-) = Q(-c)$.

Let us find on which floors our maximal pencil containing $\Gamma$ can be positioned. We note that two successive angles of rotation, $\varphi_k$ and $\varphi_{k+1}$, are related by $\varphi_{k+1} = 2\alpha - \varphi_k$, where $\alpha$ is the angle between the 'horizontal' side $AB$ of the polygon and the side in which we have reflected. Since $2\alpha \equiv 0$ (mod $2\pi/q$), we obtain the congruence $\varphi_{k+1} \equiv -\varphi_k$ (mod $2\pi/q$).

Suppose that $\Gamma$ has $2N+1$ links. Then $\varphi_{2N+1} \equiv -\varphi_0 \pmod{2\pi/q}$, where $\varphi_0$ is the angle of inclination of the first link of $\Gamma$ relative to $AB$. But $\Gamma$ is periodic, hence $\varphi_{2N+1} = \varphi_0$. Consequently, $\varphi_0 \equiv -\varphi_0 \pmod{2\pi/q}$, whence

$$\varphi_0 \equiv 0 \pmod{\pi/q}.$$

Therefore only two cases are possible: 1) $\varphi_0 = 0$ and 2) $\varphi_0 = \pi/q$. We note that in case 1) $-\varphi_0 = 0 = \varphi_0$, while in case 2) $-\varphi_0 = -\pi/q \equiv \equiv \pi/q+(q-1)\cdot 2\pi/q \pmod{2\pi/q}$. Hence, in each of the cases 1) and 2) the collection of floors $\{Q(\varphi_s^+)\}$ coincides (geometrically) with the collection of floors $\{Q(\varphi_s^-)\}$.

Thus, the maximal pencil corresponding to $\Gamma$ lies on one of the surfaces $M_0$ or $M_{\pi/q}$ (Fig. 24).

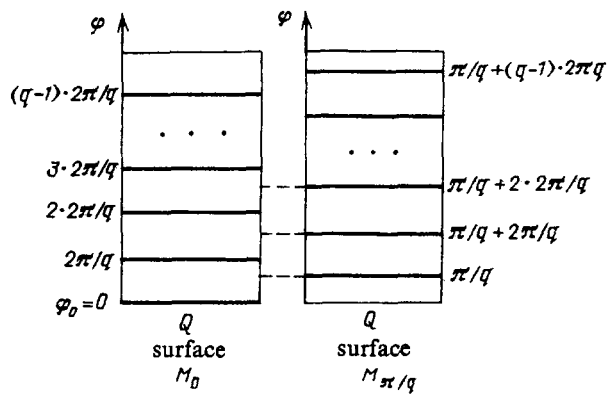



Fig. 24

This enables us to determine the number of possible directions of the links of $\Gamma$: it cannot exceed the number of floors, that is, it is at most $q$. Hence the vector $\vec{v}$ in the pair $(A, \vec{v})$ can take at most $q$ directions. Since the number of vertices of $Q$ is $n$, we find that the left boundary of the maximal pencil can hit $\partial Q$ in at most $n$ vertices. Hence the number of pencils on each of the surfaces $M_0$ and $M_{\pi/q}$ does not exceed $nq$.

Therefore the total number of isolated periodic trajectories in $Q$ is at most $2nq$, which is what we had to prove. □

*Remarks.* 1) The proof of the theorem readily implies that in an equilateral triangle there is exactly one isolated trajectory.

2) The 'transport lemma' (subsection C) implies only the finiteness of the set of all *directions* of the links of an isolated trajectory, and not the finiteness of (let alone an upper bound for) their number.

3) If $Q$ is a *convex* $n$-gon, the upper bound for the number of isolated trajectories can be reduced by a factor 2. In fact, all the angles of $Q$ are less than $\pi$, hence the number of directions of the vector $\vec{v}$ of the flight of a particle inside the polygon does not exceed $q/2$ (if $q$ points are distributed on

the boundary of a circle with equal distances between them, then on an arc of length at most $\pi$ there cannot be more than $q/2$ of these points). It is also not difficult to lower this bound to $q(n-2)$.

## §9. Isolated trajectories in acute and obtuse triangles. The bifurcation diagram of isolated trajectories (a 'hang-glider' configuration)

The method of straightening trajectories and the use of the altitude triangle (§2A) enable us to obtain a large number of isolated trajectories in both acute and obtuse triangles.

**A. Isolated trajectories in obtuse triangles.**
Let the acute angles $\alpha$ and $\beta$ of an obtuse triangle be related by

$$(5)\qquad \begin{cases} \pi/2-\beta/2<k\alpha<\pi/2\leqslant(k+1)\alpha, \\ \\ \pi/2-\alpha/2<l\beta<\pi/2\leqslant(l+1)\beta, \end{cases} \qquad (k,\ l=1,\ 2,\ \ldots).$$

***Theorem* 1.** *In the triangle $A_0B_0C_0$ with angles $\measuredangle A_0 = \alpha$, $\measuredangle C_0 = \beta$ satisfying the inequalities* (5) *there is an isolated trajectory having $2(k+l)-1$ links.*

*Proof.* Fig. 25a gives an indication as to how to proceed with the proof. For this we look at the method of straightening trajectories from a somewhat different point of view.

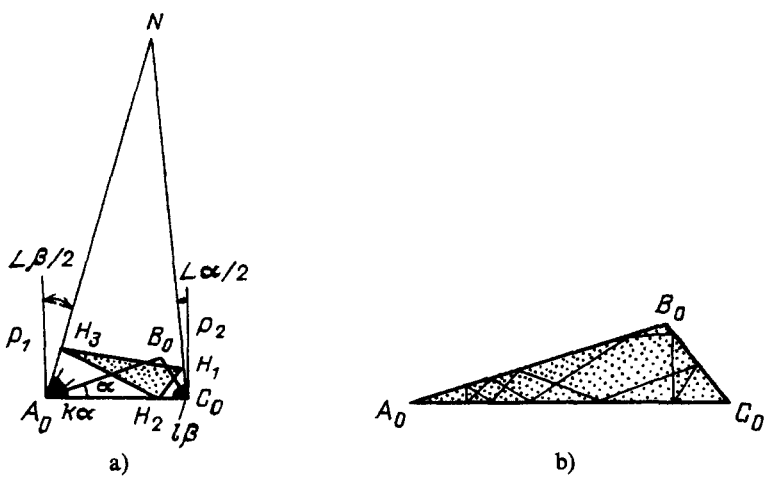



Fig. 25

We prepare a finite part of the corridor $Q_1Q_2 \ldots Q_n$ for the required periodic trajectory so that it will coincide with a part of an *acute* triangle containing its 3-link altitude trajectory (this is the essence of the construction!). When the figure $Q_1Q_2 \ldots Q_n$ is laid on the triangle $Q = A_0B_0C_0$, this 3-link trajectory becomes the required periodic trajectory $\Gamma$, as shown in Fig. 25b.

Let us realize this plan. We consider in the upper half-plane the chain of triangles

$$(6)\qquad A_{-l+1}B_{-l+1}C_0;\ \ldots;\ A_{-1}B_{-1}C_0;\ A_0B_0C_0;\ A_0B_1C_1;\ \ldots;\ A_0B_{k-1}C_{k-1},$$

in which any two neighbouring triangles have a common side and are positioned symmetrically with respect to it. For this we perform $k-1$ mirror reflections of the angle $B_0A_0C_0$ counterclockwise, and $l-1$ mirror reflections of the angle $B_0C_0A_0$ clockwise. Then the 'extreme' rays $A_0N$ and $C_0N$ form with the line $A_0C_0$ the *acute* angles $NA_0C_0 = k\alpha$ and $NC_0A_0 = l\beta$ (this follows from the assumptions), and with the perpendiculars $p_1$ and $p_2$ to the line $A_0C_0$ at $A_0$ and $C_0$ angles less than $\beta/2$ and $\alpha/2$ (this also follows from the assumptions of the theorem). The triangle $A_0NC_0$ thus obtained is acute, hence it contains a 3-link isolated trajectory, $H_1H_2H_3$, where $H_1$, $H_2$, $H_3$ are the feet of the altitudes of $\Delta A_0NC_0$.

We can immediately verify that by the conditions (5), imposed on $\alpha$ and $\beta$, the segment $H_1H_3$ lies below the vertex $B_0$, and hence the polygonal line $H_1H_2H_3$ will be contained in the figure $\Phi$ formed by the triangles in the chain (6).

Superimposing $\Phi$ on $A_0B_0C_0$ 'like an accordion' (that is, using the reverse reflections), the polygonal line $H_1H_2H_3$ becomes a $(2(k+l)-1)$-link periodic trajectory $\Gamma$ in $\Delta A_0B_0C_0$. Theorem 1 has been proved. □

***Theorem 2.*** *In an obtuse triangle with acute angles $\alpha$ and $\beta$ satisfying the inequality*

$$(7)\qquad (1-\cot(k\alpha)\cdot\tan\alpha)(1-\cot(l\beta)\cdot\tan\beta) > 1-\tan\alpha\tan\beta$$

*there is a $(2(k+l)-1)$-link isolated trajectory.*

*Proof.* We perform the same constructions as in the proof of Theorem 1. The inequality (7) ensures that the segment $H_2H_3$ lies *below* the vertex $B_0$ (being the exact condition for this segment to lie below $B_0$). This enables us to assert that the 3-link trajectory $H_1H_2H_3$ in $\Delta A_0NC_0$ becomes a $(2(k+l)-1)$-link trajectory in $\Delta A_0B_0C_0$ under the reverse reflections. □

**B. Isolated trajectories in acute triangles.**

Theorem 2, which is essentially stronger than Theorem 1, enables us to discover isolated trajectories also in acute triangles. The condition (7) on the acute angles $\alpha$ and $\beta$ is necessary and sufficient for the triangle with angles $\alpha$, $\beta$ to contain an isolated trajectory. It can be obtained by the construction given in subsection A from the 3-link trajectory in the 'large' triangle $A_0C_0N$. Condition (7) is automatically fulfilled for acute triangles. Since in an acute triangle the angle $\gamma$ is also acute, the procedure for constructing such a trajectory can be carried out three times, once for each pair of angles ($\alpha$, $\beta$), ($\alpha$, $\gamma$), and ($\beta$, $\gamma$). Moreover, (for fixed $\alpha$, $\beta$) the numbers $k$ and $l$ in (7) may vary between certain bounds: $1 \leqslant k \leqslant k_0$, $1 \leqslant l \leqslant l_0$. This implies that if an acute triangle $A_0B_0C_0$ has an isolated trajectory with $2p+1$ links, then it

has also trajectories with *three, five*, ..., $2p-1$ links (to obtain a trajectory with $2p'+1$ links, $p' < p$, we must perform a smaller number of reflections around one of the vertices $A_0$ or $C_0$, or a smaller number of reflections around both of them). Besides, these trajectories may have various combinatorial types, depending on the pair of vertices around which the reflections are performed and on their number.

*Example.* In an acute triangle $A_0B_0C_0$ with angles $A_0$ and $C_0$ close to $\pi/2$ there are many isolated trajectories. To find them we must follow the procedure in subsection A, using the side $B_0A_0$ as base and performing any number of mirror reflections around $B_0$, up to a number $k$ determined by the inequality (7).

**C. The bifurcation diagram of isolated trajectories (a 'hang-glider' configuration).** The set of points in the square $K$ (see §7) which represent triangles in which there are isolated trajectories is depicted in Fig. 26. It splits into two parts, having, however, an interior structure: 1) the triangle in $K$ with vertices $(0, \pi/2)$, $(\pi/2, \pi/2)$, $(\pi/2, 0)$ splits into the triangle with vertices $(\pi/4, \pi/4)$, $(\pi/4, 0)$, $(0, \pi/4)$ and an infinite number of right-angled trapeziums in which the numbers 3, 5, 7, ... indicate the number of links of the periodic trajectories in the corresponding acute triangles; and 2) the square with vertices (0, 0),

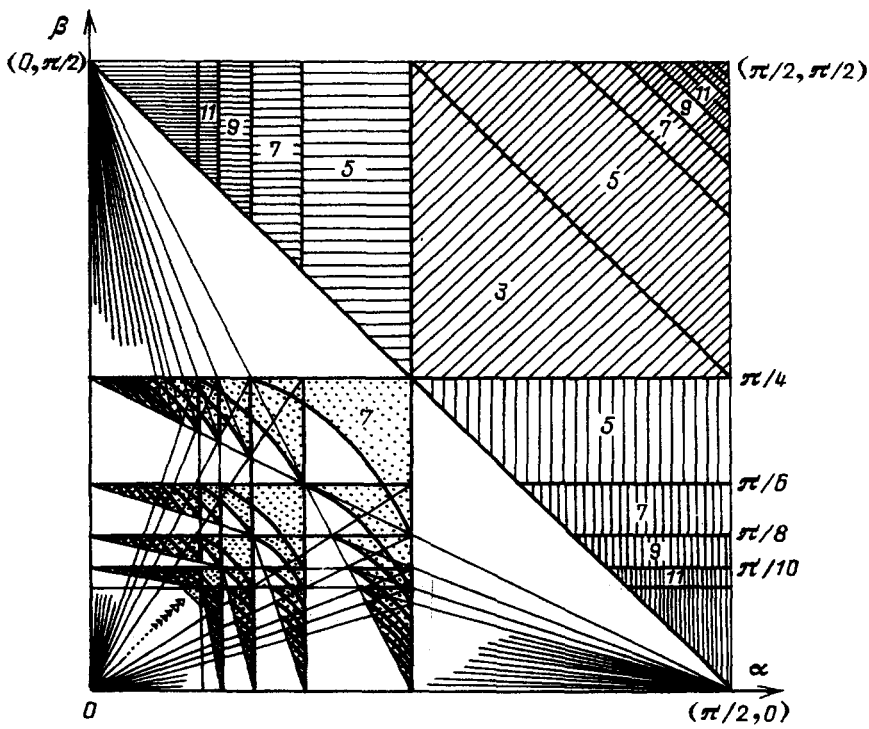



Fig. 26

$(0, \pi/4)$, $(\pi/4, \pi/4)$, $(\pi/4, 0)$, inside which we have drawn curvilinear triangles, whose boundaries are given by the trigonometric equations obtained from (7) by replacing $>$ by $=$. In their shape these curvilinear triangles resemble hang-gliders, and therefore we have called them 'hang-glider' configurations.

Moreover, in Fig. 26 we have also depicted a set of points taken from Fig. 18.

The periodic trajectories are sensitive to the shape of the triangle. The reason for the appearance of a periodic trajectory is a crossing of the boundary of a hang-glider (or of a trapezium, in the case of an acute triangle).

*Remark.* If we put $k = l = 2$ in (7), we obtain an isolated trajectory with seven links in an obtuse triangle. We can show by a simple argument that in an obtuse triangle there are no isolated trajectories with three or five links.

## §10. The density of *F*-triangles in a neighbourhood of (0, 0)

We call all triangles in which the construction of §9 can be realized *F-triangles*, in honour of Fagnano, whose problem served as the starting point for realizing this construction.

In this section we are interested in the obtuse triangles depicted in $\boldsymbol{K}$ as hang-gliders (Fig. 27). All hang-gliders are decomposed into infinitely many series (I), (II), ..., each of which is determined by a 'leading' hang-glider, with vertex $(\pi/(2k), \pi/(2k))$, $k = 2, 3, 4, \ldots$, from which 'wings' of finer hang-go down and to the left. The hang-gliders in a single series intersect, and a transition from one to the other indicates that the triangle in question contains periodic trajectories of various types.

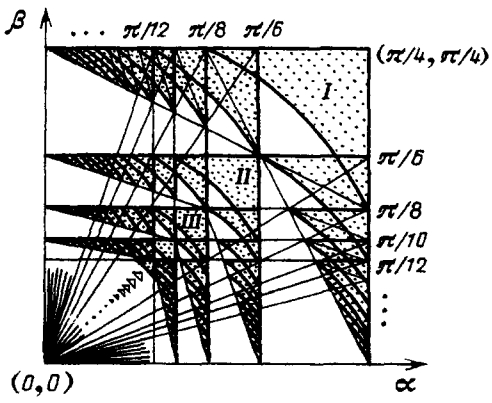



Fig. 27

A problem that can be posed is that of computing the relative measure of the set of points of the hang-gliders in the square with side $\pi/4$, that is, of finding the ratio of the shaded area in Fig. 27 to the area $\pi/4 \times \pi/4$ of the square. Another problem (with which we will deal) is the computation of the relative measure of hang-gliders in a neighbourhood of (0, 0).

**A. The 'density' of hang-gliders at (0, 0).**
We consider the hang-gliders in a given series with high number. The diagonal $XZ$ divides the rectangle $XYZT$ which encloses the downward directed 'wings' of the leading hang-glider into two triangles; all of the 'wing' lies above this diagonal $XZ$. Since the hang-gliders in the 'wing' overlap (see Fig. 27), they occupy more than half the area of $\Delta XYZ$, hence they occupy more than a quarter of the area of the rectangle $XYZT$ (Fig. 28).

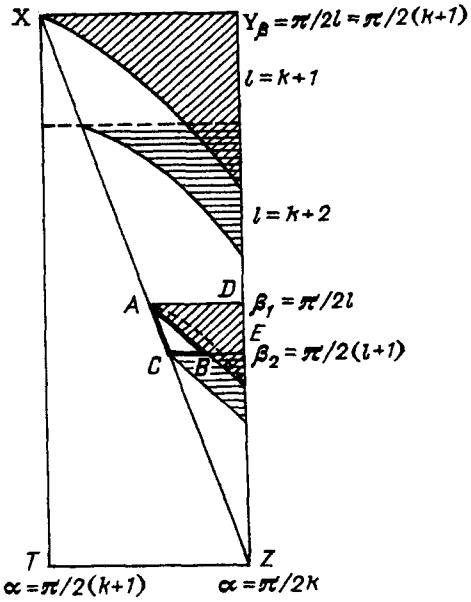



Fig. 28

From the inequality (7) (§9A) we can extract the equation of the curvilinear boundary of a hang-glider:

$$(8)\qquad \begin{aligned}(1-\cot(k\alpha)\cdot\tan\alpha)(1-\cot(l\beta)\cdot\tan\beta) &= 1-\tan\alpha\cdot\tan\beta,\\ \tan\alpha\cdot\tan\beta\cdot(1+\cot(k\alpha)\cdot\cot(l\beta)) &= \tan\alpha\cdot\cot(k\alpha)+\tan\beta\cdot\cot(l\beta).\end{aligned}$$

We consider small $\alpha$ and $\beta$, namely

$$(9)\qquad \pi/(2(k+1))<\alpha<\pi/(2k),\qquad \pi/(2(l+1))<\beta<\pi/(2l).$$

***Lemma.*** $\alpha+\beta = \pi/(2l)+\pi/(2k)-\pi/(2kl)+O(k^{-2}l^{-1}+k^{-1}l^{-2}+k^{-3}+l^{-3})$ *as* $k, l \to \infty$.

*Proof.* We substitute the values of $\alpha$ and $\beta$ from (9) into (8), having first expanded the trigonometric functions in Taylor series:

$$\begin{aligned}&[\pi/(2k)+O(k^{-2})]\,[\pi/(2l)+O(l^{-2})]\,[1+(\pi/2-l\beta)+O((\pi/2-l\beta)^3)\,(\pi/2-k\alpha+\\ &\quad +O((\pi/2-k\alpha)^3)] = [\pi(2k)+O(k^{-2})]\,[\pi/2-k\alpha+O((\pi/2-k\alpha)^3)]+\\ &\qquad +[\pi/(2l)+O(l^{-2})]\,[\pi/2-l\beta+O((\pi/2-l\beta)^3)],\end{aligned}$$

hence

$$\frac{\pi}{4kl}\{1+(\pi/2-l\beta)(\pi/2-k\alpha)\}=\frac{\pi}{2k}\left(\frac{\pi}{2}-k\alpha\right)+\frac{\pi}{2l}\left(\frac{\pi}{2}-l\beta\right)+R,$$

where

$$R=O\Big(k^{-2}l^{-1}+k^{-1}l^{-2}+[(\pi/2-l\beta)^3(\pi/2-k\alpha)+(\pi/2-k\alpha)^3(\pi/2-l\beta)]k^{-1}l^{-1}+$$
$$+k^{-2}\left(\frac{\pi}{2}-k\alpha\right)+l^{-2}\left(\frac{\pi}{2}-l\beta\right)+\left(\frac{\pi}{2}-k\alpha\right)^3k^{-1}+\left(\frac{\pi}{2}-l\beta\right)^3l^{-1}\Big).$$

We set $x=\pi/2-k\alpha$, $y=\pi/2-l\beta$. After the necessary cancellations we obtain

$$\frac{\pi}{2kl}+\frac{\pi}{2kl}xy=\frac{\pi}{2}\left(\frac{1}{k}+\frac{1}{l}\right)-(\alpha+\beta)+R, \tag{10}$$

Since $0<x<\frac{\pi}{2}\left(1-\frac{k}{k+1}\right)=\frac{\pi}{2}\frac{1}{k+1}$, $0<y<\frac{\pi}{2}\frac{1}{l+1}$, we have $xy=O(k^{-1}l^{-1})$, and from (10) we obtain

$$\alpha+\beta=\frac{\pi}{2k}+\frac{\pi}{2l}-\frac{\pi}{2kl}-\frac{\pi}{2kl}\cdot O(k^{-1}l^{-1})+R=\frac{\pi}{2k}+\frac{\pi}{2l}-\frac{\pi}{2kl}+$$
$$+O(k^{-2}l^{-2}+k^{-2}l^{-1}+k^{-1}l^{-2}+l^{-4}k^{-2}+l^{-2}k^{-4}+k^{-4}+l^{-4})=$$
$$=\frac{\pi}{2k}+\frac{\pi}{2l}-\frac{\pi}{2kl}+O(k^{-2}l^{-1}+k^{-1}l^{-2}+k^{-3}+l^{-3}),$$

which is what we had to prove. □

***Corollary.*** *For $k>l$ we have*

$$\alpha+\beta=\frac{\pi}{2k}+\frac{\pi}{2l}-\frac{\pi}{2kl}+O(k^{-3}).$$

We denote the quantity $O(k^{-3})$ by $\delta$.

***Theorem.*** *The relative measure* $\mu(0,0)$ *of hang-gliders in a neighbourhood of* $(0,0)$ *is equal to* $1/3$.

*Proof.* We compute from Fig. 28 (with the necessary precision) the measure of the unshaded ('white') portion $\Delta XYZ$. We consider $\Delta ABC$, in which

$$AC\cong CB=\frac{\pi}{2l}-\frac{\pi}{2(l+1)}=\frac{\pi}{2l(l+1)}.$$

Its area differs from the area of the curvilinear triangle $ABC$ (which we really want to compute!) by the quantity $O(\delta/l^2)$, where $\delta$ is the maximal

clearance between the curvilinear segment $AB$ and the line segment $AB$ (Fig. 29). The corollary following the lemma implies that $\delta = O(k^{-3})$.

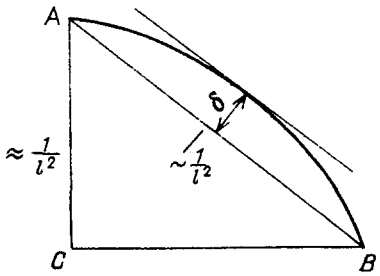



Fig. 29

The area of $\Delta ABC$ is equal to $S_{\Delta ABC} = \frac{1}{2} \cdot \frac{\pi^2}{4l^2(l+1)^2}$, and the area of the triangle $XYZ$ is equal to $S_{\Delta XYZ} = \frac{1}{2} \cdot \frac{\pi}{2(k+1)} \cdot \frac{\pi}{2(k+1)}$; therefore the ratio of the 'white' area in $\Delta XYZ$ to the area of $\Delta XYZ$ is equal to

$$\frac{S_{\text{white}}}{S_{\Delta XYZ}} = \lim_{k\to\infty} \frac{\sum_{l=k+1}^{\infty} \frac{1}{l^2(l+1)^2}}{\frac{1}{k(k+1)^2}} = \lim_{k\to\infty} \frac{\int_k^{\infty} \frac{dl}{l^4}}{\frac{1}{k^3}} = \frac{\frac{1}{3k^3}}{\frac{1}{k^3}} = \frac{1}{3}.$$

Hence

$$\frac{S_{\text{black}}}{S_{\Delta XYZ}} = 1 - \frac{1}{3} = \frac{2}{3}, \text{ and } \mu(0,0) = \frac{S_{\text{black}}}{S_{XYZT}} = \frac{1}{2}\,\frac{S_{\text{black}}}{S_{\Delta XYZ}} = \frac{1}{3},$$

which is what we had to prove. □

**B. The directional 'density' of hang-gliders.**

We can naturally introduce the following function $\rho(\alpha/\beta)$, the density at (0, 0) taken by the hang-gliders in the direction $\alpha/\beta$. Let $K(\alpha, \beta; a)$ be the area of those hang-gliders falling in the square with side $a$ and centre at $(\alpha, \beta)$ (Fig. 30). We put

$$\rho(\alpha/\beta) = \lim_{a\to 0} \lim_{t\to 0} \frac{K(\alpha, \beta; a)}{(ta)^2}.$$

The theorem just proved asserts that

$$\mu(0,0) = \int_0^1 \int_0^1 \rho(\alpha/\beta)\, d\alpha\, d\beta = \frac{1}{3}.$$

We can show that $\rho(1) = 1/2$, that is, the relative measure at (0, 0) of the 'leading' hang-gliders in all series is equal to 1/2. The function $\rho$ behaves as depicted in Fig. 31.

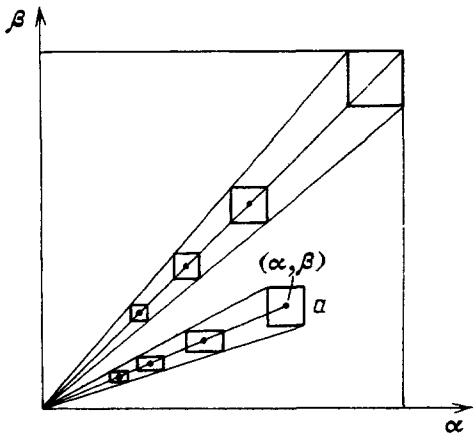



Fig. 30

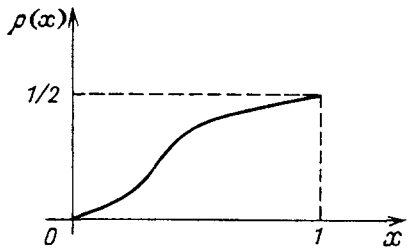



Fig. 31

## §11. Generalization of the construction of isolated trajectories in obtuse triangles

The inequalities (5), as well as the inequality (7) in §9 generalizing them, allow us to construct the diagram of $F$-triangles. However, the construction from §9 can be generalized in the following way.

Suppose we construct from a triangle $\Delta_0$ with vertices $A$, $B$, $C$ and acute angles $\alpha$, $\beta$, using $AC$ as base, a chain of triangles

$$\Delta_{-2l},\ \Delta_{-2l+1}, \ldots, \Delta_{-1},\ \Delta_0,\ \Delta_1, \ldots, \Delta_{2k-1},\ \Delta_{2k}, \tag{11}$$

which intersect the horizontal line $AC$ and the upper half-plane, and where neighbouring triangles in the chain (11) have a common side and are positioned symmetrically with respect to this side.

Since the result of two successive reflections in non-parallel lines $l_1$, $l_2$ is equal to rotation through twice the angle between these lines, the triangle $\Delta_{-2l}$ is rotated through an angle $n_1\alpha + n_2\beta + n_3\pi$ with respect to $\Delta_0$; similarly, $\Delta_{2k}$ is rotated through $m_1\alpha + m_2\beta + m_3\pi$ with respect to $\Delta_0$.

We assume that the vectors $(n_1, n_2)$ and $(m_1, m_2)$ are not collinear, while the triangles $\Delta_{-2l}$ and $\Delta_{2k}$ have sides that are *perpendicular* to the line $AC$. If, moreover, *$AC$ does not contain vertices* of triangles in the chain (11) except $A$ and $C$, it turns out that by an arbitrarily small change of $\alpha$ and $\beta$ we can obtain a triangle having isolated trajectories. For this the sides of $\Delta_{-2l}$ and $\Delta_{2k}$ perpendicular to $AC$ must be inclined to each other by a small angle such that their extensions intersect at a point $N$. As a result there arises an acute triangle $XYN$ ($X$, $Y$ are points on $AC$), in which we have to take the 3-link altitude trajectory and 'displace' it into the original triangle $ABC$.

If we apply this construction to a right-angled triangle with an acute angle of $\pi/8$, we obtain the following assertion.

***Assertion.*** *In a triangle $ABC$ whose angles $\alpha$, $\beta$ satisfy the inequalities $\pi/8-0.001 < \alpha < \pi/8$, $\pi/3 < \beta < 3\alpha$ there is a 15-link isolated trajectory.*

The proof follows by considering Fig. 32.

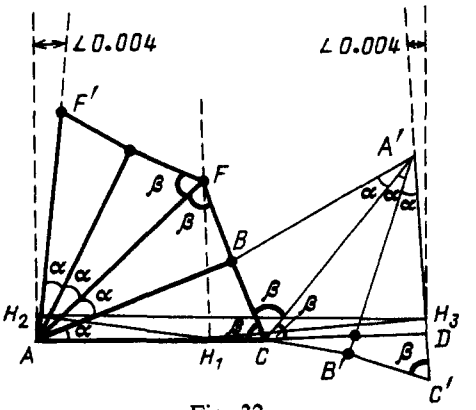



Fig. 32

Another example of a 15-link isolated trajectory, based on the same idea, is sketched in Fig. 33.

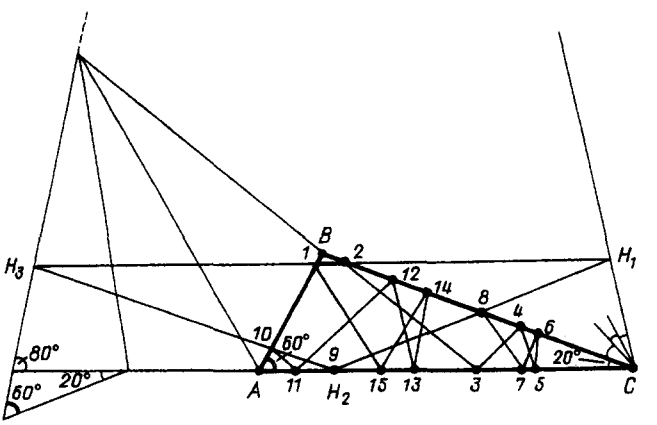



Fig. 33

## §12. Stable and unstable billiard trajectories in plane Weyl chambers

In this section we classify all stable and unstable periodic trajectories in the plane Weyl chambers: a rectangle, right-angled triangles with an angle of 45° or 30°, and an equilateral triangle. It turns out that in the first three Weyl chambers there are no stable trajectories, while in the equilateral triangle there are infinitely many combinatorial types of stable trajectories, which can be described by a single geometrical method.

## A. Periodic trajectories in a rectangle.

### A1. Notation.

We consider a partition of the plane into equal rectangles. The periodic trajectories can be very simply described in terms of the rectangular lattice thus obtained.

For our convenience we introduce the following labelling of this lattice. We take one of its nodes to be the coordinate origin, and take the coordinate axes in the directions of the lines of the lattice. We choose the units along the axes in such a way that all lattice nodes have integer coordinates. Then each lattice rectangle will have vertices with coordinates $(i, j)$, $(i+1, j)$, $(i, j+1)$, $(i+1, j+1)$, with $i, j \in \mathbb{Z}$. We call the pair $(i, j)$ the *coordinates* of this rectangle (Fig. 34).

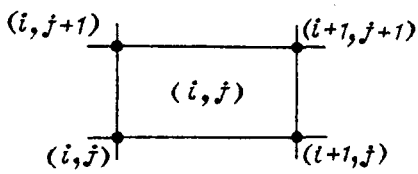



Fig. 34

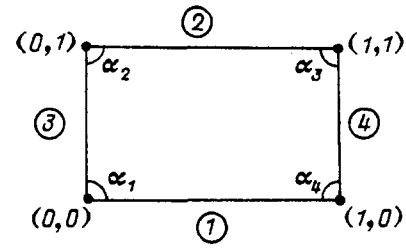
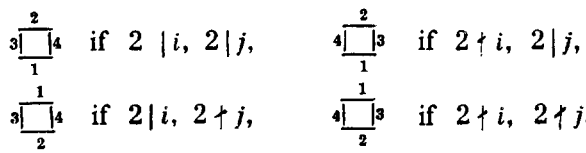



Fig. 35

We mark the sides and angles of the rectangle (0, 0) as in Fig. 35. Starting from this we can mark the sides of all lattice rectangles in a unique way so that any two rectangles with a common side are symmetrically positioned with respect to this side (taking into account the markings of the sides). The possibility of such a marking lies in the fact that at each lattice node an even number of sides (namely, 4) cross.

This marking has the form




After straightening, a periodic trajectory in the rectangle (0, 0), launched from the side (1), becomes a segment joining two corresponding points in two lattice rectangles. We can easily see that there are no isolated trajectories in a rectangle: such a trajectory must emanate perpendicular to a side of it, but we know already that a 'perpendicular' trajectory has an even number of links. Thus, after straightening, each periodic trajectory launched from the side (1) of the rectangle (0, 0) becomes a segment with endpoints $(\varepsilon, 0)$ and $(i+\varepsilon, j)$, where $i, j \in \mathbb{Z}$, $2 \mid i$, $2 \mid j$, $j > 0$, and $(\varepsilon, 0)$ is the starting point. The converse is also true. Under the condition $\mathrm{GCD}(i, j) = 2$, a trajectory going from the rectangle (0, 0) to the rectangle $(i, j)$ cannot be contracted.

**A2. *Theorem* 1.** *There are no stable periodic trajectories in a rectangle.*

Theorem 1 will be proved in subsections A3–A7.

**A3. Singular points and pencils.**
We fix a direction of periodic trajectories, that is, a rectangle $(x, y)$, where $x, y \in \mathbb{Z}$, $\mathrm{GCD}(x, y) = 2$, $y > 0$. Now we give a general definition of singular point.

*Definition.* A point on the boundary of a polygon $Q$ is called *singular for a given direction* if the trajectory launched from it in this direction hits a vertex of $Q$.

It is easily seen that the points on the side (1) of the rectangle (0, 0) dividing this side into $y/2$ equal pieces are *singular*. All remaining points on this side are non-singular, and we obtain $y/2$ pencils of periodic trajectories.

We note that after the folding the segments of the trajectory joining the points $(\varepsilon, 0)$ and $(x/2+\varepsilon, y/2)$ and the points $(x/2+\varepsilon, y/2)$ and $(x+\varepsilon, y)$ become pieces of the billiard trajectory which are positioned symmetrically relative to the centre of the rectangle. Hence, *every periodic billiard trajectory in a rectangle is symmetric relative to the centre of the rectangle.*

All the $y/2$ pencils, to one of which the trajectory $[(\varepsilon, 0), (x+\varepsilon, y)]$ belongs, are launched from the side (1) at an angle $\alpha$ for which $\tan\alpha = k \cdot y/x$, where $k$ is the ratio of the lengths of the ratio of the sides (3) and (1) of the rectangle. The same number of pencils are launched from the side (2) at the angle $\alpha$.

A *maximal pencil* is a pencil all trajectories of which are launched from a segment $(e, f)$ on some side, for certain $e$ and $f$ which are singular points on this side.

***Lemma* 1.** *Any two maximal pencils launched from the side* (1) *or* (2) *at the same angle coincide up to the direction of the motion.*

*Proof.* To prove the lemma we use the following almost obvious assertion.

***Assertion* 1.** *Suppose that in a polygon $Q$ a maximal pencil of periodic trajectories is launched from a segment $(e, f)$ on a side $l$ at a given angle. Then an arbitrary incontractible trajectory in this pencil can hit the same segment $(e, f)$ again at the same angle only in the following cases:*

1) *the direction of the pencil is a possible direction of an isolated trajectory;*
2) *the pencil consists of 'perpendicular' trajectories.*

For a rectangle the only exceptional pencil can be the pencil launched perpendicular to the side (1). For this pencil the assertion of the lemma can be established directly. Concerning the other directions we find that an incontractible trajectory launched in the direction $(x, y)$ intersects the sides (1) and (2) at an angle $\varphi = \arctan(k \cdot y/x)$, in total $y$ times. However, only $y/2$ pencils are launched from such a side at the angle $\varphi$ (this was discussed above). This also implies the assertion of the lemma. □

**A4. Forms for trajectories and contours.**

Let $P_0, P_1, \ldots, P_m$ be a corridor of lattice rectangles, where $P_0$ and $P_m$ are compatible with (0, 0) by a shift of the lattice (taking into account the marks of the sides). This corridor can be given by a code-word $W = l_{i_1} l_{i_2} \ldots l_{i_m}$ in the alphabet $\{l_1, l_2, l_3, l_4\}$, where $i_j$ is the index of the common side of $P_{j-1}$ and $P_j$, $j = 1, 2, \ldots, m$. The word $W$ obviously has *even* length.

Now let $Q$ be an *arbitrary* quadrangle. We mark its sides in the same order as we did for the rectangle (0, 0), and we construct from $W$ the sequence of quadrangles $Q_0 = Q, Q_1, Q_2, \ldots, Q_m$, with marked sides, in which $Q_{j-1}$ and $Q_j$ ($j = 1, \ldots, m$) have common side $i_j$ relative to which they are symmetrically positioned (taking into account the marks of the sides).

Since $W$ has even length, the quadrangles $Q$ and $Q_m$ are identically oriented, and we can talk about the inclination of $Q_m$ to $Q$.

***Assertion* 2.** *Up to $2\pi$ the inclination of $Q_m$ to $Q$ is a linear combination $k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 + k_4\alpha_4$, where $k_i \in \mathbb{Z}$ and $\alpha_i$ are the angles of $Q$ (with angles marked according to* Fig. 35). *The coefficients $k_1, k_2, k_3, k_4$ are unique, up to addition of one and the same number.*

Thus we have obtained a completely determined element of the set $\mathbb{Z}^4/(1, 1, 1, 1)\mathbb{Z}$, which we call a *form* of $W$ and denote by $\mathscr{A}(W)$.

We note that the description given above of the corridor of rectangles can be specified by a continuous path $\mathfrak{f}$ starting and ending in lattice rectangles compatible (by a shift of the lattice) with (0, 0), not passing through lattice nodes, and intersecting the sides of the lattice only finitely many times.

*Definition.* The *form* $\mathscr{A}(\mathfrak{f})$ *of* $\mathfrak{f}$ is the form $\mathscr{A}(W)$.

In particular, we have defined the form of a path consisting of straightened billiard trajectories. Moreover, the following lemma is obvious.

***Lemma* 2.** *A periodic trajectory $\Gamma$ is stable if and only if its form is zero:* $\mathscr{A}(\Gamma) = 0$.

Also obvious is the following lemma.

***Lemma* 3.** *Let $\mathfrak{f}_1, \mathfrak{f}_2$ be two curves for which a form is defined; let $(-\mathfrak{f}_1)$ be the curve $\mathfrak{f}_1$ with the opposite direction, and $\mathfrak{f}_1 + \mathfrak{f}_2$ the sum of the curves $\mathfrak{f}_1$ and $\mathfrak{f}_2'$, where $\mathfrak{f}_2'$ is obtained from $\mathfrak{f}_2$ by a shift of the lattice and starts in the rectangle in which $\mathfrak{f}_1$ ends. Then*

$$\mathscr{A}(\mathfrak{f}_1+\mathfrak{f}_2) = \mathscr{A}(\mathfrak{f}_1) + \mathscr{A}(\mathfrak{f}_2), \qquad \mathscr{A}(-\mathfrak{f}_1) = \mathscr{A}(\mathfrak{f}_1).$$

The marks of the sides of the lattice rectangles induce marks of the lattice nodes, that is, marks of the angles of the partition with vertex at a node. In fact (Fig. 36),

the number 1 is given to vertices $(i, j)$ for which $2 \mid i$, $2 \mid j$;
the number 2 is given to vertices $(i, j)$ for which $2 \mid i$, $2 \nmid j$;
the number 3 is given to vertices $(i, j)$ for which $2 \nmid i$, $2 \nmid j$;
the number 4 is given to vertices $(i, j)$ for which $2 \nmid i$, $2 \mid j$.

*Assertion* 3. *Let* $\mathfrak{f}$ *be a closed Jordan contour for which a form is defined, that is, its beginning (which is also its end) lies in a rectangle* $(i, j)$ *with* $2 \mid i$, $2 \mid j$. *Let* $q_1$, $q_2$, $q_3$, $q_4$ *be the numbers of lattice nodes with indices* 1, 2, 3, 4 *(respectively) lying inside* $\mathfrak{f}$. *Then*

$$\mathscr{A}(\mathfrak{f}) = \pm 4(q_1\alpha_1 + q_2\alpha_2 + q_3\alpha_3 + q_4\alpha_4)$$

*where* $+/-$ *is chosen if the contour goes in the counterclockwise/clockwise direction.*

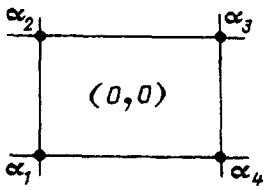



Fig. 36

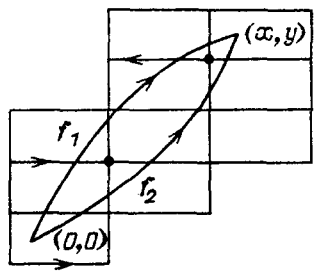



Fig. 37

**A5. Transference of stability through a singular point.**
A rectangle has the following property: every trajectory parallel to a given periodic trajectory is either itself periodic or singular (hits a vertex), and the number of singular points on the boundary of the rectangle is finite.

*Definition.* Let $x_0$ be a singular point for a given direction of trajectories, and lying on a side of a rectangle. We say that *stability is not transferred through* $x_0$ if the trajectories launched from boundary points close to $x_0$ and on both sides of $x_0$ cannot be simultaneously stable.

This definition immediately implies the following assertion.

*Assertion* 4. *If for every direction of trajectories stability is not transferred through all vertices, then stability is not transferred through all singular points.*

**A6. Transference of stability through a vertex.**
*Assertion* 5. *In a rectangle, for an arbitrary direction stability is not transferred through any vertex.*

*Proof.* We consider two incontractible periodic trajectories $\mathfrak{f}_1$ and $\mathfrak{f}_2$ launched from small left and right neighbourhoods of the vertex 1 of the rectangle (0, 0). Both trajectories end in a rectangle $(x, y)$, $\mathrm{GCD}(x, y) = 2$, $y > 0$. We consider the path $\mathfrak{f} = \mathfrak{f}_1 + (-\mathfrak{f}_2)$ (Fig. 37). $\mathfrak{f}$ is a closed Jordan contour enclosing two lattice nodes, $(x, y)$ and $(x/2, y/2)$. Hence, by Assertion 3, $\mathscr{A}(\mathfrak{f}) \neq 0$. But $\mathscr{A}(\mathfrak{f}) = \mathscr{A}(\mathfrak{f}_1 + (-\mathfrak{f}_2)) = \mathscr{A}(\mathfrak{f}_1) - \mathscr{A}(\mathfrak{f}_2)$, hence $\mathscr{A}(\mathfrak{f}_1) \neq \mathscr{A}(\mathfrak{f}_2)$, and the trajectories $\mathfrak{f}_1$ and $\mathfrak{f}_2$ cannot be simultaneously stable. □

*Corollary. In a rectangle, stability is not transferred through any singular point.*

**A7. Proof of Theorem 1.**
We consider an arbitrary direction $\vec{v}$ of periodic trajectories launched from a fixed side of a rectangle. Two cases are possible.

1) *There are singular points* for $\vec{v}$ on this side. Then, by Lemma 1, the periodic trajectories in the direction $\vec{v}$ are either all stable or all unstable. The corollary in subsection A6 implies that *all trajectories are unstable.*

2) We now assume that a side of the rectangle *does not contain singular points.* This is only possible if the trajectories are launched either perpendicular to a side, or at an angle not exceeding the angle $\varphi$ of inclination of the diagonal of the rectangle with its side (Fig. 38).

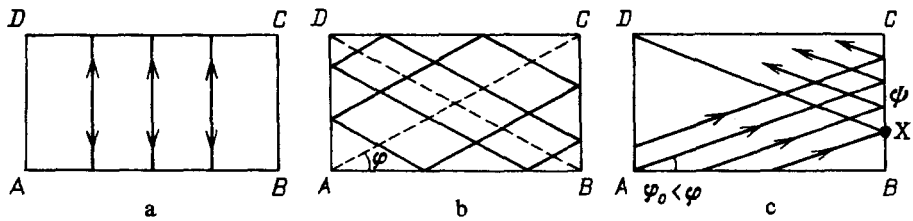



Fig. 38

In the first case, and also if the inclination *is exactly* $\varphi$, the instability can be directly proved (Fig. 38a, b). If the angle of inclination of the trajectory is *less* than $\varphi$, we consider the side of the rectangle in which the trajectory undergoes its first reflection (in Fig. 38c this is $BC$), as well as the angle $\psi$ of reflection in this side. It is obvious that the trajectories launched from the new side at the angle $\psi$ include all trajectories considered earlier; however, there are singular points for them (in Fig. 38c the point $A'$). So we are in case 1), and everything has been proved. □

**B. Periodic trajectories in right-angled triangles with angles 45°, 45°, 90°.**
**B1. Notation.**
We consider the partition of $\mathbb{R}^2$ into unit squares, with a coordinate system and marks of sides and corners of the lattice squares introduced (see subsection A). In each square $(i, j)$ we draw the diagonal joining the vertices 2 and 4. This gives a partition of $\mathbb{R}^2$ into triangles with angles 45°, 45°, 90°, which can also be obtained from one such ('basis') triangle by all possible reflections in all the sides (Fig. 39).

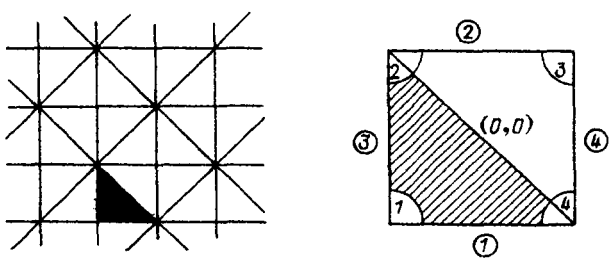



Fig. 39

The marks of the sides and vertices of the squares induce the following marks of the sides and vertices of the triangles:

sides: $a = (1) = (4)$, $b = (2) = (3)$, $c$ is the diagonal;
vertices: (I) = 1 = 3, (II) = 2, (III) = 4.

Of the two triangles into which the square $(i, j)$ is split we call the triangle containing the segment $[(i, j), (i+1, j)]$ the *basis triangle*, and say that it is the triangle $(i, j)$. The marks of the sides and vertices of the triangles $(i, j)$, for various $i$ and $j$, are depicted in Fig. 40. Here,

the index (I) is given to vertices $(i, j)$ with $2 \mid (i-j)$;
the index (II) is given to vertices $(i, j)$ with $i-j \equiv -1 \pmod 4$;
the index (III) is given to vertices $(i, j)$ with $i-j \equiv 1 \pmod 4$.

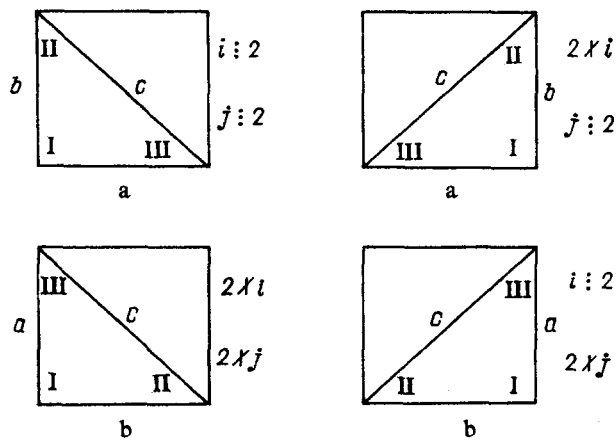



Fig. 40

The billiard trajectories will be launched from the side $a$ in the triangle (0, 0). It is easy to see that *isolated* trajectories can only go to $a$ at an angle of 45° or 90°. By directly considering these trajectories we convince ourselves that the triangle (45°, 45°, 90°) does not contain isolated trajectories.

Thus, after straightening, a periodic trajectory in the triangle (0, 0) becomes a segment joining points $(\varepsilon, 0)$ and $(i+\varepsilon, j)$, where $0 < \varepsilon < 1$, and the triangle $(i, j)$ must go into (0, 0) by a shift of the lattice (taking into account the marks of the sides), whence $2 \mid i$, $2 \mid j$. Conversely, every such trajectory not passing through lattice nodes gives a periodic trajectory in the triangle (0, 0) by folding. Here, the trajectory is incontractible if $\mathrm{GCD}(i, j) = 2$.

**B2.** ***Theorem 2.*** *The triangle* (45°, 45°, 90°) *does not contain stable periodic billiard trajectories.*

Theorem 2 will be proved in subsections B3–B7.

**B3. Singular points and pencils.**
We fix a direction of periodic billiard trajectories, that is, we fix a triangle $(x, y)$ with $\mathrm{GCD}(x, y) = 2$, $y > 0$.

Trajectories in this direction are launched from the side $a$ at an angle $\alpha = \arctan(y/x)$. As is easily seen, the singular points divide the side $a$ of the triangle (0, 0) into $y/2$ equal parts, so we have $y/2$ maximal pencils of periodic trajectories.

We note that a straightened periodic trajectory launched from the side $a$ of the triangle (0, 0) at an angle $\alpha \neq 45°, 90°$ can intersect the sides $a$ and $b$ at the same angle $\alpha$ only if these sides (under development) are parallel to the side $a$ of (0, 0). Therefore a trajectory going from a point $(\varepsilon, 0)$ to a point $(x+\varepsilon, y)$, where $\arctan(y/x) \neq 45°, 90°$, is symmetric relative to the axis of symmetry of the triangle if and only if it passes through the point $(i+\varepsilon, j)$, where the triangle $(i, j)$ is oriented in the opposite way to (0, 0). The only such triangle can be $(x/2, y/2)$. It is oriented in the necessary manner if $4 \mid (x-y)$. By considering the singular directions of 45° and 90°, we arrive at the following result.

***Assertion* 6.** *The trajectories that are symmetric relative to the axis of symmetry of the triangle* (0, 0) *are the trajectories launched in the direction* $(x, y)$, *where* $\mathrm{GCD}(x, y) = 2$, $y > 0$, $4 \mid (x-y)$, *and also the trajectory launched from the midpoint of the side a at an angle of* 90°.

As in the case of rectangles, the following holds.

***Lemma* 4.** *Any two maximal pencils launched from the sides a and b at the same angle coincide, up to the direction of motion.*

*Proof.* We have $y/2$ pencils launched from the side $a$ of the triangle (0, 0) in the direction of the triangle $(x, y)$. The same number of pencils are launched from $b$ at the same angle $\arctan(y/x)$. But an arbitrary incontractible trajectory intersects $a$ and $b$ at the angle $\alpha = \arctan(y/x)$ at least $y$ times. Therefore, according to Assertion 6, all the $y$ pencils launched from the two legs of the triangle (0, 0) at the angle $\alpha$ coincide up to the direction of motion. Only the singular directions can be exceptional: those at an angle of 45° or 90° to a side of the triangle. However, a direct verification shows that the assertion of the lemma is true also in this case. The lemma has been proved. □

**B4. Forms for trajectories and contours.**
As in the case of a rectangle we consider a sequence of triangles $T_0, T_1, \ldots, T_m$ of the partition in which neighbouring triangles have a common side and are positioned symmetrically about it, while $T_0$ and $T_m$ become (0, 0) by a shift of the lattice (taking into account the marks of the sides). Up to the choice of $T_0$, this sequence can be given by a code-word $W = i_1 i_2 \ldots i_m$ in the alphabet $\{a, b, c\}$, where $i_j$ is the index of the common side of $T_{j-1}$ and $T_j$ $(j = 1, \ldots, m)$.

Further, we choose an *arbitrary* triangle $T$, mark its sides as we did for (0, 0), and construct from the word $W$ the sequence of triangles $T^0, T^1, \ldots, T^m$ with marked sides. Here, $T$ and $T^m$ are identically oriented, and, up to a multiple of $\pi$, the inclination of $T^m$ to $T$ is equal to $k_1\alpha_{\mathrm{I}}+k_2\alpha_{\mathrm{II}}+k_3\alpha_{\mathrm{III}}$, where $\alpha_{\mathrm{I}}, \alpha_{\mathrm{II}}, \alpha_{\mathrm{III}}$ are the angles of $T$, marked as those of (0, 0), and $k_1, k_2, k_3 \in \mathbb{Z}$ are defined up to addition of one and the same number.

*Definition.* We call the expression $k_1\alpha_{\mathrm{I}}+k_2\alpha_{\mathrm{II}}+k_3\alpha_{\mathrm{III}}$, regarded as an element of the group $\mathbb{Z}^3/(1, 1, 1)\mathbb{Z}$, the *form of the sequence* $T_0, T_1, \ldots, T_m$ (*of the word* $W$).

As in the case of a rectangle, the sequences or words considered above can be given by means of continuous paths $\mathfrak{f}$ joining two lattice triangles which coincide with (0, 0) by a shift of the lattice, not passing through lattice nodes, and intersecting its sides only finitely many times.

*Definition.* The *form* $\mathscr{A}(\mathfrak{f})$ *of the path* $\mathfrak{f}$ is the form of the corresponding word.

***Lemma* 5.** *A trajectory* $\Gamma$ *is stable if and only if* $\mathscr{A}(\mathfrak{f}) = 0$.

The lemma follows from the definition of stability of a trajectory and of the form $\mathscr{A}(\mathfrak{f})$. □

***Assertion* 7.** *Let* $\mathfrak{f}$ *be a closed Jordan contour for which the form is defined, and let* $q_1, q_2, q_3$ *be the numbers of lattice nodes with indices* (I), (II), (III) *lying inside this contour. Then*

$$\mathscr{A}(\mathfrak{f}) = \pm(4q_1\alpha_{\mathrm{I}}+8q_2\alpha_{\mathrm{II}}+8q_3\alpha_{\mathrm{III}}),$$

*where* $+/-$ *is chosen if the contour goes in the counterclockwise/clockwise direction.*

**B5. Transference of stability through vertices.**

***Assertion* 8.** *In the triangle* (45°, 45°, 90°), *for an arbitrary direction stability does not transfer through vertices.*

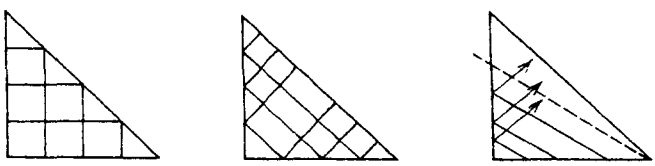
*Proof.* 1) Let $\Gamma_1$ and $\Gamma_2$ be two incontractible periodic trajectories launched from small left and right neighbourhoods of the vertex of the right angle of the triangle (0, 0) in the same direction. Both trajectories end in one triangle $(x, y)$ with $\mathrm{GCD}(x, y) = 2$, $y > 0$. Then the closed contour $\mathfrak{f} = \Gamma_2+(-\Gamma_1)$ encloses the two lattice nodes $(x, y)$ and $(x/2, y/2)$, therefore $\mathscr{A}(\mathfrak{f}) \neq 0$. But $\mathscr{A}(\mathfrak{f}) = \mathscr{A}(\Gamma_2)-\mathscr{A}(\Gamma_1)$, so the trajectories $\Gamma_1$ and $\Gamma_2$ cannot both be stable.

2) Suppose now that $\Gamma_1$ and $\Gamma_2$ emanate parallel to each other near a vertex of the angle 45°, but on different sides of it. In view of the symmetry we may assume this vertex to be (III). As in the case 1), the trajectories end in a triangle $(x, y)$ with $\mathrm{GCD}(x, y) = 2$, $y > 0$, and the closed contour $\mathfrak{f} = \Gamma_2+(-\Gamma_1)$ also encloses two lattice nodes, $(x+1, y)$ and $(x/2+1, y/2)$. Therefore $\mathscr{A}(\mathfrak{f}) = \mathscr{A}(\Gamma_2)-\mathscr{A}(\Gamma_1) \neq 0$ and stability is not transferred through the vertex of the angle 45°.

**B6. Proof of Theorem 2.**
We consider an arbitrary direction of periodic trajectories, launched from a leg of the triangle (45°, 45°, 90°). Two cases are possible.

1) If *there is a singular point* for this direction on the leg, then Lemma 4 and Assertion 8 imply that all trajectories going in this direction are unstable.

2) Suppose now that the leg *does not contain singular points.* Then the trajectories are launched either perpendicular to the leg, or at an angle not exceeding 45°. In the first case, and also for angle of incidence exactly 45°, instability can be directly verified (Fig. 41a, b).



a b c

Fig. 41

If the angle at which the trajectories are launched is less than 45°, then we choose a direction of motion of these trajectories in such a way that their first reflection occurs in the other leg. We then take this leg to be the initial leg, and choose as initial direction the one in which the trajectories move after their first reflection. Then the trajectories emanating from the new leg include all trajectories considered earlier, but now we may have singular points on the leg, that is, we have reduced this case to the previous one. The theorem has been proved. □

**C. Periodic trajectories in an equilateral triangle.**
**C1. Notation.**
We consider the partition of $\mathbb{R}^2$ into equilateral triangles with side lengths 1, and obtain a triangular lattice. We take one of the triangles to be the basis one; more precisely, we take one of its vertices to be the coordinate origin, while the axes are directed along its sides emanating from this vertex (Fig. 42). In this affine coordinate system, all lattice nodes have integer coordinates.

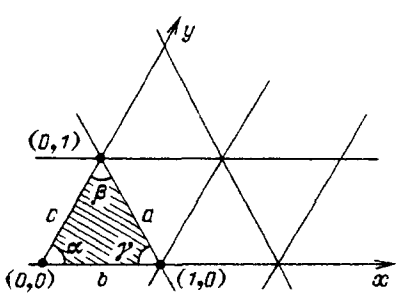



Fig. 42

The triangles in the partition can be split into two groups. One group, the *main one*, consists of the triangles that are compatible with the basis triangle by shifts. The vertices of such a triangle have coordinates $(i, j)$, $(i+1, j)$, $(i, j+1)$. The pair $(i, j)$ is called the *coordinates* of this triangle (the coordinates of the basis triangle are (0, 0)). The other group of triangles has vertices $(i+1, j)$, $(i, j+1)$, $(i+1, j+1)$.

We denote the sides and angles of the basis triangle by the letters $a$, $b$, $c$ and $\alpha$, $\beta$, $\gamma$, as indicated in Fig. 42. Starting from this we can 'carry over' the notation for sides and angles in a unique way to all of the lattice (this is possible because the number of sides passing through a single vertex is 6, which is even). Here, the side $[(i, j), (i+1, j)]$ of the triangle $(i, j)$ receives the letter $b$ if $i-j \equiv 0 \pmod 3$, the letter $a$ if $i-j \equiv 1 \pmod 3$, and the letter $c$ if $i-j \equiv -1 \pmod 3$.

**C2. Pencils of periodic trajectories in an equilateral triangle.**
We shall launch our billiard trajectories from the side $b$ of the triangle (0, 0). It is obvious that an isolated trajectory can be launched from $b$ only at an angle of 60° or 90°. Verification (and the reasoning in §8) shows that there is always a *single* isolated trajectory, which has three links (the altitude triangle).

After straightening, the other trajectories from a segment with ends $(\varepsilon, 0)$ and $(i+\varepsilon, j)$, where $i, j \in \mathbb{Z}$, $j > 0$, $0 < \varepsilon < 1$, and the triangle $(i, j)$ must be compatible with (0, 0) by a shift of the lattice (taking into account the marks of the sides), that is, $3 \mid (i-j)$. Conversely, it is obvious that all such segments not passing through a lattice node are straightened periodic trajectories. A trajectory is *incontractible* if $\mathrm{GCD}(i, j, (i-j)/3) = 1$.

We fix a direction of periodic trajectories, that is, a triangle $(x, y)$ with $\mathrm{GCD}(x, y, (x-y)/3) = 1$, $y > 0$. Then *either* $x \equiv y \not\equiv 0 \pmod 3$, and the singular points divide the starting side into $y$ equal parts, *or* $x \equiv y \equiv 0 \pmod 3$, and the singular points divide the starting side into $y/3$ equal parts. All the remaining points are non-singular. In the first case we obtain $y$, and in the second case $y/3$, maximal pencils of periodic trajectories in the direction $(x, y)$. We note that in the triangle (0, 0) such trajectories are launched at an angle

$$\alpha = \arctan \frac{\sqrt{3}}{2(x/y)+1}.$$

We note that if $x \equiv y \equiv 0 \pmod 3$, then the segment $[(\varepsilon, 0), (x+\varepsilon, y)]$ of the straightened trajectory going from the point $(x/3+\varepsilon, y/3)$ to $(2x/3+\varepsilon, 2y/3)$ can be obtained from the segment going from $(\varepsilon, 0)$ to $(x/3+\varepsilon, y/3)$ by rotation through 120° about the centre of the triangle (0, 0) in the counterclockwise (clockwise) direction if $x/3-y/3 \equiv 1$ $(x/3-y/3 \equiv -1) \pmod 3$. The same applies for the segments $[(2x/3+\varepsilon, 2y/3), (x+\varepsilon, y)]$ and $[(x/3+\varepsilon, y/3), (2x/3+\varepsilon, 2y/3)]$. Therefore, *if* $x \equiv y \equiv 0 \pmod 3$, *then the periodic trajectories are invariant under rotation of the equilateral triangle through* 120° *about its centre.*

However, if $x \equiv y \not\equiv 0 \pmod 3$, then the trajectory *is not invariant* under such a rotation. In fact, for the trajectory $[(\varepsilon, 0), (x+\varepsilon, y)]$ to be invariant it is necessary that it hits all sides of the triangle at equal angles at identically positioned points, that is, that the straightened trajectory passes through the points $(i_1+\varepsilon, j_1)$, $(i_2+\varepsilon, j_2)$, and $(i_3+\varepsilon, j_3)$, where $i_k - j_k \equiv k \pmod 3$, $k = 1, 2, 3$, and this can only happen if $x \equiv y \equiv 0 \pmod 3$.

Thus, we have $y/3$ $(y)$ pencils of periodic trajectories launched at an angle $\alpha$, $\tan\alpha = \sqrt{3}/(2(x/y)+1)$, from a side of the triangle, and $y$ $(3y)$ pencils in total for all sides. However, it is obvious that in general every incontractible trajectory in the direction $(x, y)$ intersects the sides of the triangle $(0, 0)$ $y$ times at an angle $\alpha$ (if $\alpha \neq 60°$). We obtain as a result the following assertion.

***Assertion* 9.** *If $x \equiv y \equiv 0 \pmod 3$, then all pencils launched from a side of the triangle at an angle $\alpha = \arctan\sqrt{3}/(2(x/y)+1)$ coincide, up to direction of motion, while if $x \equiv y \not\equiv 0 \pmod 3$, then there are three distinct pencils passing into each other under rotation of the triangle through* 120° *about its centre.*

We note that in this way the condition $x \equiv y \equiv 0 \pmod 3$ is determined by the shape of the trajectory (whether it is invariant or not under this rotation of the triangle), and does not depend on how the trajectory is straightened.

**C3. Forms for trajectories and contours.**

We consider a sequence of lattice triangles $T_0, T_1, \ldots, T_m$, where $T_0$ and $T_m$ belong to the main group and any two neighbouring triangles have a common side. We regard two such sequences as *equivalent* if they differ by a shift of the lattice (the marks for the sides not necessarily corresponding). Let $T_0 = (0, 0)$; then $T_m$ becomes $T_0$ under a transformation $\tau \circ m$, where $m$ is a shift of the lattice and $\tau$ is a transformation of $T_0$ mapping it onto itself.

*Definition.* $\tau$ is called the *type of the sequence* $T_0, T_1, \ldots, T_m$.

We note that the side $b$ of $(0, 0)$ is parallel to the side $\tau(b)$ of $T_m$.

Now let $T$ be an arbitrary triangle. We mark its sides as we did for $T_0$, and construct the sequence $T_{(0)} = T, T_{(1)}, \ldots, T_{(m)}$ of triangles with marked sides, in which $T_{(j-1)}$ and $T_{(j)}$ $(j = 1, \ldots, m)$ have a common side (with the same mark as the common side of $T_{j-1}$ and $T_j$), and positioned symmetrically with respect to it (taking into account side marks). The triangles $T$ and $T_{(m)}$ have the same orientation, and we may speak of the *inclination of $T_{(m)}$ to $T$*, that is, of the inclination of the side $\tau(b)$ of $T_{(m)}$ to the side $b$ of $T$.

***Assertion.*** *The inclination of $\tau(b)$ to $b$ is, up to $\pi$, a linear form $\mathscr{A} = k\alpha + m\beta + n\gamma$, where $\alpha$, $\beta$, $\gamma$ are the angles of $T$, marked as for* (0, 0), *and the coefficients $k, m, n \in \mathbb{Z}$ are determined up to addition of one and the same number.*

*Remark.* For those sequences $T_0, T_1, \ldots, T_m$, where $T_0$ and $T_m$ are compatible with $(0, 0)$ by a shift of the lattice, with corresponding sides, the definition

of the form coincides with the definition of the form for the triangle (45°, 45°, 90°).

As in the previous cases, sequences of triangles may be specified by paths, and forms may be defined for these paths. The following lemma is obvious.

***Lemma.*** *A trajectory* $\Gamma$ *is stable if and only if* $\mathscr{A}(\Gamma) = 0$.

As in subsections A, B, for two paths $\mathfrak{f}_1$ and $\mathfrak{f}_2$ we can introduce the idea of the path $\mathfrak{f}_1+\mathfrak{f}_2$ (word for word). Let $\tau$ and $\mathscr{A}$ be the *type* and *form* of paths.

***Assertion.*** $\tau(\mathfrak{f}_1+\mathfrak{f}_2)=\tau(\mathfrak{f}_1)\tau(\mathfrak{f}_2)$, $\tau(-\mathfrak{f}_1) = (\tau(\mathfrak{f}_1))^{-1}$, $\mathscr{A}(\mathfrak{f}_1+\mathfrak{f}_2) = \mathscr{A}(\mathfrak{f}_1)+\tau(\mathfrak{f}_1)\mathscr{A}(\mathfrak{f}_2)$, *where, by definition,* $\tau(k\alpha+m\beta+n\gamma) = k\tau(\alpha)+m\tau(\beta)+n\tau(\gamma)$.

*Remark.* For the usual sequences of triangles, such as those for 45°, 45°, 90°, the type $\tau$ = id, and the first two formulae are tautologies.

**C4. Stable trajectories.**

***Theorem* 3.** *The trajectories in the triangle* (0, 0) *for which* $x \equiv y \equiv 0 \pmod 3$ *are stable.*

*Proof.* Let $\Gamma = [(\varepsilon, 0), (x+\varepsilon, y)]$ be a straightened incontractible trajectory, and $\Gamma_1 = [(\varepsilon, 0), (x/3+\varepsilon, y/3)]$ its first third. Then $\Gamma = \Gamma_1+\Gamma_1+\Gamma_1$, hence $\mathscr{A}(\Gamma) = \mathscr{A}(\Gamma_1)+\tau\mathscr{A}(\Gamma_1)+\tau^2\mathscr{A}(\Gamma_1)$, where $\tau = \tau(\Gamma_1)$. The type $\tau$ of $\Gamma_1$ is not the identity (this would contradict the incontractibility of $\Gamma$), that is, $\tau$ is rotation through 120°. Let $\mathscr{A} = \mathscr{A}(\Gamma_1) = k\alpha+m\beta+n\gamma$. Then $\tau\mathscr{A} = k\cdot\tau(\alpha)+m\cdot\tau(\beta)+n\cdot\tau(\gamma)$, $\tau^2\mathscr{A} = k\cdot\tau^2(\alpha)+m\cdot\tau^2(\beta)+n\cdot\tau^2(\gamma)$, whence $\mathscr{A}(\Gamma) = k(\alpha+\tau(\alpha)+\tau^2(\alpha))+m(\beta+\tau(\beta)+\tau^2(\beta))+n(\gamma+\tau(\gamma)+\tau^2(\gamma))$. Since $\tau$ is rotation through 120°, either $\tau = (\alpha\beta\gamma) \in S_3$ or $\tau = (\alpha\gamma\beta) \in S_3$, hence $\alpha+\tau(\alpha)+\tau^2(\alpha) \equiv 0$, $\beta+\tau(\beta)+\tau^2(\beta) \equiv 0$, $\gamma+\tau(\gamma)+\tau^2(\gamma) \equiv 0$, and hence $\mathscr{A}(\Gamma) = 0$, that is, $\Gamma$ is a stable trajectory. □

**C5. Unstable trajectories.**

In the sequel we show the following property of an equilateral triangle. Suppose that a periodic trajectory is launched from a certain point in the triangle in a certain direction. Then every trajectory launched from any other point of the triangle in the same direction is periodic (provided it does not hit a vertex); moreover, these trajectories are either *all invariant* under rotation through 120° about the centre of the triangle, or are *all non-invariant.*

***Assertion.*** *For non-invariant trajectories stability is not transferred through vertices of the triangle.*

*Proof.* Let $\Gamma_1$, $\Gamma_2$ be incontractible trajectories launched in the same direction from points located on different sides of the triangle and close to a vertex (we may assume this vertex to be (0, 0)). Both trajectories end in a triangle $(x, y)$, $3 \mid (x-y)$, $y > 0$; moreover, by non-invariance, $3 \nmid x$, $3 \nmid y$. In this case

the closed contour $\Gamma = \Gamma_2 + (-\Gamma_1)$ contains only *one* lattice node, $(x, y)$, and hence $\mathscr{A}(\Gamma) \neq 0$. Since $\Gamma_1$ and $\Gamma_2$ are straightened billiard trajectories, we have $\tau(\Gamma_1) = \tau(\Gamma_2) = \mathrm{id}$, so $\mathscr{A}(\Gamma) = \mathscr{A}(\Gamma_1) - \mathscr{A}(\Gamma_2) \neq 0$, and thus $\Gamma_1$ and $\Gamma_2$ cannot both be stable. □

***Corollary.*** *For non-invariant trajectories stability is not transferred through any singular point.*

*Proof.* If $x_0$ is a singular point, and $\Gamma_1$ and $\Gamma_2$ are parallel trajectories launched from different sides of $x_0$, then after some reflections in one and the same side of the triangle they hit the boundary of the triangle on different sides of a vertex. Hence they cannot both be stable. □

***Theorem 4.*** *In an equilateral triangle, non-invariant trajectories are unstable.*

*Proof.* We consider all non-invariant trajectories emanating from a fixed side of the triangle in a given direction. If *there are singular points on this side* for this direction, then the previous corollary and remarks made earlier imply that *all* trajectories in this direction are unstable.

Suppose now that *there are no singular points on the side.* As is easily seen, the trajectories are then launched from the side at an angle not exceeding 60° (in fact, less than 60°, since there are invariant trajectories with angle 60°). We now consider the side of the triangle in which the trajectories are reflected for the first time, and the direction in which they go after this reflection. Clearly, the trajectories launched from the new side in the new direction include all the trajectories previously considered. However, already for this direction there are singular points on the side of the triangle (Fig. 43), and we find that the trajectories under consideration are unstable. □

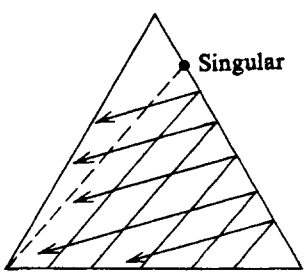



Fig. 43

**D. Periodic trajectories in a triangle with angles 30°, 60°, 90°.**

**D1. Notation.**

A right-angled triangle with acute angles 30°, 60° is 'one-half' of an equilateral triangle, and can naturally be complemented first to an equilateral triangle, and then also to a lattice of equilateral triangles. Therefore we can

immediately start with a partition of $\mathbb{R}^2$ into equilateral triangles, with marking of sides and vertices compatible with the system of coordinates introduced in subsection C.

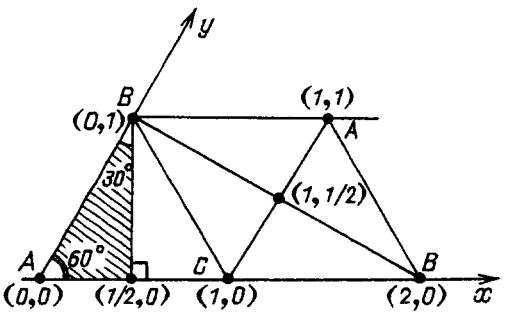



Fig. 44

In each equilateral triangle $ABC$ in the partition we draw the altitude from the vertex $B$. As a result we obtain a partition of the plane into triangles with angles 30°, 60°, 90°, which can also be obtained from a single triangle using repeated reflections in its sides (Fig. 44). Taking into account the coordinate system introduced, the lattice nodes corresponding to the angles 30° and 60° have integer coordinates. The right angles correspond to nodes with coordinates

$$\begin{cases} (i+1/2,\ j), & \text{where } i,\ j \in \mathbb{Z},\ i-j \equiv 0 \pmod 3; \\ (i,\ j+1/2), & \text{where } i,\ j \in \mathbb{Z},\ i-j \equiv 1 \pmod 3; \\ (i+1/2,\ j+1/2, & \text{where } i,\ j \in \mathbb{Z},\ i-j \equiv -1 \pmod 3). \end{cases}$$

Our billiard trajectories will be launched from the side $l = [(0, 0), (1/2, 0)]$ of the 'basis' triangle $ABC$ with vertices (0, 0), (1/2, 0), (0, 1) (in Fig. 44, the leg $AC$). It is easily seen that isolated periodic trajectories can only be launched at angles of 30°, 60° or 90° with $l$. However, we can immediately verify that in these directions there cannot be isolated trajectories.

Thus, after straightening, each periodic trajectory launched from $l$ becomes segment $[(\varepsilon, 0), (x+\varepsilon, y)]$, where $0 < \varepsilon < 1/2$, $x, y \in \mathbb{Z}$, $y > 0$, ending in the triangle with vertices $(x, y)$, $(x+1/2, y)$, $(x, y+1)$, so $3 \mid (x-y)$. A trajectory is incontractible under the condition $\mathrm{GCD}(x, y, (x-y)/3) = 1$. The angle $\alpha$ at which the periodic trajectories of the basis triangle are launched is uniquely determined by this triangle with vertices $(x, y)$, $(x+1/2, y)$, $(x, y+1)$, and is equal to

$$\alpha = \arctan \frac{\sqrt{3}}{2(x/y)+1}.$$

Since all the vertices in the partition have coordinates of the form $(i/2, j/2)$, with $i, j \in \mathbb{Z}$, the singular points of the chosen direction must be among the

points (0, 0), $(1/k, 0)$, $(2/k, 0)$, ..., (1/2, 0), where $k = y/2 \cdot \mathrm{GCD}(x, y)$. These latter points are called *quasi-singular*. Moreover, the points (0, 0), $(2/k, 0)$, $(4/k, 0)$, ... are clearly singular, since lines drawn from them at an angle $\alpha$ with $l$ pass through integer points, that is, through lattice nodes (vertices).

**D2. Pencils of periodic trajectories.**

***Assertion* 1.** *If $3 \mid y$, then all quasi-singular points of the direction $\alpha$ are singular. Moreover, all $y/3$ pencils of periodic trajectories launched from $l$ at an angle $\alpha$ coincide, up to the direction of motion.*

*Proof.* The segment $[(n/k, 0), (x+n/k, y)]$, where $n = 0, 1, \ldots, [k/2]$, $x, y \in \mathbb{Z}$, passes through three points with coordinates of the form $(i/2, j/2)$, $i, j \in \mathbb{Z}$. More precisely, these points are $(u+\xi, v+\eta)$, $(u+x/3+\xi, v+y/3+\eta)$, $(u+2x/3+\xi, v+2y/3+\eta)$, where $u, v \in \mathbb{Z}$, $\xi, \eta$ are 0 or 1/2, and $\xi+\eta > 0$. Since $x/3-y/3 \not\equiv 0 \pmod 3$, there is among these points precisely one lattice node, and hence the point $(n/k, 0)$ is singular.

Further, each incontractible straightened trajectory intersects the sides of the triangle opposite to the angles 30° and 90° $y$ times at an angle $\alpha$. Since at the first intersection (with $l$) $y/3$ pencils of trajectories may be launched, and at the second $2y/3$ pencils, for angles $\alpha$ different from 30°, 60°, 90° we deduce, using earlier assertions, that all these $y/3+2y/3 = y$ pencils coincide, up to the direction of motion. For the singular angles 30°, 60° and 90° the truth of the proposition can be directly verified. □

***Assertion* 2.** *Let $3 \nmid y$. We divide the side $l$ of $ABC$ into $y$ equal parts, and its hypotenuse into $2y$ equal parts. From the $3y$ segments thus obtained, we launch pencils of periodic trajectories at an angle $\alpha$. Then there are precisely three distinct pencils among these $3y$ pencils.*

*Proof.* That there are at least three follows from the fact that an incontractible trajectory emanating at an angle $\alpha$ intersects the sides under consideration $y$ times at an angle $\alpha$.

We now take an equilateral triangle, divide each side of it into $y$ parts, and launch $3y$ pencils of trajectories at an angle $\alpha$ to a side. As we have shown in subsection C, all these pencils are maximal, and there are only three distinct pencils among them. Dividing each pencil into halves, we obtain six distinct pencils. Now we draw an altitude in the equilateral triangle and reflect in it one half of the triangle (together with all trajectories in it) onto the other. Clearly, we obtain the triangle 30°, 60°, 90°, and in it six pencils of trajectories emanating at an angle $\alpha$. It is also clear that these six pencils form three pairs of coincident pencils, and we obtain three distinct pencils. □

**D3. Forms of trajectories and transference of stability.**

As in the case of the triangle 45°, 45°, 90° we can introduce, in a similar manner, the *form* $\mathscr{A}$ of a sequence of triangles in the partition, as well as of the curves generating such sequences, in particular of straightened periodic

billiard trajectories. As in all previous cases, a periodic trajectory is stable if its form is equal to zero. It is easy to obtain the following results.

***Assertion* 3.** *Let $\Gamma$ be a closed contour for which the form $\mathscr{A}$ is defined, and let $q_1$, $q_2$, $q_3$ be the numbers of lattice nodes corresponding to the angles* 30°, 60°, 90°, *lying inside the contour. Then*

$$\mathscr{A}(\Gamma) = -(12q_1 \cdot \alpha_{30} + 6q_2 \cdot \alpha_{60} + 4q_3 \cdot \alpha_{90}).$$

***Assertion* 4.** *In the triangle* 30°, 60°, 90° *stability is not transferred through a vertex.*

*Proof.* Let $\Gamma_1$, $\Gamma_2$ be two incontractible trajectories launched in the same direction from points close to the vertex (0, 0) (the angle 60°) of the basis triangle. Both trajectories end in a triangle with vertices $(i, j)$, $(i+1/2, j)$, $(i, j+1)$, where $3 \mid (i-j)$, $\mathrm{GCD}(i, j, (i-j)/3) = 1$.

If $i \equiv j \not\equiv 0 \pmod 3$, then there is only *one* lattice node inside the contour $\Gamma = \Gamma_2 + (-\Gamma_1)$ corresponding to the angles 30° and 60°, and then $\mathscr{A}(\Gamma) \neq 0$.

If, on the other hand, $i \equiv j \equiv 0 \pmod 3$, then there are *two* vertices inside $\Gamma$ corresponding to the angle 60° and one for the angle 30°, namely $(i/3, j/3)$, $(2i/3, 2j/3)$, $(i, j)$. Since vertices corresponding to right angles have coordinates $(u, v)$ with $2u, 2v \in \mathbb{Z}$, but $u \notin \mathbb{Z}$ or $v \notin \mathbb{Z}$, the only possible vertices with right angles lying inside $\Gamma$ are $(i/6, j/6)$ and $(5i/6, 5j/6)$ (and these points need not be vertices). Consequently, also in this case $\mathscr{A}(\Gamma) \neq 0$.

Since $\mathscr{A}(\Gamma_2) - \mathscr{A}(\Gamma_1) = \mathscr{A}(\Gamma) \neq 0$, the trajectories $\Gamma_1$ and $\Gamma_2$ cannot both be stable. Thus, *stability is not transferred through a vertex with angle* 60°. We can similarly prove that *stability is not transferred through a vertex with angle* 30°.

It remains to show that *stability is not transferred through the vertex with a right angle.* Let $\Gamma_1$, $\Gamma_2$ be two incontractible trajectories launched in the same direction from points close to this vertex but lying on different legs. Again we consider the contour $\Gamma = \Gamma_2 + (-\Gamma_1)$. If it encloses only vertices with a right angle, then $\mathscr{A}(\Gamma) \neq 0$, and stability is not transferred. However, if it encloses a vertex $x_0$ with angle 30° or 60°, then $\Gamma_1$ and $\Gamma_2$ must pass near $x_0$ on different sides of $x_0$. Since stability is not transferred through $x_0$, $\Gamma_1$ and $\Gamma_2$ cannot both be stable.

Thus we have completely proved Assertion 4. □

***Assertion* 5.** *Let $\Gamma_1$, $\Gamma_2$ be two periodic trajectories launched from the side of the triangle* 30°, 60°, 90° *opposite the angle* 30° *or* 90° *at an angle* $\alpha = \arctan\sqrt{3}/(2(x/y)+1)$, *where* $x, y \in \mathbb{Z}$, $3 \mid (x-y)$, $\mathrm{GCD}(x, y) = 1$, $y > 0$. *If there are 'quasi-singular' points between the starting points of these trajectories, if at least one of the points is singular, and if there are at most four 'quasi-singular' points, then $\Gamma_1$ and $\Gamma_2$ cannot both be stable.*

*Proof.* If the conditions of the assertion hold, the contour $\Gamma = \Gamma_2 + (-\Gamma_1)$ encloses at most two vertices corresponding to angles 30° and 60°, hence

$\mathscr{A}(\Gamma) \neq 0$ and $\mathscr{A}(\Gamma_1) \neq \mathscr{A}(\Gamma_2)$. This means that $\Gamma_1$ and $\Gamma_2$ cannot both be stable. □

**D4. Absence of stable trajectories.**

***Theorem 5.*** *In the triangle* 30°, 60°, 90° *all periodic trajectories are unstable.*

*Proof.* 1) We consider all trajectories launched from the side $l$ at an angle $\alpha = \arctan\sqrt{3}/(2(x/y)+1)$, where $x, y \in \mathbb{Z}$, $y > 0$, $3 \mid x$, $3 \mid y$, $\mathrm{GCD}(x, y, (x-y)/3) = 1$. Assertion 1 implies that all these trajectories belong to one pencil. Hence, if there are singular points on $l$ for the given angle $\alpha$, then, since stability is not transferred through singular points, we find that *all trajectories* emanating at an angle $\alpha$ are *unstable*.

2) Now we suppose that $3 \mid (x-y)$, but $x \equiv y \not\equiv 0 \pmod 3$. We divide the leg $l$ and the hypotenuse into equal segments of length $1/(2y)$ (the points of subdivision are quasi-singular for the direction $\alpha$), and launch from each such segment a pencil of trajectories making an angle $\alpha$ with the side. Let us assume that some trajectory is stable. Then Assertion 5 implies that the number of pencils coinciding with this pencil does not exceed $\lceil y/5 \rceil + \lceil 2y/5 \rceil$ (here, $\lceil z \rceil = z$ for $z \in \mathbb{Z}$ and $\lceil z \rceil = [z] + 1$ for $z \notin \mathbb{Z}$, where $[\cdot]$ is the integer part). On the other hand, by Assertion 2 exactly $y$ pencils coincide with it. The inequality $y \leqslant \lceil y/5 \rceil + \lceil 2y/5 \rceil$ implies that $y \leqslant 2$. Therefore, for $y > 2$ the pencil under consideration is in fact unstable.

Thus, it remains to prove the assertion for three directions, for which in case 1 there are no singular points on $l$ and in case 2 we have $y \leqslant 2$, that is, there are at most two singular points. As in subsections A, B, C we can reduce the proof to cases already taken care of, by considering the reflected pencils. Theorem 5 has been proved. □

## §13. A criterion for the stability of periodic trajectories in a regular hexagon

In this section we prove the following theorem.

***Theorem 6.*** *A billiard trajectory $\gamma$ in a regular 6-gon is periodic if and only if one (and then also all) of its links forms an angle*

$$\alpha = \arctan\frac{1}{\sqrt{3}(2(x/y)-1)}$$

*with the side with which it has a point in common; here, $x$ and $y$ are coprime integers and $y > 0$.*

*A periodic trajectory $\gamma$ launched at this angle $\alpha$ from a side of the hexagon is stable if and only if it is invariant under rotations of the hexagon through* 120° *and $y$ is divisible by* 3.

To prove the theorem we introduce some concepts and notation. The proof is given as a chain of individual assertions.

A. Let $P$ be a regular 6-gon. We successively mark its sides, beginning with an arbitrary one, by $l_1$, $l_2$, $l_3$, $l_4$, $l_5$, $l_6$. Using these marks, we denote the angles of $P$ by $\alpha_1$, $\alpha_2$, $\alpha_3$, $\alpha_4$, $\alpha_5$, $\alpha_6$, as shown in Fig. 45. Let $P_0 = P, P_1, P_2, \ldots, P_m$ be a sequence of regular hexagons in which neighbouring terms have a common side and are positioned symmetrically with respect to it. It is obvious that we can naturally mark the sides of $P_j$ $(1 \leqslant j \leqslant m)$ by $l_1$, $l_2$, $l_3$, $l_4$, $l_5$, $l_6$ in such a manner that the symmetry of $P_{j-1}$ and $P_j$ $(1 \leqslant j \leqslant m)$ includes also the correspondence of side marks. We note that a 6-gon that occurs several times in this sequence may have different side marks for different occurrences. Let $l_{i_j}$ be the mark of the common side of $P_{j-1}$ and $P_j$ $(1 \leqslant j \leqslant m)$. Then clearly the sequence $P_0, P_1, P_2, \ldots, P_m$ is uniquely determined by its code-word $W = l_{i_1} l_{i_2} \ldots l_{i_m}$.

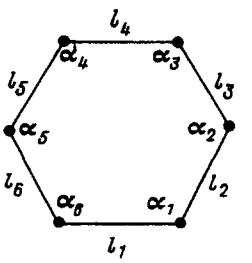



Fig. 45

It is obvious that $P_m$ can be mapped onto $P$ by a shift $M$. To map $P_m$ onto $P$ taking into account the side marks it is necessary to apply additionally a transformation $\tau$ of the 6-gon $P$ $(\tau \in S(P))$. We call this $\tau$ the *type of the sequence* $P_0, P_1, P_2, \ldots, P_m$. We can represent $\tau$ in a natural way as a permutation of the set $\{l_1, \ldots, l_6\}$ or of the set $\{\alpha_1, \ldots, \alpha_6\}$. In this form it is uniquely determined by $W$, hence it is also called the *type of this word.*

Let $P_0, P_1, P_2, \ldots, P_m$ be a sequence of regular 6-gons in which neighbouring terms have a common side with respect to which they are symmetrically positioned and $P_0$ is obtainable from $P$ by a shift. By applying this shift to $P$ we obtain an induced marking of the sides of $P_0$, and then in a natural manner we can associate with the sequence $P_0, P_1, P_2, \ldots, P_m$ a code-word $W$ of length $m$ in the alphabet $\{l_1, \ldots, l_6\}$.

Let $W_1$ and $W_2$ be code-words in the alphabet $\{l_1, \ldots, l_6\}$, $P = P_0, P_1, \ldots, P_m$ a sequence of regular 6-gons with word $W_1$, and $P_m, P_{m+1}, \ldots, P_{m+n}$ a sequence of regular 6-gons with word $W_2$. We denote by $W_1 \square W_2$ the word corresponding to the sequence $P_0, P_1, \ldots, P_m, P_{m+1}, \ldots, P_{m+n}$.

We let $*$ denote the concatenation of words, and for a transformation $\tau \in S(P)$ (considered as a permutation on the set $\{l_1, \ldots, l_6\}$) and a word $W = l_{i_1} l_{i_2} \ldots l_{i_m}$ in the alphabet $\{l_1, \ldots, l_6\}$ we let $\tau W$ denote the word $\tau(l_{i_1})\tau(l_{i_2}) \ldots \tau(l_{i_m})$.

***Assertion* 1.** *The word $W_1 \square W_2$ is uniquely determined by the words $W_1$ and $W_2$, namely $W_1 \square W_2 = W_1 * (\tau_1 W_2)$, where $\tau_1$ is the type of $W_1$.*

*Proof.* Each of the 6-gons $P_m, P_{m+1}, \ldots, P_{m+n}$ has two markings of its sides: one as a term of the sequence $P_0, P_1, \ldots, P_{m+n}$ and one as a term of the sequence $P_m, P_{m+1}, \ldots, P_{m+n}$. To prove the equality $W_1 \square W_2 = W_1 * (\tau_1 W_2)$ it suffices to prove that the second marking becomes the first when each mark $l_i$ $(1 \leqslant i \leqslant 6)$ is replaced by $\tau_1(l_i)$. For $P_m$ this follows from the definitions. In fact, let $M$ be the shift mapping $P_m$ onto $P$. Then for $i = 1, \ldots, 6$ we have: $M(\{\textit{side } l_i \textit{ of } P_m \textit{ with the second marking}\}) = \{\textit{side } l_i \textit{ of } P\}$, $M(\{\textit{side } l_i \textit{ of } P_m \textit{ with the first marking}\}) = \{\textit{side } \tau_1^{-1}(l_i) \textit{ of } P\}$. Hence: $\{\textit{side } \tau_1(l_i) \textit{ of } P_m \textit{ with the first marking}\} = \{\textit{side } l_i \textit{ with the second marking}\}$. This is what we needed to prove. For the 6-gon $P_{m+j}$ $(1 \leqslant j \leqslant n)$ this assertion can be proved by induction with respect to $j$. Thus we have proved the assertion. $\square$

***Assertion* 2.** *Let $W_1$ and $W_2$ be words in the alphabet $\{l_1, \ldots, l_6\}$, $\tau_1$ and $\tau_2$ their types, and $\tau$ an arbitrary element of $S(P)$. Then*

a) $\mathrm{type}(W_1 * W_2) = \tau_2\tau_1$;

b) $\mathrm{type}(\tau W_1) = \tau\tau_1\tau^{-1}$;

c) $\mathrm{type}(W_1 \square W_2) = \tau_1\tau_2$.

*Proof.* a) Let $P = P_0, P_1, \ldots, P_m, \ldots, P_{m+n}$ be a sequence of regular 6-gons corresponding to the word $W_1 * W_2$, with $P_0, P_1, \ldots, P_m$ a subsequence corresponding to the word $W_1$. We take $P_m$ with side marks inherited from the sequence $P_0, P_1, \ldots, P_m, \ldots, P_{m+n}$ as the new initial 6-gon. Then the sequence of regular 6-gons $P_m, P_{m+1}, \ldots, P_{m+n}$ corresponds to the word $W_2$, and the shift $M_2$ mapping $P_{m+n}$ onto $P_m$ maps the side $l_i$ $(1 \leqslant i \leqslant 6)$ of the first to the side $\tau_2^{-1}(l_i)$ of the second. Moreover, since the sequence $P_0, \ldots, P_m$ with initial 6-gon $P_0$ corresponds to $W_1$, the shift $M_1$ mapping $P_m$ onto $P_0$ maps the side $l_i$ of $P_m$ to the side $\tau_1^{-1}(l_i)$ of $P_0$ $(1 \leqslant i \leqslant 6)$. Thus, the shift $M_1 \circ M_2$ maps $P_{m+n}$ onto $P_0$, and such that the side $l_i$ $(1 \leqslant i \leqslant 6)$ of $P_{m+n}$ is mapped to the side $(\tau_1^{-1} \circ \tau_2^{-1})(l_i)$ of $P_0$. This implies that

$$\mathrm{type}(W_1 * W_2) = (\tau_1^{-1} \circ \tau_2^{-1})^{-1} = \tau_2 \circ \tau_1.$$

b) Let $P_0 = P, P_1, \ldots, P_m$ be a sequence of regular 6-gons corresponding to the word $W_1$. We mark the sides of the initial 6-gon $P$ anew: each of the letters $l_1, \ldots, l_6$ is replaced by $\tau(l_1), \ldots, \tau(l_6)$, respectively. Here, clearly, we have also changed the induced marks of the sides of $P_1, P_2, \ldots, P_m$, that is, the sequence $P_0, P_1, \ldots, P_m$ now corresponds to the word $\tau W_1$. Let $M$ be the shift mapping $P_m$ onto $P$. Then for $i = 1, \ldots, 6$ we have: $M(\{\textit{old side } l_i \textit{ of } P_m\}) = \{\textit{old side } \tau_1^{-1}(l_i) \textit{ of } P\}$. However, $\{\textit{old side } l_i \textit{ of } P_m\} = \{\textit{new side } \tau(l_i) \textit{ of } P_m\}$, $\{\textit{old side } \tau_1^{-1}(l_i) \textit{ of } P\} = \{\textit{new side } (\tau \circ \tau_1^{-1})(l_i) \textit{ of } P\}$. So, $M(\{\textit{new side } l_i \textit{ of } P_m\}) = \{\textit{new side } (\tau \circ \tau_1^{-1} \circ \tau^{-1})(l_i) \textit{ of } P\}$. Hence,

$$\mathrm{type}(\tau W_1) = (\tau \circ \tau_1^{-1} \circ \tau^{-1})^{-1} = \tau \circ \tau_1 \circ \tau^{-1}.$$

c) From Assertion 1 and the assertions proved above we obtain

$$\mathrm{type}(W_1 \square W_2) = \mathrm{type}(W_1 * (\tau_1 W_2)) = \mathrm{type}(\tau_1 W_2) \circ \mathrm{type}(W_1) = \\ = (\tau_1 \circ \tau_2 \circ \tau_1^{-1}) \circ \tau_1 = \tau_1 \circ \tau_2. \ \square$$

***Corollary* 1.** *Let $\mathfrak{g}$ be the subgroup of $S(P)$ consisting of six transformations: the identity, two rotations through 120°, and three symmetries with respect to the lines passing through opposite vertices of $P$ (in terms of permutations on $\{l_1, ..., l_6\}$ we have:*

$$\mathfrak{g} = \{\mathrm{id}, (l_1 l_3 l_5)(l_2 l_4 l_6), (l_5 l_3 l_1)(l_6 l_4 l_2), \\ (l_1 l_4)(l_2 l_3)(l_5 l_6), (l_1 l_6)(l_2 l_5)(l_3 l_4), (l_1 l_2)(l_3 l_6)(l_4 l_5)\}).$$

*Then the types of the words in the alphabet $\{l_1, ..., l_6\}$ take values in $\mathfrak{g}$ only.*

*Proof.* Assertion 2a) and the finiteness of the group $S(P)$ imply that the set of all possible types of words in the alphabet $\{l_1, ..., l_6\}$ is a subgroup of $S(P)$, generated by the types of the one-letter words. However,

$$\begin{aligned} \mathrm{type}(l_1) &= \mathrm{type}(l_4) = (l_1 l_4)(l_2 l_3)(l_5 l_6), \\ \mathrm{type}(l_2) &= \mathrm{type}(l_5) = (l_1 l_6)(l_2 l_5)(l_3 l_4), \\ \mathrm{type}(l_3) &= \mathrm{type}(l_6) = (l_1 l_2)(l_3 l_6)(l_4 l_5), \end{aligned}$$

and these three transformations generate $\mathfrak{g}$. $\square$

***Corollary* 2.** *Let $W_1$, $W_2$, $W_3$ be arbitrary words in the alphabet $\{l_1, ..., l_6\}$. Then $(W_1 \square W_2) \square W_3 = W_1 \square (W_2 \square W_3)$.*

*Proof.* Let $\tau_1$, $\tau_2$ be the types of $W_1$ and $W_2$. Then from Assertions 1 and 2 we obtain:

$$(W_1 \square W_2) \square W_3 = (W_1 \square W_2) * (\mathrm{type}(W_1 \square W_2) W_3) = \\ = W_1 * (\tau_1 W_2) * ((\tau_1 \tau_2) W_3) = W_1 * \tau_1 (W_2 * (\tau_2 W_3)) = \\ = W_1 \square (W_2 * (\tau_2 W_3)) = W_1 \square (W_2 \square W_3). \ \square$$

*Remarks.* 1) If the elements of $S(P)$ are regarded as permutations of the set $\{\alpha_1, ..., \alpha_6\}$ of angles of $P$, then $\mathfrak{g}$ is the collection of elements of $S(P)$ that map the 'odd' angles $\alpha_1$, $\alpha_3$, $\alpha_5$ to the 'odd' angles, and the 'even' angles $\alpha_2$, $\alpha_4$, $\alpha_6$ to the 'even' ones.

2) The type of a word of *odd* length is a *symmetry*, and the type of a word of *even* length is a *rotation* through 120° or the identity.

**B.** Let $H$ be an arbitrary convex 6-gon. We mark its sides, starting with some side, counterclockwise by $l_1$, $l_2$, $l_3$, $l_4$, $l_5$, $l_6$; subsequently we mark its angles by $\alpha_1$, $\alpha_2$, $\alpha_3$, $\alpha_4$, $\alpha_5$, $\alpha_6$, in accordance with Fig. 45. From the word $W = l_{i_1} l_{i_2} ... l_{i_m}$ in the alphabet $\{l_1, ..., l_6\}$ we construct the sequence of 6-gons $H_0 = H, H_1, ..., H_m$, with sides marked by $l_1, l_2, ..., l_6$, in which $H_{j-1}$ and $H_j$ $(1 \leqslant j \leqslant m)$ have a common side $l_{i_j}$ with respect to which they are symmetrically positioned (taking into acocunt the side marks). On the side $l_1$ of $H$ and $\tau(l_1)$ of $H_m$ (here $\tau = \mathrm{type}(W)$) we choose the direction corresponding

to counterclockwise traversal of the boundary of the corresponding 6-gon. By $\mathscr{A}(H)$ we denote the inclination of the oriented side $\tau(l_1)$ of $H_m$ to the oriented side $l_1$ of $H$, measured in the counterclockwise direction.

***Assertion* 3.** *The inclination $\mathscr{A}(H)$ is a linear combination $k_1\alpha_1+k_2\alpha_2+k_3\alpha_3+$ $+k_4\alpha_4+k_5\alpha_5+k_6\alpha_6$ of the numbers $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$ (the quantities corresponding to the angles of $H$) with integer coefficients $k_1, k_2, k_3, k_4, k_5, k_6$. These coefficients are uniquely determined by the word $W$, up to addition of one and the same number.*

*Proof.* We orient the sides of $H_0, H_1, \ldots, H_m$ in such a way that the direction of the side $l_i$ $(1 \leqslant i \leqslant 6)$ of $H_j$ $(0 \leqslant j \leqslant m)$ corresponds to traversal of the boundary of $H_j$ in the *counterclockwise* direction if $i$ and $j$ have different parity, and in the *clockwise* direction if $i$ and $j$ have the same parity. Here, clearly, the orientations of the common sides of $H_{j-1}$ and $H_j$ $(1 \leqslant j \leqslant m)$ are compatible; moreover, for the sides $l_1$ of $H$ and $\tau(l_1)$ of $H_m$ this orientation coincides with the one introduced before. It is further obvious that with this orientation the inclination of one side of a 6-gon in the sequence to another side of it is $\alpha_i$ or $-\alpha_i$, where $\alpha_i$ is the angle between these sides. This implies that the inclination of the side $l_i$ of $H_j$ to the side $L_{i'}$ of $H_{j'}$ $(1 \leqslant i, i' \leqslant 6;$ $0 \leqslant j, j' \leqslant m)$ has the form $k_1\alpha_1+ \ldots +k_6\alpha_6$, where $k_1, \ldots, k_6$ are integer coefficients depending on $i, i', j, j'$ but independent of $H$. In particular, the inclination $\mathscr{A}(H)$ has this form. So, let $\mathscr{A}(H) = k_1\alpha_1+ \ldots +k_6\alpha_6$, with $k_1, \ldots, k_6$ integers. For an arbitrary integer $n$ we have

$$(k_1+n)\alpha_1+\ldots+(k_6+n)\alpha_6=k_1\alpha_1+\ldots+k_6\alpha_6+n(\alpha_1+\ldots+\alpha_6).$$

Since $n(\alpha_1+ \ldots +\alpha_6) = 4\pi n$ and $\mathscr{A}(H)$ is determined up to $2\pi s$, $s \in \mathbb{Z}$, the collection of coefficients $k_1+n, \ldots, k_6+n$ also gives $\mathscr{A}(H)$.

Conversely, let $\mathscr{A}(H) = k_1'\alpha_1+ \ldots +k_6'\alpha_6$, with $k_1', \ldots, k_6'$ integers. We put $\tilde{k} = k_6'-k_6$. Then, clearly,

$$(k_1-k_1'+\tilde{k})\alpha_1+\ldots+(k_5-k_5'+\tilde{k})\alpha_6\equiv 0 \pmod{2\pi}$$

independently of the choice of $H$. Since we can choose a 6-gon $H$ in which the angles $\alpha_1, \ldots, \alpha_6$ and the number $\pi$ are rationally independent, we thus have

$$k_1-k_1'+\tilde{k}=\ldots=k_5-k_5'+\tilde{k}=0\,(=k_6-k_6'+\tilde{k}),$$

that is, the collection $k_1', \ldots, k_6'$ has the form $k_1+\tilde{k}, \ldots, k_6+\tilde{k}$.

We have thereby completely proved Assertion 3. □

C. Thus, we obtain a well-defined element of the group $\mathbb{Z}^6/(1, 1, 1, 1, 1, 1)\mathbb{Z}$, namely $(k_1, k_2, \ldots, k_6)+(1, 1, 1, 1, 1, 1)\mathbb{Z}$, corresponding to the word $W$.

*Definition.* We call this element the *form* of $W$, and denote it by $\mathscr{A}(W)$. We will also write it as $k_1\alpha_1+ \ldots +k_6\alpha_6$.

*Definition.* Let $\mathscr{A}(W) = k_1\alpha_1 + \ldots + k_6\alpha_6$. The element $(k_1+k_3+k_5, k_2+k_4+k_6)+(1, 1)\mathbb{Z}$ of the group $\mathbb{Z}^2/(1, 1)\mathbb{Z}$ is clearly uniquely determined by $\mathscr{A}(W)$. We denote it by $\mathscr{A}'(W)$, and call it the *stroke-form* of $W$.

***Assertion 4.*** *Let $W_1$, $W_2$ be two arbitrary words in the alphabet $\{l_1, l_2, \ldots, l_6\}$, and let $\tau_1$ be the type of $W_1$. Then*

$$\mathscr{A}(W_1 \square W_2) = \mathscr{A}(W_1) + \tau_1\mathscr{A}(W_2), \quad \mathscr{A}'(W_1 \square W_2) = \mathscr{A}'(W_1) + \mathscr{A}'(W_2)$$

*(where for an arbitrary $k_1\alpha_1 + \ldots + k_6\alpha_6 \in \mathbb{Z}^6/(1, \ldots, 1)\mathbb{Z}$ we put*

$$\tau_1(k_1\alpha_1 + \ldots + k_6\alpha_6) = k_1\tau_1(\alpha_1) + \ldots + k_6\tau_1(\alpha_6)).$$

*Proof.* First we prove that $\mathscr{A}(W_1 * W_2) = \mathscr{A}(W_1) + \tau_1\mathscr{A}(\tau_1^{-1}W_2)$.

Let $H$ be an arbitrary 6-gon with sides and angles marked in accordance with Fig. 45. Starting from it we construct the sequence of 6-gons $H_0 = H, H_1, \ldots, H_m, \ldots, H_{m+n}$ corresponding to the word $W_1 * W_2$, and let $H_0, H_1, \ldots, H_m$ be the subsequence corresponding to $W_1$. The sides and angles of the 6-gons $H_1, H_2, \ldots, H_{m+n}$ have the marks induced by those of the sides and angles of $H$. We take $H_m$ with marks of sides $l_1' = \tau_1(l_1), \ldots, l_6' = \tau_1(l_6)$ and marks of angles $\alpha_1' = \tau_1(\alpha_1), \ldots, \alpha_6' = \tau_1(\alpha_6)$ as the new initial 6-gon. We note that $H_m$ with these side and angle marks is regularly oriented, that is, traversal of its boundary along $l_1', \ldots, l_6'$ is in the *counterclockwise* direction.

Let $W$ be the word corresponding to $H_m, H_{m+1}, \ldots, H_{m+n}$ with the new initial 6-gon, $\tau$ its type, and $\mathscr{A}(W) = k_1\alpha_1 + \ldots + k_6\alpha_6$ its form. Then the inclination of the side $\tau(l_1')$ of $H_{m+n}$ to the side $l_1'$ of $H_m$ is equal to

$$k_1\alpha_1' + \ldots + k_6\alpha_6' = k_1\tau_1(\alpha_1) + \ldots + k_6\tau_1(\alpha_6) = \tau_1\mathscr{A}(W)$$

(the sides are oriented in the direction of counterclockwise traversal of the boundary of the corresponding 6-gon). Moreover, the inclination of the side $l_1' = \tau_1(l_1)$ of $H_m$ to the side $l_1$ of $H$ is equal to $\mathscr{A}(W_1)$. This implies that the inclination of the side $\tau(l_1')$ of $H_{m+n}$ to the side $l_1$ of $H$ is equal to $\mathscr{A}(W_1) + \tau_1\mathscr{A}(W)$.

We note that under transition from the old side marks to the new ones each side $l_i$ $(1 \leqslant i \leqslant 6)$ of the 6-gons $H_m, H_{m+1}, \ldots, H_{m+n}$ has the mark $(\tau_1^{-1}(l_i))'$; then $W = \tau_1^{-1}W_2$, and by Assertion 2b, $\tau = \tau_1^{-1}\tau_2\tau_1$. Further, let $\tau(l_1') = l_i'$; then $\tau(l_1) = l_i$, and we obtain

$$\tau(l_1') = l_i' = \tau_1(l_i) = \tau_1(\tau(l_1)) = [\tau_1(\tau_1^{-1}\tau_2\tau_1)](l_1) = (\tau_2\tau_1)(l_1).$$

Since $\tau_2\tau_1$ is the type of the word $W_1 * W_2$, the inclination of the side $\tau(l_1') = (\tau_2\tau_1)(l_1)$ of $H_{m+n}$ to the side of $l_1$ of $H$ is equal to $\mathscr{A}(W_1 * W_2)$. Hence

$$\mathscr{A}(W_1 * W_2) = \mathscr{A}(W_1) + \tau_1\mathscr{A}(W_2) = \mathscr{A}(W_1) + \tau_1\mathscr{A}(\tau_1^{-1}W_2).$$

From Assertion 1 and the formula already obtained we find that

$$\mathscr{A}(W_1 \square W_2) = \mathscr{A}(W_1 * \tau_1 W_2) = \mathscr{A}(W_1) + \tau_1\mathscr{A}(\tau_1^{-1}(\tau_1 W_2)) = \\ = \mathscr{A}(W_1) + \tau_1\mathscr{A}(W_2).$$

Let $\mathscr{A}(W_1) = k_1\alpha_1 + \ldots + k_6\alpha_6$, $\mathscr{A}(W_2) = k_1'\alpha_1 + \ldots + k_6'\alpha_6$, then

$$\mathscr{A}(W_1 \square W_2) = k_1\alpha_1 + \ldots + k_6\alpha_6 + k_1'\tau_1(\alpha_1) + \ldots + k_6'\tau_1(\alpha_6).$$

Earlier we have noted that the transformation $\tau_1$, like all elements of $\mathfrak{g}$, maps even angles to even angles, and odd angles to odd angles; therefore

$$\mathscr{A}'(W_1 \square W_2) = ((k_1+k_3+k_5)+(k_1'+k_3'+k_5'),\ (k_2+k_4+k_6)+ \\ +(k_2'+k_4'+k_6'))+(1,1)Z = \mathscr{A}'(W_1)+\mathscr{A}'(W_2).$$

Assertion 4 has been proved. □

**D.** We again return to the regular 6-gon $P$ with sides and angles marked in accordance with Fig. 45. We reflect this 6-gon in all its sides, and then repeat this operation with all reflected 6-gons, and so on to infinity. As a result we obtain a partition of the plane into regular 6-gons. We introduce coordinates in the plane as follows: as coordinate origin we take the vertex of $P$ at the angle $\alpha_6$, the $x$-axis has direction along the side $l_1$ of $P$, and the $y$-axis makes an angle of 150° with the $x$-axis in the counterclockwise direction (Fig. 46). As unit segment on the $x$-axis we take three times the side of $P$, and as unit the $y$-axis the side of $P$ multiplied by $\sqrt{3}$. In this coordinate system each 6-gon in the partition can be obtained from $P$ by a shift by a vector $(i, j)$ with $i, j \in \mathbb{Z}$. We call the pair of numbers $(i, j)$ the *coordinates of the 6-gon.*

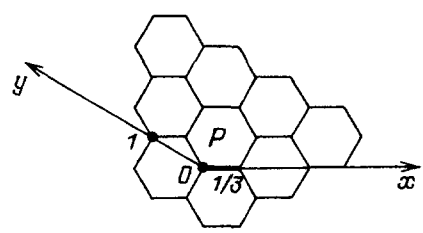



Fig. 46

Each vertex of a 6-gon in the partition has coordinates of the form $(i, j)$ or $(i+1/3, j)$, where $i, j \in \mathbb{Z}$; conversely, all such points are coordinates of vertices of 6-gons in the partition. Let $P_0 = P, P_1, \ldots, P_m$ be a sequence of 6-gons in the partition such that two neighbouring 6-gons have exactly one side in common. Then we mark the sides and angles of these 6-gons as for $P$, in the way described above. Here, a 6-gon in the partition that occurs in different sequences, or even twice in the same sequence, may have different side and angle marks. Nevertheless, *the mark of an angle at a fixed vertex of the 6-gon has constant parity*, namely, the angle at $(i, j)$ is always 'even', and the angle at $(i+1/3, j)$ is always 'odd' (here $i, j \in \mathbb{Z}$). The proof of this is trivial.

We consider all possible periodic billiard trajectories in $P$. Without loss of generality we may assume that all these trajectories are launched from the side $l_1$. With each such trajectory we have associated a word $W$ in the alphabet $\{l_1, ..., l_6\}$ (the sequence of sides of $P$ in which a point moving along the trajectory is reflected). Let $P_0 = P, P_1, ..., P_m$ be a sequence of 6-gons in the partition corresponding to $W$. Mapping the $j$-th link of the trajectory into $P_{j-1}$ (that is, applying to this link the transformation mapping $P$ onto $P_{j-1}$, taking account of the side marks induced by the sequence $P_0 = P, P_1, ..., P_m$), we obtain a line segment, the 'straightening' of the trajectory.

Since all sides of the 6-gons in the partition have angles of inclination to the side $l_1$ of $P$ equal to 0° or 60°, the following *singular directions* of billiard trajectories are distinguished: those at an angle of 30° or 60° to $l_1$. A trajectory launched at such an angle can hit some side at the same angle so that under development of the trajectory this side will not be parallel to the side $l_1$ of $P$ (in particular, only in such directions can the *odd-link* trajectories go). Moreover, *perpendicular* trajectories (that is, trajectories having links perpendicular to a side of a 6-gon) can only be launched at angles of 30° or 90° to $l_1$. By considering these directions we find that $P$ always contains *two* odd-link periodic trajectories, and both have three links; one passes through the midpoints of the sides $l_1, l_3, l_5$, the other through the midpoints of the sides $l_2, l_4, l_6$. Moreover, $P$ contains three pencils of perpendicular trajectories; these have two links and reflect in opposite sides of the 6-gon.

Let $L$ be an *even-link* periodic billiard trajectory launched from a point $A(\varepsilon, 0)$ $(0 < \varepsilon < 1/3)$ on the side $l_1$ of $P$. After straightening, it becomes a segment not passing through vertices of 6-gons in the partition of the plane, and joining $A$ to the point $B = (i+\varepsilon, j)$ of the 6-gon $(i, j)$, where $i, j \in \mathbb{Z}$, $j > 0$. The type of the word of the trajectory (that is, the type of the word corresponding to the sequence of 6-gons in the partition through which the segment $AB$ passes) is equal to id. Conversely, suppose that the segment $AB$, where $A = (\varepsilon, 0)$, $B = (i+\varepsilon, j)$, $i, j \in \mathbb{Z}$, $j > 0$, $0 < \varepsilon < 1/3$, does not pass through vertices of 6-gons in the partition. Let $P, P_1, ..., P_m$ be the sequence of 6-gons in the partition through which $AB$ passes, and let $W$ be its corresponding word. Then $AB$ is the development of a billiard trajectory if and only if $W$ has type id. If this is true, the trajectory has an even number of links and word $W$.

Let us assume that the type of $W$ is not equal to id. We now consider the segments $AC$ and $AD$, where $C = (2i+\varepsilon, 2j)$, $D = (3i+\varepsilon, 3j)$. The words $W \square W$ and $W \square W \square W$ correspond to the sequences of 6-gons through which $AC$ and $AD$ pass. Let $\tau$ be the type of $W$. Then type$(W \square W) = \tau^2$, type$(W \square W \square W) = \tau^3$. Each element of the group $\mathfrak{g}$ of types has order 1, 2, or 3, and thus we find that if $AB$ is not the development of a periodic billiard trajectory, then $AC$ or $AD$ is.

Suppose that the billiard trajectory $L$ is launched from the point $(\varepsilon, 0)$ $(0 < \varepsilon < 1/3)$ in the direction of the 6-gon $(x, y)$ (with $x$ and $y$ coprime integers, $y > 0$), that is, along the vector $(x, y)$ in the chosen coordinate system. Then this trajectory either hits a vertex of $P$ or is periodic. In the second case we call it *k-fold* ($k = 1$, 2, or 3) if $k$ is the smallest natural number for which the segment $[(\varepsilon, 0), (kx+\varepsilon, ky)]$ is the development of this trajectory. It is obvious that *the multiplicity of a trajectory is the order of the type* $\tau$ *of the word* $W$ *corresponding to the segment* $[(\varepsilon, 0), (x+\varepsilon, y)]$. We call $W$ the *short word* corresponding to the trajectory. Then $W\square \ldots \square W$ ($k$ times) is the usual word corresponding to $L$.

***Assertion* 5.** *The billiard trajectory $L$ is invariant under the transformation $\tau$ of the 6-gon $P$. Conversely, if a trajectory $L$ is invariant under a transformation $\sigma \in \mathfrak{g}$, $\sigma \neq \mathrm{id}$, then $\sigma = \tau$ or $\sigma = \tau^{-1}$. The only exception is the trajectory launched at an angle of 30° from the side $l_1$ of $P$: it is 3-fold but invariant under the action of the whole group $\mathfrak{g}$.*

*Proof.* We mark the points $B_0(\varepsilon, 0)$, $B_1(x+\varepsilon, y)$, ..., $B_k(kx+\varepsilon, ky)$, where $k$ is the multiplicity of $L$. Then we can easily see that under the transformation $\tau^{j-1}$ the part of $L$ whose development is $B_{j-1}B_j$ $(1 \leqslant j \leqslant k)$ is obtained from the part whose development is $B_0B_1$. We denote these parts of $L$ by $L_1, \ldots, L_k$. Then $\tau(L_1) = L_2, \ldots, \tau^{k-1}(L_1) = L_k$, and finally $\tau^k(L_1) = \mathrm{id}(L_1) = L_1$. This implies that $L_j = \tau(L_{j-1})$, $2 \leqslant j \leqslant k$, and $\tau(L_k) = L_1$, so

$$\tau(L)=\tau(L_1\cup\ldots\cup L_k)=\tau(L_1)\cup\ldots\cup\tau(L_k)=L_2\cup\ldots\cup L_k\cup L_1=L,$$

that is, $L$ is *invariant* under $\tau$.

Conversely, suppose that $L$ is invariant under a transformation $\sigma \in \mathfrak{g}$, $\sigma \neq \mathrm{id}$. Let $AB$ be the segment that is the straightening of the part of $L$ from its beginning to the link which is followed by the link obtained from the first one by application of $\sigma$ ($A = (\varepsilon, 0)$, $0 < \varepsilon < 1/3$). If the direction of $L$ is *non-singular*, then $B = (x+\varepsilon, y)$, where $x$, $y$ are integers, $y > 0$. Moreover, by construction the word corresponding to $AB$ has type $\sigma^{-1}$. Therefore, *if $x$, $y$ are coprime*, then $\sigma^{-1} = \tau$, and since the trajectory is invariant under $\tau$, it is also invariant under $\sigma = \tau^{-1}$. On the other hand, *if $x$, $y$ are not coprime*, then we can easily see that $\mathrm{GCD}(x, y) = 2$, $\tau$ has order 3, and $\tau^2 = \sigma^{-1}$. Thus, $\sigma = (\tau^2)^{-1} = \tau$, and we may refer to the case proved above. By separately considering the trajectories launched in singular directions we obtain the unique exceptional trajectory mentioned in the conditions of the assertion. $\square$

***Assertion* 6.** *Let $L$ be an even-link periodic trajectory, and $\widehat{W}$ its word. Then $L$ is stable if and only if the form of $\widehat{W}$ is equal to zero.*

*Proof.* Let $H$ be an abitrary convex 6-gon, with sides and angles marked in accordance with Fig. 45. Starting from it we construct the sequence of 6-gons

$H_0 = H, H_1, \ldots, H_m$ corresponding to the word $\widehat{W}$. Since type($\widehat{W}$) = id, the angle $\varphi$ associated with $H$ by the form $\mathscr{A}(\widehat{W})$ is the angle of inclination of $H_m$ to $H$, measured in the counterclockwise direction. Then counterclockwise rotation through $\varphi$ gives the linear part of an affine orthogonal transformation $t = t(\widehat{W})$ mapping $H$ onto $H_m$, with corresponding sides and angles. As is well known, an even-link trajectory $L$ is stable if and only if the linear part of $t$ is equal to id (that is, $\varphi \equiv 0 \pmod{2\pi}$) for any 6-gon $H$. Since the form corresponding to $\widehat{W}$ is unique, the latter condition is equivalent to $\mathscr{A}(\widehat{W}) = 0$. □

***Corollary* 1.** *A contractible trajectory is stable if and only if the incontractible trajectory corresponding to it is stable.*

*Proof.* Let $\widetilde{L}$ be the billiard trajectory that is actually the periodic trajectory $L$ traversed $n$ times. The word $\widetilde{W} = W * \ldots * W$ ($n$ times) corresponds to it, where $W$ is the word corresponding to $L$. Since type($\widetilde{W}$) = id, $\mathscr{A}(\widetilde{W}) = \mathscr{A}(W * \ldots * W) = \mathscr{A}(W \square \ldots \square W) = n \cdot \mathscr{A}(W)$. Thus $\mathscr{A}(\widetilde{W}) = 0 \Leftrightarrow \mathscr{A}(W) = 0$, that is, the trajectories $L$ and $\widetilde{L}$ are simultaneously stable or unstable. □

***Corollary* 2.** *For a trajectory $L$ to be stable it is necessary that the stroke-form of its short word $W$ is equal to zero. For 3-fold trajectories this condition is also sufficient.*

*Proof.* Let $L$ be a stable trajcetory. Then $\mathscr{A}(\widetilde{W}) = 0$, where $\widetilde{W} = W \square \ldots \square W$ ($k$ times, where $k$ is the multiplicity of $L$) is the word corresponding to $L$. Hence $\mathscr{A}'(\widetilde{W}) = 0$. But $\mathscr{A}'(\widetilde{W}) = k\mathscr{A}'(W)$, so $\mathscr{A}'(W) = 0$.

Conversly, let $\mathscr{A}'(W) = 0$ and $L$ be 3-fold ($k = 3$). Then the type $\tau$ of $W$ has order 3, that is, it is a rotation through 120° about the centre of $P$. This implies that $\alpha_1, \tau(\alpha_1), \tau^2(\alpha_1)$, as well as $\alpha_3, \tau(\alpha_3), \tau^2(\alpha_3)$ and $\alpha_5, \tau(\alpha_5), \tau^2(\alpha_5)$, are the angles $\alpha_1, \alpha_3, \alpha_5$, written in some order. Also, $\alpha_2, \tau(\alpha_2), \tau^2(\alpha_2)$; $\alpha_4, \tau(\alpha_4), \tau^2(\alpha_4)$; $\alpha_6, \tau(\alpha_6), \tau^2(\alpha_6)$ are the angles $\alpha_2, \alpha_4, \alpha_6$, written in some order. Let $\mathscr{A}(W) = k_1\alpha_1 + \ldots + k_6\alpha_6$. Then by Assertion 4,

$$\begin{aligned}\mathscr{A}(W\square W\square W) &= \mathscr{A}(W) + \tau\mathscr{A}(W) + \tau^2\mathscr{A}(W) = \\ &= k_1(\alpha_1 + \tau(\alpha_1) + \tau^2(\alpha_1)) + \ldots + k_6(\alpha_6 + \tau(\alpha_6) + \tau^2(\alpha_6)) = (k_1 + k_3 + k_5)\alpha_1 + \\ &+ (k_2 + k_4 + k_6)\alpha_2 + (k_1 + k_3 + k_5)\alpha_3 + (k_2 + k_4 + k_6)\alpha_4 + (k_1 + k_3 + k_5)\alpha_5 + \\ &+ (k_2 + k_4 + k_6)\alpha_6,\end{aligned}$$

hence

$$\mathscr{A}'(W\square W\square W) = (3(k_1 + k_3 + k_5),\ 3(k_2 + k_4 + k_6)) + (1,\ 1)\mathbb{Z}.$$

Since $\mathscr{A}'(W \square W \square W) = 0$, we have $k_1 + k_3 + k_5 = k_2 + k_4 + k_6$, from which we obtain

$$\mathscr{A}(W\square W\square W) = 0,$$

and hence $L$ is stable. □

E. We fix a pair of coprime integers $(x, y)$, $y > 0$, and consider all billiard trajectories launched from the side $l_1$ of $P$ in the direction $(x, y)$ (with respect to the chosen coordinate system). Each such trajectory is either periodic or hits a vertex of $P$. In the latter case we call the point of $l_1$ from which the trajectory is launched *singular* for the direction $(x, y)$. To find all singular points for the direction $(x, y)$ we have to project all vertices of the 6-gons in the partition of the plane lying in the half-plane $\{(i, j) : j > 0\}$ onto the line $\{(i, j) : j = 0\}$ along this direction. Since $x$ and $y$ are coprime, the projections of the vertices of the form $(i, j)$, $i, j \in \mathbb{Z}$, are the points $(n/y, 0)$, $n \in \mathbb{Z}$, and the projections of the vertices of the form $(i+1/3, j)$, $i, j \in \mathbb{Z}$, are the points $(1/3+n/y, 0)$, $n \in \mathbb{Z}$.

If $y$ is divisible by 3 the two sets of projections coincide. The points $(n/y, 0)$, $0 \leqslant n \leqslant y/3$, of these sets belong to the segment $[(0, 0), (1/3, 0)]$. They divide this segment (the *side* $l_1$ *of* $P$) into $y/3$ equal parts. The trajectories launched from one such part form a pencil: they all have the same word, the same multiplicity, the same length, corresponding links of them are parallel, and they are all simultaneously stable or unstable.

If $y$ is not divisible by 3, then the sets of projections of the 'even' and the 'odd' vertices of the 6-gons in the partition do not intersect. The segment $[(0, 0), (1/3, 0)]$ contains the points $(n/y, 0)$, $0 \leqslant n \leqslant [y/3]$, of the first set, and the points $(1/3-n/y, 0)$, $0 \leqslant n \leqslant [y/3]$, of the second set. These points divide the segment $[(0, 0), (1/3, 0)]$ into $2[y/3]+1$ parts, where adjacent parts have different lengths but all parts with even index and all parts with odd index have the same length.

***Assertion* 7.** *Let $(x, y)$ be a pair of positive coprime integers. If $y$ is divisible by 3, then all trajectories launched from the side $l_1$ of $P$ in the direction $(x, y)$ and not hitting a vertex of it are 1- or 3-fold. If $y \equiv 2 \pmod 3$, then the trajectories launched from the 1-st, 3-rd, ..., $(2[y/3]+1)$-th segments of $l_1$ (into which $l_1$ is divided by the singular points of the direction $(x, y)$) are 2-fold, while the trajectories launched from the remaining segments are 1- or 3-fold. If $y \equiv 1 \pmod 3$, then, on the contrary, the trajectories launched from the 1-st, 3-rd, ..., $(2[y/3]+1)$-th segments are 1- or 3-fold, and all the remaining trajectories are 2-fold.*

*Proof.* Let $(\varepsilon, 0)$ $(0 < \varepsilon < 1/3)$ be a point from which a trajectory not hitting a vertex of $P$ is launched in the direction $(x, y)$. We consider the segment $AB$, $A = (\varepsilon, 0)$, $B = (x+\varepsilon, y)$. The short word corresponding to the trajectory is the word corresponding to the sequence of 6-gons of the partition through which $AB$ passes. Since the multiplicity of a trajectory is the type of its short word, a trajectory is odd- or even-fold depending on whether an even or odd number of sides of the 6-gons in the partition intersect $AB$ (not counting the side $l_1$ of $P$). We first assume that the point $(\varepsilon, 0)$ is very close to the point $(0, 0)$, so there is no singular point for the direction $(x, y)$ between them. Let $\Gamma$ be the curve that has beginning and end in common

with $AB$ and successively passes through the 6-gons in the partition that form a sequence with word $\underbrace{l_5 \square l_5 \square \ldots \square l_5}_{y \text{ times}} \square \underbrace{(l_3 \square l_2) \square \ldots \square (l_3 \square l_2)}_{x \text{ times}}$ (Fig. 47). The fact that precisely three sides of 6-gons pass through each vertex of the partition implies the following result.

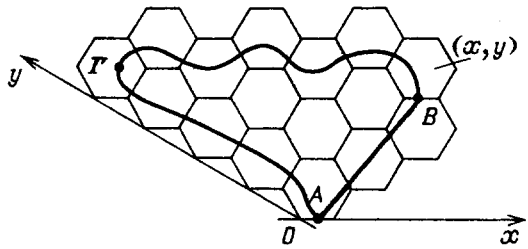



Fig. 47

***Lemma.*** *Let $\Gamma$ be a closed Jordan contour not passing through vertices of 6-gons in the partition and intersecting their sides only finitely many times. Then the number of sides intersected by $\Gamma$ has the same parity as the number of vertices of the partition inside $\Gamma$.*

Let $n$ be the number of sides of 6-gons intersected by $AB$. The contour $\Gamma$ intersects $y+2x$ sides, hence the closed contour $\gamma = (-\Gamma)+AB$ intersects $y+2x+n$ sides. We note that if $\gamma$ encloses a vertex $(i+1/3, j)$, $i, j \in \mathbb{Z}$, then it also encloses the vertex $(i, j)$. Hence the lemma implies that $y+2x+n \equiv l \pmod 2$, that is, $y+n \equiv l \pmod 2$, where $l$ is the number of vertices $(i, j)$, $i, j \in \mathbb{Z}$, inside $\gamma$ for which the corresponding vertex $(i+1/3, j)$ is outside $\gamma$. Clearly, $l$ is the number of 'odd' vertices $(i+1/3, j)$, $i, j \in \mathbb{Z}$, $0 < j \leqslant y$, that are projected onto the $x$-axis along the direction $(x, y)$ into points of the half-interval $((0,0), (1/3,0)]$. Through a point $(\tilde{\varepsilon}, 0)$ $(0 < \tilde{\varepsilon} \leqslant 1/3)$ we draw the line parallel to the vector $(x, y)$. Since $x$ and $y$ are coprime, all odd vertices of 6-gons in the partition lying on this line (if there are such) can be obtained from each other by shifts by vectors $m \cdot (x, y)$, $m \in \mathbb{Z}$. This implies that this line either contains no odd vertices or that only one such vertex can satisfy the condition $0 < j \leqslant y$. Our previous reasoning concerning singular points for the direction $(x, y)$ implies that $l = y/3$ if $y$ is divisible by 3, and $l = [y/3]+1$ otherwise. In the first case we have $y+n \equiv y/3 \pmod 2$, so $n$ is an *even* number, and thus the trajectory launched from $(\varepsilon, 0)$ in the direction $(x, y)$ is *odd-fold*. If, on the other hand, $y = 3\tilde{y}+\nu$, where $\tilde{y} \in \mathbb{Z}$, $\nu = 1$ or $\nu = 2$, then $l = \tilde{y}+1$, and $y+n \equiv \tilde{y}+1 \pmod 2$. Since $y-\tilde{y} = (3\tilde{y}+\nu)-\tilde{y} = 2\tilde{y}+\nu \equiv \nu \pmod 2$, we have $n \equiv \nu+1 \pmod 2$. Thus, the trajectory launched from $(\varepsilon, 0)$ in the direction $(x, y)$ is 1- or 3-fold if $\nu = 1$, and 2-fold if $\nu = 2$.

Now let $(\varepsilon_0, 0)$ be an interior point of the side $l_1$ of $P$ which is singular for the direction $(x, y)$. Let $A(\varepsilon_1, 0)$ and $B(\varepsilon_2, 0)$ be points on $l_1$ between which there are no singular points other than $(\varepsilon_0, 0)$. We denote the points $(x+\varepsilon_1, y)$ and $(x+\varepsilon_2, y)$ by $\tilde{A}$ and $\tilde{B}$, respectively. We assume that the segments $A\tilde{A}$ and $B\tilde{B}$ intersect $n_1$ and $n_2$ sides of 6-gons in the partition, not counting the side $l_1$ of $P$. Then the closed polygonal line $A\tilde{A}\tilde{B}BA$ intersects $n_1+n_2$ sides. All the vertices of 6-gons in the partition lying inside this polygonal line are distributed on the interval $[(\varepsilon_0, 0), (x+\varepsilon_0, y)]$. Moreover, *if $y$ is divisible by* 3, then there are *two* such vertices, an even and an odd one, and *if $y$ is not divisible by* 3, then there is only *one* such vertex. By the lemma, $n_1+n_2$ is even or odd, depending on whether $y$ is or is not divisible by 3. Hence, trajectories launched in the direction $(x, y)$ from adjacent segments (we have in mind the segments into which $[(0, 0), (1/3, 0)]$ is divided by the singular points) have multiplicities with equal parity if $y$ is divisible by 3 (this follows from the fact that the trajectories are odd-fold, proved above). If $y$ is not divisible by 3, then these trajectories have multiplicities of different parities.

Thus, Assertion 7 has been completely proved. □

***Corollary.*** *In Assertion 7 the condition $x > 0$ is redundant.*

*Proof.* Let $(x, y)$ be a pair of coprime integers, $y > 0$, $x \leqslant 0$. In $P$ we change the marking of sides according to the scheme of Fig. 48. Now our coordinate system in the plane changes. We can easily show that the 6-gon $(x, y)$ has coordinates $(y-x, y)$ in the new coordinate system. Since $y-x \geqslant y > 0$, Assertion 7 holds for the numbers $(x, y)$. □

Fig. 48

**F** . We consider trajectories launched from the side $l_1$ of $P$ in the direction $(x, y)$ and not hitting a vertex of this 6-gon. Here, $x$ and $y$ are coprime integers, $y > 0$, and $y$ is divisible by 3. By Assertion 7, each such trajectory is either 1- or 3-fold. They form $y/3$ pencils: each pencil consists of the trajectories launched from the segment $(((n-1)/y, 0), (n/y, 0))$, $n = 1, 2, \ldots, y/3$. Let $\varepsilon$ be an arbitrary number in the segment $(0, 1/y)$. We denote by $L_0, L_1, \ldots, L_{y/3-1}$ the trajectories launched in the direction $(x, y)$ from the points $(\varepsilon, 0), (1/y+\varepsilon, 0), \ldots, (1/3-1/y+\varepsilon, 0)$, respectively.

***Assertion 8.*** *Under the action of some transformation of $P$ in $\mathfrak{g}$, for any $s \in \{1, 2, \ldots, y/3-1\}$ the trajectory $L_s$ becomes the trajectory $L_0$ (up to the choice of the first link).*

*Proof.* The straightened trajectory $L_0$ is the segment $AB$, where $A = (\varepsilon, 0)$, $B = (kx+\varepsilon, ky)$. Here, $k$ is the multiplicity of the trajectory: $k = 1$ or $k = 3$. This segment intersects the line $\{(i, j) : j = m\}$, $m \in \mathbb{Z}$, $0 < m \leqslant y$, at the point $(m(x/y)+\varepsilon, m)$. Since $x$ and $y$ are coprime, the fractional parts $\{m(x/y)\}$, $m = 1, 2, ..., y$, coincide with the numbers $0, 1/y, ..., (y-1)/y$, taken in some order. In particular, for any $s \in \{1, 2, ..., y/3-1\}$ there is an integer $m$, $0 < m < y$, such that $(m(x/y)+\varepsilon, m) = (i+s/y+\varepsilon, m)$, where $i \in \mathbb{Z}$. We denote the point $(i+s/y+\varepsilon, m)$ by $C$. We let $\tilde{l}$ be the segment of $L_0$ lying inside the hexagon $(i, m)$. It is obvious that displacement by the vector $-(\overrightarrow{i,m})$ takes the segment $\tilde{l}$ into the first link of $L_s$. This implies that the first link of $L_s$ becomes the link of $L_0$ corresponding to $\tilde{l}$ under the action of the transformation $\tau$ which is the type of the sequence of hexagons in the partition through which $AC$ passes. But this means that under the action of $\tau$ ($\tau \in \mathfrak{g}$) the trajectory $L_s$ becomes $L_0$, up to the choice of the first link. Since we have taken $s$ as any number of the set $\{1, 2, ..., y/3-1\}$, the assertion has been proved. □

***Corollary.*** *The periodic trajectories launched in the direction $(x, y)$ are either all 1-fold or all 3-fold. Moreover, they are either all stable or all unstable.*

*Proof.* The trajectories in a pencil all have the same word, hence their multiplicities are equal, and they are also either all stable or all unstable. Moreover, under transformations of $P$ and change of the first link the multiplicity of a trajectory does not change (since, as follows from Assertion 5, the multiplicity of a trajectory is uniquely determined by the transformations of $\mathfrak{g}$ under which the trajectory is invariant), and stability and instability are preserved. It now remains to refer to Assertion 8. □

***Assertion 9.*** *Let $P_0, P_1, ..., P_m$ be a sequence of 6-gons in the partition of the plane with word $W$, where $P_m = P_0$, and suppose that there is a closed Jordan curve $\Gamma$ beginning in $P_0$ and passing successively through the 6-gons $P_0, P_1, ..., P_m$. Then $\mathscr{A}'(W) = \pm(3k_0, 3k_l)+(1, 1)\mathbb{Z}$, where $k_0$ and $k_l$ are the numbers of 'odd' and 'even' vertices of the 6-gons in the partition lying inside $\Gamma$, respectively; the sign + or − is chosen depending on the direction of traversal of $\Gamma$, clockwise or counterclockwise.*

*Proof.* Let the words $W_1$ and $W_2$ correspond to the contour $\Gamma$, described *counterclockwise* or *clockwise*, respectively. Then $W_1 \square W_2 = W_1 * \overline{W}_1$, where $\overline{W}_1$ is the word $W_1$ written in reverse order. The word $W_1 * \overline{W}_1$ has almost the same properties as the empty word. In particular, $\mathscr{A}'(W_1 * \overline{W}_1) = 0$. However,

$$\mathscr{A}'(W_1 * \overline{W}_1) = \mathscr{A}'(W_1 \square W_2) = \mathscr{A}'(W_1) + \mathscr{A}'(W_2).$$

Thus, $\mathscr{A}'(W_1) = -\mathscr{A}'(W_2)$, that is, it suffices to prove the assertion under the assumption that $\Gamma$ is traversed in the *counterclockwise* direction.

We carry out the proof by induction on the number of vertices of the 6-gons in the partition lying inside $\Gamma$. If there are no such vertices inside $\Gamma$, then we can clearly reduce $W$ to the empty word by eliminating the subword $W_1 * \overline{W}_1$, and then $\mathscr{A}(W) = 0$ and $\mathscr{A}'(W) = 0$.

Suppose that the assertion is satisfied for all contours enclosing fewer than $n$ vertices; let $\Gamma$ enclose exactly $n$ vertices of 6-gons in the partition. Let $x$ be one such vertex, and $H_0, H_1, H_2, H_3 = H_0$ the sequence of 6-gons having $x$ as a vertex, taken in the *clockwise* direction around $x$, and having the word $\widetilde{W} = l_1 l_6 l_1$ if $x$ is an *even* vertex and the word $\widetilde{W} = l_2 l_1 l_2$ if $x$ is an *odd* vertex. Also, let $P'_0, P'_1, \ldots, P'_{m'}$ be a sequence of 6-gons without self-intersection and not going outside $\Gamma$, and, moreover, such that $P'_0 = P_0$, $P'_{m'} = H_0$. We denote the word corresponding to this sequence by $W_1$, and the word corresponding to the reverse sequence by $W_{-1}$. Then we can easily see that there is a closed Jordan contour $\gamma$ beginning in $P_0$ and successively passing through the 6-gons

$$P_0, P_1, \ldots, P_m = P_0, P'_1, \ldots, P'_{m'} = H_0, H_1, H_2, H_3 = H_0,$$
$$P'_{m'-1}, \ldots, P'_0 = P_0,$$

that is, the word $W_0 = W \square W_1 \square \widetilde{W} \square W_{-1}$ corresponds to it. The contour $\gamma$ is described counterclockwise, and it encloses the same vertices of 6-gons in the partition as $\Gamma$, except for $x$. By the induction hypothesis, $\gamma$ and its word $W_0$ satisfy the assertion to be proved. However,

$$\mathscr{A}'(W_0) = \mathscr{A}'(W) + \mathscr{A}'(W_1) + \mathscr{A}'(\widetilde{W}) + \mathscr{A}'(W_{-1}) =$$
$$= \mathscr{A}'(W) + \mathscr{A}'(\widetilde{W}) + \mathscr{A}'(W_1 \square W_{-1}) = \mathscr{A}'(W) + \mathscr{A}'(\widetilde{W}) + \mathscr{A}'(W_1 * \overline{W}_1) =$$
$$= \mathscr{A}'(W) + \mathscr{A}'(\widetilde{W}).$$

We can immediately calculate that $\mathscr{A}(l_1 l_6 l_1) = -3\alpha_6$, $\mathscr{A}(l_2 l_1 l_2) = -3\alpha_1$, so $\mathscr{A}'(\widetilde{W}) = (-3, 0) + (1, 1)\mathbb{Z}$ if $x$ is an *odd* vertex and $\mathscr{A}(\widetilde{W}) = (0, -3) + (1, 1)\mathbb{Z}$ if $x$ is an *even* vertex. Since $\mathscr{A}'(W) = \mathscr{A}'(W_0) - \mathscr{A}'(\widetilde{W})$, the assertion to be proved holds for $\Gamma$ too. This completes the proof. $\square$

From Assertion 9 we can derive the following two corollaries.

***Corollary* 1.** *A periodic trajectory launched in a direction* $(x, y)$, *where* $x$, $y$ *are coprime integers,* $y > 0$, *and* $y$ *is not divisible by* 3, *is unstable.*

*Proof.* Let $A(\varepsilon, 0)$ be the point from which the trajectory is launched, $B$ the point $(x+\varepsilon, y)$, $W$ the *short* word corresponding to the trajectory (that is, the word corresponding to the sequence of 6-gons through which the segment $AB$ passes), and $W_{-1}$ the word corresponding to the reverse sequence (or to the

segment $BA$). Moreover, let

$$\widetilde{W}=\begin{cases}\underbrace{l_5\,\square\, l_5\,\square\ldots\square\, l_5}_{y\text{ times}}\,\square\,\underbrace{(l_3\,\square\, l_2)\,\square\ldots\square\,(l_3\,\square\, l_2)}_{x\text{ times}} & \text{if } x>0,\\ \underbrace{l_5\,\square\ldots\square\, l_5}_{y\text{ times}} & \text{if } x=0,\\ \underbrace{l_5\,\square\ldots\square\, l_5}_{y\text{ times}}\,\square\,\underbrace{(l_5\,\square\, l_6)\,\square\ldots\square\,(l_5\,\square\, l_6)}_{(-x)\text{ times}} & \text{if } x<0.\end{cases}$$

It is obvious that the sequence of 6-gons with word $\widetilde{W}$ and starting with $P$ ends with the 6-gon $(x, y)$. Therefore there is a closed Jordan contour beginning in $P$ and corresponding to the word $\widetilde{W}\,\square\, W_{-1}$. By Assertion 9, $\mathscr{A}'(\widetilde{W}\,\square\, W_{-1}) = (3k_1, 3k_2)+(1, 1)\mathbb{Z}$, where $k_1$, $k_2$ are certain integers. Since $\mathscr{A}'(\widetilde{W}\,\square\, W_{-1}) = \mathscr{A}'(\widetilde{W})+\mathscr{A}'(W_{-1})$ and $\mathscr{A}'(W)+\mathscr{A}'(W_{-1}) = \mathscr{A}'(W\,\square\, W_{-1}) = \mathscr{A}'(W*\overline{W}) = 0$, we have $\mathscr{A}'(W) = -\mathscr{A}'(W_{-1}) = \mathscr{A}'(\widetilde{W})-\mathscr{A}'(\widetilde{W}\,\square\, W_{-1})$. Further, by direct calculation we see that

$$\mathscr{A}(l_5)=2\alpha_5+\alpha_6,\quad \mathscr{A}(l_3\square l_2)=-3\alpha_1-3\alpha_2,\quad \mathscr{A}(l_5\square l_6)=3\alpha_5+3\alpha_6,$$

implying that

$$\mathscr{A}'(l_5)=(2, 1)+(1, 1)\mathbb{Z}=(1, 0)+(1, 1)\mathbb{Z},\quad \mathscr{A}'(l_3\square l_2)=\mathscr{A}'(l_5\square l_6)=0.$$

Then $\mathscr{A}'(\widetilde{W}) = (y, 0)+(1, 1)\mathbb{Z}$ and $\mathscr{A}'(W) = \mathscr{A}'(\widetilde{W})-\mathscr{A}'(\widetilde{W}\,\square\, W_{-1}) = (y-3k_1, -3k_2)+(1, 1)\mathbb{Z}$. Since $y$ is not divisible by 3, we have $y-3k_1 \neq -3k_2$, that is, $\mathscr{A}'(W) \neq 0$ and, by Corollary 2 to Assertion 6, the trajectory under consideration is unstable. $\square$

***Corollary* 2.** *A* 3*-fold periodic billiard trajectory launched in a direction* $(x, y)$, *where* $x$, $y$ *are coprime integers,* $y > 0$, *and* $y$ *is divisible by* 3, *is stable.*

*Proof.* We first consider the case when $x > 0$. By Corollary 2 to Assertion 6, it suffices to consider the case when the trajectory is launched from a point $A(\varepsilon, 0)$, where $0 < \varepsilon < 1/y$. Let $W$ be the short word of this trajectory, that is, the word corresponding to the sequence of 6-gons through which the segment $AB$ passes, where $B = (x+\varepsilon, y)$, and let $W_{-1}$ be the word corresponding to the reverse sequence. Then

$$\mathscr{A}'(W\square W_{-1})=\mathscr{A}'(W*\overline{W})=0,$$

$$\mathscr{A}'(W)=\mathscr{A}'(W\square W_{-1})-\mathscr{A}'(W_{-1})=-\mathscr{A}(W_{-1}).$$

Let $\widetilde{AB}$ be a curve beginning at $A$ and ending at $B$, not having interior points in common with the segment $AB$, and successively passing through the 6-gons $P_0 = P, P_1, \ldots, P_m = (x, y)$, corresponding to the word $\widetilde{W}=\underbrace{(l_5\,\square\, l_5\,\square\ldots\square\, l_5)}_{y\text{ times}}\,\square\,\underbrace{(l_3\,\square\, l_2)\,\square\ldots\square\,(l_3\,\square\, l_2)}_{x\text{ times}}$. Then $\gamma = \widetilde{AB}+BA$ is a

closed Jordan contour with word $\widetilde{W} \square W_{-1}$, traversed in the clockwise direction (Fig. 47). As is clear from the proof of Assertion 7, this contour encloses $y/3$ more even vertices than odd ones, hence, by Assertion 9,

$$A'(\widetilde{W} \square W_{-1}) = -\left(0,\ 3 \cdot \frac{y}{3}\right) + (1, 1)\mathbb{Z} = (y, 0) + (1, 1)\mathbb{Z}.$$

Moreover, in the proof of the previous corollary we calculated that $\mathscr{A}'(\widetilde{W}) = (y, 0) + (1, 1)\mathbb{Z}$. Then

$$\mathscr{A}'(W) = -\mathscr{A}'(W_{-1}) = -(\mathscr{A}'(\widetilde{W} \square W_{-1}) - \mathscr{A}'(\widetilde{W})) = 0,$$

and by Corollary 2 to Assertion 6, the trajectory under consideration is stable (since it has multiplicity 3).

Now suppose that $x \leqslant 0$. As in the proof of the corollary to Assertion 7 we change the marks of the sides of the hexagon $P$ using the scheme of Fig. 48. Then we have also changed the coordinate system, and in the new coordinates the direction $(x, y)$ becomes $(y-x, y)$. Since $y-x \geqslant y > 0$, the stability of a 3-fold trajectory launched in the direction $(x, y)$ (in the ordinary coordinates) follows from what we proved above. $\square$

*Remark.* In marking the sides of the 6-gon according to the new scheme the orientation has changed, and, as we can easily see, the forms and stroke-forms change sign. However, this does not influence the course of reasoning.

***Assertion* 10.** *Suppose that the two trajectories launched in the same direction from points close to a vertex of $P$ but on different sides of it are both* 1-*fold. Then they cannot both be stable.*

*Proof.* Since $P$ is a regular 6-gon, without loss of generality we may assume that the vertex in question is the one at the angle $\alpha_6$. We denote the points from which the trajectories are launched by $A_1$ and $A_2$, with $A_1 \in l_1$, $A_2 \in l_6$, and let $(x, y)$ be the direction in which they go, with $y > 0$ and $x, y$ coprime integers. Since the trajectories are 1-*fold*, their developments are the segments $A_1B_1$ and $A_2B_2$, where $B_1 = A_1 + (x, y)$, $B_2 = A_2 + (x, y)$ (Fig. 49).

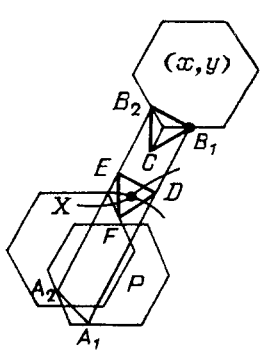



Fig. 49

Let $W_1$ and $W_2$ be the words corresponding to the sequences of 6-gons through which $A_1B_1$ and $A_2B_2$ pass, and let $W_{-1}$ and $W_{-2}$ be the words corresponding to the reverse sequences. It is obvious that for stability of the first, respectively second, trajectory it is necessary and sufficient that $\mathscr{A}(W_1) = 0$, respectively $\mathscr{A}(W_2) = 0$.

In fact, for the first trajectory this follows from Assertion 6, and to apply Assertion 6 to the second trajectory we first have to rename the sides of $P$ so that $l_6$ has the mark $l_1$. However, the form corresponding to the segment $A_2B_2$ with the new marks differs from the form $\mathscr{A}(W_2)$ only by permutation of coefficients.

So, we may assume that both trajectories are stable, that is, $\mathscr{A}(W_1) = \mathscr{A}(W_2) = 0$. Let $\widetilde{W}$ be the word corresponding to the closed polygonal line $A_1B_1B_2A_2A_1$. It is obvious that $\widetilde{W} = W_1 \square W_{-2}$, and since $\mathrm{type}(W_1) = \mathrm{type}(W_2) = \mathrm{id}$ (since $W_1$ and $W_2$ are words corresponding to periodic trajectories), we obtain

$$0=\mathscr{A}(W_2*\overline{W}_2)=\mathscr{A}(W_2\square W_{-2})=\mathscr{A}(W_2)+\ \mathrm{type}(W_2)\mathscr{A}(W_{-2})=\mathscr{A}(W_{-2}),$$

whence

$$\mathscr{A}(\widetilde{W})=\mathscr{A}(W_1\square W_{-2})=\mathscr{A}(W_1)+\mathrm{type}(W_1)\mathscr{A}(W_{-2})=0.$$

Now clearly $y$ is divisible by 3 (otherwise the trajectory launched from $A_1$ would be unstable, by Corollary 1 to Assertion 9). Since $A_1$ and $A_2$ are close to the vertex (0, 0), we may assume that the intervals ((0, 0), $A_1$) and ((0, 0), $A_2$) do not contain points that are *singular* for the direction $(x, y)$. Hence all the vertices of 6-gons in the partition lying inside the polygonal line $A_1B_1B_2A_2A_1$ belong to the line $\{(i, j) : i/j = x/y\}$. It is easy to see that there are only *two* such vertices: one *even*, this is $(x, y)$, and one *odd*, this is either $(x/3, y/3)$, or $(2x/3, 2y/3)$, depending on whether the residue when $x$ is divided by 3 is 1 or 2 (here we have used the fact that $y$ is divisible by 3). We denote the odd vertex by $X$.

We choose points $C$, $D$, $E$, $F$ such that the following conditions hold: $D$ lies inside $A_1B_1$, and $E$ inside $A_2B_2$; the points $X$ and $F$ are inside the closed polygonal line $A_1DEA_2A_1$ and $(x, y)$ and $C$ inside $DB_1B_2ED$; the closed polygonal line $B_1B_2CB_1$ goes round the point $(x, y)$ once in the *counterclockwise* direction, and passes through only three 6-gons having $(x, y)$ as vertex; the closed polygonal line $DEFD$ goes round $X$ in a similar way (Fig. 49).

We let $L_1$ be the segment $A_1D$, $L_2$ be $DB_1$, $L_0$ be the polygonal line $B_1B_2CB_1$, $\widetilde{L}_2$ be $B_1CB_2ED$, $L^0$ be $DEFD$, and $\widetilde{L}_1$ be $DEA_2A_1$. We use the same letters to denote the words corresponding to the sequences of 6-gons in the partition through which these polygonal lines and segments pass. Then the word $L_1 \square L_2 \square L_0 \square \widetilde{L}_2 \square \widetilde{L}_1$ corresponding to the polygonal line $A_1B_1B_2CB_1CB_2EDEA_2A_1$ can clearly be obtained from the word $\widetilde{W}$ by inserting the words $U*\overline{U}$ and $V*\overline{V}$, where $U$ corresponds to $B_2CB_1$ and $V$

to $ED$. Hence,

$$\mathscr{A}(L_1\square L_2\square L_0\square \tilde{L}_2\square \tilde{L}_1)=\mathscr{A}(\mathscr{W})=0.$$

On the other hand,

$$\mathscr{A}(L_1\square L_2\square L_0\square \tilde{L}_2\square \tilde{L}_1)=\mathscr{A}(L_1\square L_2)+\operatorname{type}(L_1\square L_2)\mathscr{A}(L_0)+ \\ +[\operatorname{type}(L_1\square L_2)\circ\operatorname{type}(L_0)]\mathscr{A}(\tilde{L}_2\square \tilde{L}_1).$$

It is obvious that $L_1\square L_2$ corresponds to $A_1B_1$, that is, $L_1\square L_2 = W_1$. Then $\operatorname{type}(L_1\square L_2) = \mathrm{id}$ and $\mathscr{A}(L_1\square L_2) = 0$. Moreover, by construction $L_0 = l_6l_1l_6$, so $\operatorname{type}(L_0) = (\alpha_2\alpha_4)(\alpha_1\alpha_5) =: \sigma$, $\mathscr{A}(L_0) = 3\alpha_6$, and hence

$$0=\mathscr{A}(L_1\square L_2)+\operatorname{type}(L_1\square L_2)\mathscr{A}(L_0)+[\operatorname{type}(L_1\square L_2)\circ\operatorname{type}(L_0)]\mathscr{A}(\tilde{L}_2\square \tilde{L}_1)= \\ =\mathscr{A}(L_0)+\sigma\mathscr{A}(\tilde{L}_2\square \tilde{L}_1).$$

This implies that $\sigma\mathscr{A}(\tilde{L}_2\square \tilde{L}_1) = -\mathscr{A}(L_0) = -3\alpha_6$ and $\mathscr{A}(\tilde{L}_2\square \tilde{L}_1) = \sigma^2\mathscr{A}(\tilde{L}_2\square \tilde{L}_1) = -3\sigma(\alpha_6) = -3\alpha_6$. Then

$$\mathscr{A}(L_1\square L_2\square \tilde{L}_2\square \tilde{L}_1)=\mathscr{A}(L_1\square L_2)+\operatorname{type}(L_1\square L_2)\cdot\mathscr{A}(\tilde{L}_2\square \tilde{L}_1)=-3\alpha_6.$$

The word $L_2\square \tilde{L}_2$ corresponds to the closed polygonal line $DB_1CB_2ED$, inside which there are no vertices of hexagons in the partition. Hence this word can be reduced to the empty word by cancelling subwords of the form $L*\bar{L}$, and since $\operatorname{type}(\nu) = \mathrm{id}$, $\mathscr{A}(L*\bar{L}) = 0$, we have

$$\mathscr{A}(L_1\square \tilde{L}_1)=\mathscr{A}(L_1\square L_2\square \tilde{L}_2\square \tilde{L}_1)=-3\alpha_6.$$

We can similarly reduce $L_1\square L^0\square \tilde{L}_1$, which corresponds to the polygonal line $A_1DFEDEA_2A_1$, to the empty word, hence $\mathscr{A}(L_1\square L^0\square \tilde{L}_1) = 0$. This implies that $0 = \mathscr{A}(L_1)+\tau\mathscr{A}(L^0)+\tau\varkappa\mathscr{A}(\tilde{L}_1)$, where $\tau = \operatorname{type}(L_1)$ and $\varkappa = \operatorname{type}(L^0)$. By construction, $L^0$ consists of three letters, $\mathscr{A}(L^0) = -3\alpha_n$, where $n\in\{1, 3, 5\}$ (since $X$ is an *odd* vertex), and $\varkappa$ is a symmetry leaving the angle $\alpha_n$ fixed. Then

$$0=\mathscr{A}(L_1)-3\tau(\alpha_n)+\tau\varkappa\mathscr{A}(\tilde{L}_1).$$

Moreover, as we showed above, $-3\alpha_6 = \mathscr{A}(L_1\square \tilde{L}_1) = \mathscr{A}(L_1)+\tau\mathscr{A}(\tilde{L}_1)$, so

$$-3\tau(\alpha_n)+\tau\varkappa\mathscr{A}(\tilde{L}_1)-\tau\mathscr{A}(\tilde{L}_1)=3\alpha_6(\operatorname{mod}(\alpha_1+\ldots+\alpha_6)).$$

Let $\mathscr{A}(\tilde{L}_1) = k_1\alpha_1+\ \ldots\ +k_6\alpha_6$. Since $\varkappa(\alpha_n) = \alpha_n$, we have

$$\varkappa\mathscr{A}(\tilde{L}_1)=k_n\alpha_n+\sum_{\substack{1\le l\le 6\\ l\ne n}}k_l\cdot\varkappa(\alpha_l).$$

Then

$$\tau\varkappa\mathscr{A}(\tilde{L}_1)-\tau\mathscr{A}(\tilde{L}_1)=\sum_{\substack{1\le l\le 6\\ l\ne n}}k_l(\tau\varkappa(\alpha_l)-\tau(\alpha_l))=\sum_{\substack{1\le l\le 6\\ \alpha_l\ne\tau(\alpha_n)}}k_l'\alpha_l,$$

where $\sum_{\substack{1\le l\le 6\\ \alpha_l\ne\tau(\alpha_n)}} k_l' = 0$. Thus,

$$-3\tau(\alpha_n)+\sum_{\substack{1\le l\le 6\\ \alpha_l\ne\tau(\alpha_n)}} k_l'\alpha_l=3\alpha_6(\mathrm{mod}(\alpha_1+\ldots+\alpha_6)).$$

Since the sum of the coefficients of the form $-3\tau(\alpha_n)+\sum_l k_l'\alpha_l$ is equal to $-3$ and that of the form $3\alpha_6$ is equal to 3, the forms $-3\tau(\alpha_n)+\sum_l k_l'\alpha_l$ and $3\alpha_6-(\alpha_1+\ \ldots\ +\alpha_6) = -\alpha_1-\alpha_2-\alpha_3-\alpha_4-\alpha_5+2\alpha_6$ coincide not only up to a term of the form $m(\alpha_1+\ \ldots\ +\alpha_6)$, $m\in\mathbb{Z}$, but also coefficientwise. However, this is impossible: in the first form the coefficient of the angle $\tau(\alpha_n)$ is $-3$, and in the second it is $-1$ (because $\tau(\alpha_n)$, like $\alpha_n$, is an odd angle, and thus $\tau(\alpha_n)\ne\alpha_6$).

This contradiction shows that the assumed equality $\mathscr{A}(W_1) = \mathscr{A}(W_2) = 0$ is false. Thus, at least one of the forms $\mathscr{A}(W_1)$, $\mathscr{A}(W_2)$ is not equal to zero, and then at least one of the trajectories whose developments are $A_1B_1$ and $A_2B_2$ is unstable. □

***Assertion* 11.** *The* 1*-fold periodic billiard trajectories in the hexagon P are unstable.*

*Proof.* Let $(x, y)$, where $x$, $y$ are coprime integers, $y > 0$, be a direction along which a 1-*fold* trajectory may be launched from the side $l_1$ of $P$. We note that the 1-fold trajectories launched in this direction are either all *stable* or all *unstable*: if $y$ is divisible by 3 this follows from the corollary to Assertion 8, and if $y$ is not divisible by 3, then all such trajectories are unstable by Corollary 1 to Assertion 9.

We first prove the *instability of* 1*-fold trajectories* launched in the direction $(x, y)$ for which there are interior singular points on $l_1$.

If $y$ is not divisible by 3, then Corollary 1 to Assertion 9 implies that the trajectory is unstable. If $y$ is divisible by 3, then by the corollary to Assertion 8, all periodic trajectories launched in the direction $(x, y)$ are either 3-fold or 1-fold. In the first case there is nothing to prove (we do not consider 3-fold trajectories), and we consider the second case separately.

So, let $q$ be an interior point of the side $l_1$ of $P$ which is singular for the direction $(x, y)$, and let $B_1$, $B_2$ be two non-singular points on $l_1$ between which there are no singular points for the direction $(x, y)$ except $q$. We launch billiard trajectories $\Gamma_1$ and $\Gamma_2$ from the points $B_1$ and $B_2$ in the direction $(x, y)$; by hypothesis these are 1-fold. From $q$ we launch a trajectory $\Gamma$ in the same direction. Then the first reflections of $\Gamma$, $\Gamma_1$, and $\Gamma_2$ are in the *same* side of $P$, and $\Gamma$ subsequently hits the vertex $X$ of $P$. At this moment the trajectories $\Gamma_1$ and $\Gamma_2$ hit points $A_1$ and $A_2$ lying on *different* sides of $P$ emanating from the vertex $X$.

We denote by $\gamma_1$ and $\gamma_2$ the billiard trajectories launched from $A_1$ and $A_2$, respectively, in the direction opposite to the direction in which $\Gamma_1$ and $\Gamma_2$ approached $A_1$ and $A_2$. The trajectories $\gamma_1$ and $\gamma_2$ differ from $\Gamma_1$ and $\Gamma_2$ only in their direction and choice of initial points, hence they are 1-*fold* (by Assertion 5, 1-fold periodic trajectories are characterized by the fact that they are not invariant under non-identity transformations in $\mathfrak{g}$, hence the fact that a trajectory is 1-fold does not depend on the direction of motion along it and on the choice of initial point). Moreover, the trajectories launched from the segments $(A_1, X)$ and $(A_2, X)$ in the same direction as $\gamma_1$ and $\gamma_2$ do not hit a vertex of $P$. Therefore, by Assertion 10, $\gamma_1$ and $\gamma_2$ cannot both be stable. Then also $\Gamma_1$ and $\Gamma_2$ cannot both be stable, and by the remark made above, all 1-fold trajectories launched in the direction $(x, y)$ are unstable. Thus we have proved this particular case of the assertion.

Suppose now that the side $l_1$ does not contain singular points for the direction $(x, y)$. Then all trajectories launched from $l_1$ in the direction $(x, y)$ (which are all 1-fold), reflect first in one and the same side, $l_i$, $i \neq 1$. When the point $q$ on $l_1$ approaches the endpoints of this side, the end of the first link of the trajectory launched from $q$ in the direction $(x, y)$ approaches certain points $u$ and $v$ on $l_i$. Suppose that one of these, say $u$, is not a vertex of $P$. Then the side $l_i$ of $P$ has an interior singular point for the direction $-(x, y)$, namely $u$. Therefore, by what we proved earlier, all 1-fold trajectories launched from $l_i$ in the direction $-(x, y)$ are unstable. In particular, the trajectories launched from the interval $(u, v)$ are unstable. However, up to the choice of the direction of motion and the initial point, all such trajectories coincide with trajectories launched from $l_1$ in the direction $(x, y)$. Thus, all such trajectories are also unstable.

Finally, if $u$ and $v$ are vertices of $P$, then we can easily see that the trajectories are launched at an angle of 30°, 60° or 90° to $l_1$. It is not difficult to verify that the trajectories launched at an angle of 30° or 60° to $l_1$ are 3-fold, while the instability of the trajectories launched perpendicular to $l_1$ is obvious.

Thus we have completely proved Assertion 11. □

## Conclusion

Here we give some assertions not included in the main text. We also pose some questions and state some conjectures which arose during our investigation of periodic billiard trajectories in polygons.

1. The authors of [3] posed the following problem (Problem 8): *do there exist in a given triangle infinitely many pairwise inequivalent (having distinct combinatorial types) periodic billiard trajectories*?

While the answer to this question in its full generality is not known, by using the results of the present paper it is nevertheless possible to state that

*the average number of distinct types of periodic trajectories in a triangle is infinite.*

We can strengthen this assertion by adding the word 'stable' (in the proof, it suffices to consider isolated trajectories in acute triangles). There are infinitely many types of periodic trajectories in an *equilateral* triangle (according to §12, even the number of types of *stable* trajectories is infinite), and in an *arbitrary right-angled* triangle (this follows directly from Fig. 50). The periodic trajectories in the right-angled triangle constructed in Fig. 50 are unstable. Taking into account the results of §12 concerning right-angled triangles with angles of 30° or 45°, the following question arises.

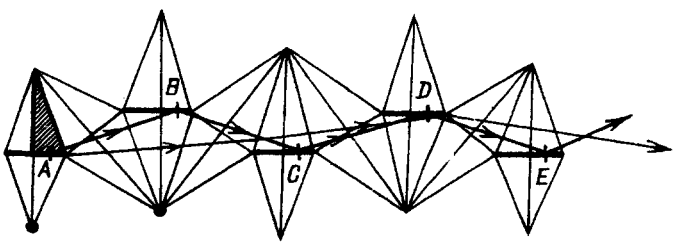



Fig. 50

***Question.*** *Does there exist at least one right-angled triangle containing stable trajectories? If so, what sort of set do they form?*

**2.** It seems that the following conjecture is true.

***Conjecture* 1.** *The set of triangles in which there are periodic trajectories contains an open everywhere dense subset in the space of all triangles.*

The next conjecture is equivalent to it.

***Conjecture* 2.** *There is an everywhere dense set of rational triangles containing stable periodic trajectories.*

Conjecture 2 may be strengthened as follows.

***Conjecture* 3.** *There is an everywhere dense set of rational triangles containing isolated trajectories.*

Similar conjectures can be naturally formulated for arbitrary polygons.

**3.** We now consider an isosceles triangle $\triangle$ with angles of 30° at the base. The following theorem holds.

***Theorem.*** (a) *If $\gamma$ is a periodic trajectory in $\triangle$ that is symmetric with respect to the altitude of this triangle, then $\gamma$ is unstable.* (b) *The triangle $\triangle$ contains periodic trajectories of at least two distinct types, corresponding to two different words $W_1$ and $W_2$.*

The proof of this theorem mainly repeats the reasoning of §13. For the words mentioned in (b) we may take the following words, of least possible length:

$$W_1 = bcbcacbcacabcbcbcacaca \text{ and } W_2 = cbcbcbacacbcacbcbacaca.$$

The stable periodic trajectories corresponding to them have 22 links and are launched at angles $\alpha_1 = \arctan(\sqrt{3}/5)$ and $\alpha_2 = \arctan(\sqrt{3}/2)$ to a lateral side of $\triangle$.

***Conjecture* 4.** *All periodic trajectories launched at a small angle to a lateral side of $\triangle$ are unstable.*

**4.** The geometrical condition of invariance of a trajectory given in the criterion for stability of a periodic trajectory in a regular 6-gon can also be stated in purely arithmetical terms, making it possible to algorithmetize the search for such trajectories.

In this paper we have considered plane billiards, except in §5. A forthcoming publication of ours will be devoted to the construction and classification of periodic trajectories in polyhedra.

## References


[1] V.I. Arnol'd, *Matematicheskie metody klassicheskoi mekhaniki*, Nauka, Moscow 1974. MR **57** # 14032.
Translation: Mathematical methods of classical mechanics, Springer-Verlag, New York 1978.

[2] M. Berger, *Géométrie*, Vols. 1–5, Nathan Information, Paris 1977. MR **81k**:51001a–e. (Especially Ch. 9.)

[3] C. Boldrighini, M. Keane, and F. Marchetti, Billiards in polygons, Ann. Probab. **6** (1978), 532–540. MR **58** # 31007b.

[4] N. Bourbaki, *Groupes et algèbres de Lie*, Masson, Paris 1981–1982, Ch. 4–6 (MR **83g**:17001) and Ch. 9 (MR **84i**:22001).

[5] L.A. Bunimovich, Ya.G. Sinai, and N.N. Chernov, Markov partitions for two-dimensional hyperbolic billiards, Uspekhi Mat. Nauk **45**:3 (1990), 97–133. = Russian Math. Surveys **45**:3 (1990), 105–152 (especially §7.3).

[6] G.A. Gal'perin and N.I. Chernov, *Billiardy i khaos* (Billiards and chaos), Izdat. Znachenie, Moscow 1991. (Matematika i kibernetika, No. 5).

[7] ——— and A.N. Zemlyakov, *Matematicheskie bil'yardy* (Mathematical billiards), Nauka, Moscow 1990. (Biblioteka Kvant, No. 77.)

[8] A. Katok, The growth rate for the number of singular and periodic orbits for a polygonal billiard, Commun. Math. Phys. **111** (1987), 151–160. MR **88g**:58162.

[9] I.P. Kornfel'd, Ya.G. Sinai, and S.V. Fomin, *Ergodicheskaya teoria*, Nauka, Moscow 1980. MR **83a**:28107.
Translation: Ergodic theory, Springer-Verlag, New York 1982.

[10] H. Masur, Closed trajectories for quadratic differentials with an application to billiards, Duke Math. J. **53** (1986), 307–314. MR **87j**:30107.

[11] Ya.G. Sinai, *Vvedenie v ergodicheskuyu teoriyu*, Erevan 1973. MR **58** # 28437j.
Translation: Introduction to ergodic theory, Princeton Univ. Press, Princeton, NJ 1976.

[12] D.V. Treshchev, On the problem of the existence of periodic trajectories of the Birkhoff billiard, Vestn. Moskov. Univ. Ser. 1 Mat. Mekh. **1987**, no. 5, 72–75. MR **90c**:58155.

[13] Ya.B. Vorobets, G.A. Gal'perin, and A.M. Stepin, Periodic billiard trajectories in polygons, Uspekhi Mat. Nauk **46**:5 (1991), 165–166.
= Russian Math. Surveys **46**:5 (1991), 204–205.

[14] A.N. Zemlyakov and A.B. Katok, Topological transivity of billiards in polygons, Mat. Zametki **18** (1975), 291–300. MR **53** # 3267.
= Math. Notes **18** (1975), 760–764.

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